Gaussian Splatting and Neural Rendering for Dynamic 3D Scene Reconstruction: A Comprehensive Survey

1 Evolution of Neural Representations: From Implicit Fields to Explicit Primitives

1.1 The Paradigm Shift from Volumetric Rendering to Explicit Rasterization

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The landscape of neural scene representation has undergone a seismic shift in recent years, moving decisively away from the computational constraints of implicit volumetric rendering toward the high-efficiency paradigm of explicit rasterization. For a significant period, Neural Radiance Fields (NeRFs) stood as the dominant methodology for novel view synthesis, captivating the research community with their ability to reconstruct complex 3D scenes with unprecedented photorealism. However, this fidelity came at a steep computational cost. The fundamental bottleneck of NeRF lies in its reliance on implicit representations, where the scene geometry and appearance are encoded within the weights of a Multi-Layer Perceptron (MLP). To render a single image, traditional NeRF pipelines must query this network millions of times per frame, integrating color and density values along rays cast through the volume. This volumetric rendering process, while mathematically elegant, results in prohibitively slow training and inference times, effectively hindering real-time applications and interactive editing [1]. The emergence of 3D Gaussian Splatting (3DGS) represents a critical inflection point, challenging the necessity of neural network queries for every pixel and introducing a paradigm where explicit, point-based primitives are rendered via standard graphics rasterization pipelines.

This transition is characterized by a fundamental change in how light transport is simulated. In the implicit regime, the scene is a continuous function mapped by neural networks, requiring dense sampling to resolve high-frequency details, which creates a severe disparity between rendering quality and computational speed. As noted in recent literature, the computational bottleneck due to volumetric rendering has been a persistent challenge hindering the widespread adoption of NeRFs in resource-constrained environments [2]. Conversely, 3D Gaussian Splatting abandons the voxel-grid or ray-marching paradigm in favor of representing the scene as a collection of learnable 3D Gaussian ellipsoids. These primitives possess explicit attributes—position, covariance, opacity, and spherical harmonic coefficients for view-dependent color—which can be directly projected onto the 2D image plane. This shift allows the utilization of highly optimized, tile-based rasterizers originally designed for polygon meshes, thereby bypassing the expensive MLP evaluations that defined the previous generation of neural renderers [1].

The efficiency gains afforded by this paradigm shift are substantial. By leveraging differentiable point-based projections, 3DGS achieves real-time rendering speeds that were previously unattainable with implicit methods. The rasterization process sorts and blends Gaussian splats in screen space, an operation that is inherently more parallelizable and hardware-friendly than the sequential ray integration required by NeRFs. Studies have demonstrated that this approach not only accelerates inference but also significantly reduces training times, enabling the rapid reconstruction of high-fidelity scenes [3]. Furthermore, the explicit nature of the Gaussian representation facilitates a level of editability and physical simulation that is notoriously difficult to achieve with implicit "black box" neural networks. Because the scene is composed of discrete, manipulatable primitives, tasks such as dynamic reconstruction, geometry editing, and physical interaction become feasible without the need for complex latent space disentanglement [1].

Despite these advantages, the transition to explicit rasterization introduces new challenges regarding memory consumption and spatial distribution that necessitate further architectural evolution. While 3DGS eliminates the query bottleneck, it often requires millions of Gaussian primitives to maintain the high fidelity associated with NeRFs, leading to substantial storage overheads. Research indicates that a significant drawback of current 3DGS implementations is the substantial number of 3D Gaussians required, which demands large amounts of memory and storage [2]. This has spurred a secondary wave of optimization focused on pruning, quantization, and efficient initialization strategies to ensure that the benefits of rasterization do not come at the expense of scalability. Techniques such as learnable mask strategies and vector quantization have been developed to reduce the number of Gaussians and compress their attributes without sacrificing visual quality, highlighting the ongoing evolution of this explicit paradigm [2].

Moreover, the shift to explicit primitives has redefined the relationship between geometry and appearance, exposing a critical limitation that motivates the subsequent exploration of hybrid models. In volumetric rendering, geometry is an emergent property of the density field, often leading to ambiguous surface definitions; in contrast, explicit Gaussian primitives offer a more direct handle on geometric structure, yet early iterations struggled with view-consistent geometry and topological integrity. Recent advancements have begun to address this by refining projection models and introducing priors to align explicit primitives with true scene geometry. For instance, the analysis of projection errors in Gaussian Splatting has led to optimal projection strategies that reduce artifacts and improve photo-realism, demonstrating that the rasterization paradigm is still maturing [4]. Additionally, the integration of depth and normal priors into the optimization process helps align these explicit primitives with true scene geometry, bridging the gap between the speed of rasterization and the structural accuracy of traditional computer graphics [5]. However, while these improvements enhance explicit methods, they also reveal the inherent trade-offs between the rendering efficiency of Gaussians and the geometric rigor of implicit surfaces, setting the stage for architectures that seek to unify these complementary strengths.

In summary, the move from implicit volumetric rendering to explicit rasterization marks a transformative era in neural rendering. It resolves the long-standing trade-off between rendering speed and image quality by replacing computationally intensive neural queries with efficient, differentiable point projections. While challenges regarding memory footprint and geometric consistency remain active areas of research, the foundational shift established by 3D Gaussian Splatting has unlocked new possibilities for real-time applications, interactive editing, and large-scale scene reconstruction, firmly establishing explicit primitives as the forefront of 3D representation technology [6]. This evolution naturally leads to the next logical step: reconciling the speed of explicit primitives with the geometric precision of implicit fields through hybrid architectures.

1.2 Hybrid Architectures Bridging Implicit Surfaces and Explicit Primitives

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While the transition to explicit rasterization established in the previous section has revolutionized rendering speeds, it inadvertently exposed a critical trade-off: the sacrifice of geometric rigor for computational efficiency. Pure 3D Gaussian Splatting (3DGS) excels at novel view synthesis but often struggles to produce watertight, topologically consistent meshes, frequently manifesting as "floaters" or ambiguous surface definitions. Conversely, implicit methods based on Signed Distance Functions (SDFs) offer mathematically precise surface representations but remain burdened by the slow ray-marching processes that 3DGS sought to eliminate. To reconcile these divergent strengths, recent research has coalesced around hybrid architectures that integrate SDFs with Gaussian primitives. These dual-branch frameworks aim to harness the real-time rasterization capabilities of explicit methods while leveraging the rigorous geometric priors of implicit representations to enforce surface fidelity, directly addressing the structural limitations highlighted at the end of the preceding discussion.

A foundational challenge in this domain is bridging the structural misalignment between the continuous zero-level sets of SDFs and the discrete, opaque nature of Gaussian clouds. Early attempts often relied on separate optimization streams, but modern hybrid approaches prioritize joint optimization where the implicit field actively guides the spatial distribution of explicit primitives. A pivotal contribution in this direction is the work presented in [7], which introduces a mechanism to incorporate an implicit SDF directly within the 3D Gaussian framework. This method utilizes a differentiable SDF-to-opacity transformation function, effectively converting signed distance values into Gaussian opacities. By doing so, it creates a unified optimization landscape where the geometric constraints of the SDF regularize the positioning and scaling of the Gaussians. This alignment ensures that the high-frequency details captured by the Gaussians adhere strictly to the underlying continuous surface defined by the SDF, thereby eliminating redundant surfaces outside the sampling range through consistency regularization.

Building upon the concept of mutual guidance, other frameworks have adopted a dual-branch architecture where an explicit surrogate mesh and an implicit SDF field are learned in parallel. The study [8] exemplifies this strategy by unifying volume rendering of the implicit SDF with surface rendering of a surrogate mesh via a shared neural shader. In this setup, the synchronized learning process drives the deformation of the explicit mesh based on functions induced from the implicit SDF. This synergy not only promotes convergence to a single, coherent underlying surface but also enables surface-guided volume sampling. Such a mechanism significantly improves sampling efficiency per ray, addressing the computational bottlenecks traditionally associated with pure volumetric rendering while maintaining the high recovery quality of implicit methods.

The integration of SDFs into explicit frameworks also addresses specific failure modes related to material properties and transparency, which pure Gaussian or pure SDF methods handle suboptimally in isolation. For scenarios involving semi-transparent or thin objects, the decoupling of geometry and opacity becomes critical. The approach detailed in [9] demonstrates how implicit representations can be adapted to handle complex light transport. By utilizing analytical solutions for ray-surface intersections within an implicit framework, this method avoids the artifacts common in coupled volumetric approaches. When extended to hybrid systems, such decoupling allows Gaussian primitives to model the radiance field while the SDF branch strictly enforces the geometric boundary, ensuring that semi-transparent layers are reconstructed with physical plausibility.

Furthermore, the robustness of hybrid architectures is enhanced by leveraging the gradient information inherent in SDFs to refine the placement of explicit primitives. The theoretical underpinnings of gradient consistency, as explored in [10], highlight the importance of parallel level sets for accurate geometry inference. In a hybrid context, these gradient constraints can be imposed on the Gaussian centers, ensuring that the discrete point cloud aligns with the normal fields of the implicit surface. This is particularly vital for reconstructing complex topologies where simple density thresholds fail. Additionally, the incorporation of occupancy aids, as discussed in [11], suggests that combining occupancy representations with SDFs can mitigate optimization biases in low-intensity areas. When applied to hybrid Gaussian-SDF systems, this principle helps stabilize the learning process in texture-less regions where visual cues are sparse, preventing the collapse of explicit primitives into ambiguous configurations.

The versatility of these hybrid models extends to dynamic and articulated scenarios as well. While many initial hybrid works focused on static scenes, the principles of anchoring explicit primitives to implicit fields provide a robust foundation for deformation. By treating the SDF as a canonical space and the Gaussians as deformable entities bound to it, systems can achieve both temporal consistency and real-time performance. The success of methods like [12] further validates that the combination of flexible 3DGS representations with neural SDFs unlocks potential for more accurate surface reconstructions in large-scale, intricate scenes. These frameworks effectively alleviate the inherent constraints of choosing a single representation, demonstrating that the future of high-fidelity 3D reconstruction lies in the synergistic fusion of implicit geometric rigor and explicit rendering efficiency. However, while hybrid architectures successfully stabilize geometry, they often inherit the initialization dependencies of their explicit components. The reliance on high-quality point clouds from Structure-from-Motion (SfM) remains a bottleneck, motivating the subsequent exploration of optimization strategies that decouple Gaussian Splatting from fragile preprocessing pipelines.

1.3 Optimization Independence and Initialization Strategies Beyond SfM

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While hybrid architectures discussed in the previous section effectively reconcile geometric rigor with rendering efficiency by fusing implicit and explicit representations, the practical deployment of pure explicit methods like 3D Gaussian Splatting (3DGS) remains hindered by a critical dependency: the requirement for high-quality point cloud initialization derived from Structure-from-Motion (SfM) algorithms such as COLMAP. This reliance creates a significant fragility in the reconstruction pipeline; SfM frequently fails in texture-less regions, under poor lighting, or with wide-baseline imagery, resulting in sparse or missing initial points that prevent 3DGS from converging. Consequently, a burgeoning area of research has emerged focused on decoupling Gaussian Splatting optimization from strict SfM prerequisites. By exploring alternative initialization schemes—including random seeding, progressive propagation, and distillation from coarse volumetric models—these methods aim to establish robust optimization landscapes that do not rely on fragile preprocessing, thereby complementing the structural stability sought by hybrid approaches.

One primary direction of investigation challenges the necessity of accurate geometric initialization altogether. Early assumptions posited that random initialization would lead to catastrophic failure compared to SfM-derived points. However, rigorous analysis has demonstrated that with carefully designed optimization strategies, random initialization can yield results comparable to, and in some cases superior to, traditional methods. For instance, the work titled [13] provides a comprehensive frequency domain analysis revealing that the failure of random initialization stems from optimization traps rather than fundamental representational limitations. By introducing a novel strategy that relaxes the accurate initialization constraint, this approach successfully trains 3D Gaussians from random point clouds, mitigating the massive performance drops previously observed. Similarly, the study [14] systematically investigates various initialization protocols and concludes that by combining improved random seeding with structure distillation from low-cost Neural Radiance Field (NeRF) models, it is possible to bypass SfM entirely while maintaining equivalent reconstruction quality.

To address the specific challenge of texture-less surfaces where SfM inherently fails to generate sufficient tie points, researchers have looked to classical Multi-View Stereo (MVS) techniques for inspiration. The method proposed in [15] introduces a progressive propagation strategy that guides the densification of 3D Gaussians without relying on a dense initial SfM cloud. Instead of the standard split-and-clone operations which struggle in untextured areas, this framework leverages priors from existing reconstructed geometry and employs patch-matching techniques to generate new Gaussians with accurate positions and orientations. This approach effectively fills the gaps left by failed SfM procedures, particularly in large-scale scenes containing walls or ceilings, resulting in significant improvements in Peak Signal-to-Noise Ratio (PSNR) over standard 3DGS on challenging datasets like Waymo.

Furthermore, the integration of geometric priors and coarse volumetric representations has proven effective in stabilizing optimization in the absence of reliable point clouds. The framework presented in [16] eliminates SfM dependency by combining a grid-based volume with the 3DGS pipeline. This hybrid approach allows for the rendering of complex urban environments, including distant skies and low-texture areas, by utilizing a grid to initialize and guide the distribution of Gaussians where SfM points are absent. Complementary to this, the research in [17] addresses the overfitting and geometric instability that occur when training with limited views and no robust initialization. By incorporating dense depth maps from pre-trained monocular estimation models as geometric guides, this method aligns Gaussian optimization with underlying scene structure, effectively mitigating floating artifacts and ensuring geometric consistency even when traditional feature matching fails.

Advanced strategies also explore the use of frequency-domain regularization and pixel-aware gradients to refine initialization and growth dynamics, ensuring that the scene representation remains coherent regardless of the starting condition. The technique described in [18] tackles over-reconstruction issues common in poorly initialized scenarios by performing coarse-to-fine densification based on frequency components. By minimizing discrepancies in the Fourier space, this method ensures that large Gaussians do not erroneously cover high-variance regions, a common artifact when initialization is suboptimal. Additionally, [19] revises the growth conditions of Gaussians by weighting gradients based on the number of pixels covered in each view. This pixel-aware mechanism promotes the growth of Gaussians in areas with insufficient initialization, such as boundaries or texture-poor zones, leading to more accurate and detailed reconstructions without the need for a perfect starting point cloud.

Finally, the exploration of feed-forward models represents a paradigm shift towards complete independence from iterative SfM pipelines, paving the way for the scalable applications discussed in the following section. The approach in [20] demonstrates that a cost volume representation built via plane sweeping can provide sufficient geometry cues to localize Gaussian centers directly from sparse inputs. By learning Gaussian primitives jointly with depth estimation through photometric supervision alone, this method achieves state-of-the-art performance with inference speeds far exceeding traditional optimization-based pipelines. Collectively, these advancements signify a maturing of the field, moving 3D Gaussian Splatting from a technique contingent on fragile preprocessing pipelines to a robust, self-contained framework. By resolving initialization bottlenecks, these methods enable the generation of massive, consistent Gaussian sets that necessitate the hierarchical Level-of-Detail and anti-aliasing mechanisms explored next to ensure efficient rendering across varying scales.

1.4 Hierarchical Levels of Detail and Anti-Aliasing Mechanisms

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By resolving the initialization bottlenecks and establishing robust optimization landscapes independent of fragile SfM pipelines, as detailed in the preceding section, 3D Gaussian Splatting (3DGS) has evolved into a self-contained framework capable of generating massive, consistent Gaussian sets. However, the sheer scale of these generated primitives introduces a new critical challenge: rendering efficiency and fidelity across varying viewing distances. The initial formulations of explicit Gaussian representations, while revolutionary for real-time novel view synthesis, face significant degradation when handling large-scale scenes or drastic camera zooms. A primary limitation stems from the fixed nature of primitive scales; as a camera zooms out, the projected size of individual Gaussians can exceed the pixel footprint, leading to severe aliasing artifacts and a dramatic drop in performance due to the excessive number of primitives falling within the viewing frustum. To address these scalability issues, recent research has pivoted towards hierarchical Level-of-Detail (LOD) mechanisms and anti-aliasing strategies that dynamically adapt the scene representation to viewing conditions, ensuring both visual fidelity and consistent frame rates—a prerequisite for the advanced semantic and interactive applications discussed in the following section.

The core innovation in this domain involves restructuring the explicit primitive space using hierarchical data structures, most notably octrees, to manage geometric complexity across different scales. Drawing inspiration from traditional mesh-based LOD techniques that have long represented vast environments with logarithmic space complexity, researchers have extended this paradigm to neural and explicit radiance fields. For instance, [21] proposes a framework where the scene is decomposed into an octree structure, allowing the system to dynamically select the appropriate level of detail based on the camera's distance. This method ensures that distant parts of the scene are rendered using coarser, aggregated Gaussian representations, while close-up views utilize fine-grained primitives. By implementing this adaptive selection, the approach effectively mitigates the rendering bottlenecks associated with complex details in large scenes, maintaining high-fidelity output without the computational penalty of processing millions of unnecessary points in zoomed-out views.

Similarly, the concept of hierarchical decomposition has been tailored to handle the specific challenges of large-scale urban and outdoor environments. [22] introduces a divide-and-conquer training strategy coupled with a block-wise detail level selection mechanism. This architecture generates different levels of detail through the compression of fused Gaussian primitives, enabling seamless fusion and fast rendering across vastly different scales. The system aggregates these block-wise details dynamically, ensuring that the rendering pipeline remains efficient regardless of the scene's extent. This capability is crucial for applications requiring navigation through extensive environments, where the density of geometric features can vary by orders of magnitude between foreground and background elements.

Beyond structural reorganization, specialized anti-aliasing mechanisms have been developed to address the signal processing limitations inherent in rasterizing discrete primitives. When the screen-space projection of a 3D Gaussian falls below the Nyquist frequency relative to the pixel grid, standard alpha blending fails to reconstruct the signal correctly, resulting in shimmering and moiré patterns. To counteract this, [23] proposes maintaining Gaussians at multiple distinct scales within the same scene representation. In this framework, high-resolution images are rendered using a dense set of small Gaussians, whereas lower-resolution or distant views utilize fewer, larger Gaussians. This multi-scale approach allows the renderer to bypass the sequential blending of excessive primitives for distant objects, achieving significant improvements in both Peak Signal-to-Noise Ratio (PSNR) and rendering speed. The method effectively decouples the representation resolution from the rendering resolution, providing a robust solution to aliasing that scales efficiently with image resolution changes.

Further advancing the integration of screen-space techniques, [24] presents a novel approach that combines ideas from Gaussian splatting and neural reconstruction by utilizing a screen-space image pyramid. In this architecture, points are rasterized into different layers of a pyramid based on their projected size, with the selection of the pyramid layer determined dynamically. This allows for the rendering of arbitrarily large points using a single trilinear write operation, followed by a lightweight neural network that reconstructs a hole-free image with details extending beyond the native splat resolution. This technique not only resolves aliasing but also addresses the "cloudy" artifacts often seen in highly detailed scenes rendered by standard 3DGS, offering a compelling blend of crisp image quality and real-time performance.

The theoretical underpinnings of these hierarchical methods draw inspiration from broader computer vision and graphics concepts, such as Laplacian pyramids and scale-space representations. The utility of multi-scale feature extraction is well-documented in tasks like semantic segmentation and object detection, where aggregating context across scales is vital for accuracy. [25] demonstrates how multi-resolution architectures can refine boundaries and preserve localization information, a principle that translates effectively to the geometric refinement required in anti-aliased rendering. Furthermore, the efficiency of handling massive datasets through incremental LOD generation, as seen in [26], highlights the potential for streaming-based approaches where level-of-detail construction and rendering occur simultaneously on the GPU. Although originally designed for point clouds, the principles of incremental octree updates and real-time intermediate rendering provide a foundational strategy for future dynamic Gaussian systems that must handle continuous data streams.

In summary, the evolution of hierarchical LOD and anti-aliasing mechanisms represents a critical maturation of explicit neural rendering, directly addressing the scalability challenges introduced by the robust, initialization-independent methods of the previous section. By leveraging octree structures, multi-scale primitive sets, and screen-space pyramids, modern methods like those described in [21], [22], and [23] have successfully bridged the gap between the high fidelity of explicit representations and the efficiency required for scalable, real-time applications. These architectural innovations ensure that Gaussian Splatting remains robust across varying viewing distances, resolving aliasing artifacts while maintaining the interactive frame rates necessary to support the next frontier of research: the transition from purely photorealistic synthesis to semantically aware and editable 3D environments.

1.5 From Radiance Fields to Semantic and Editable Feature Fields

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Building upon the structural efficiencies and anti-aliasing mechanisms established in hierarchical LOD systems, the trajectory of neural scene representation has undergone a further profound transformation: shifting from the primary objective of photorealistic novel view synthesis to the more ambitious goal of creating semantically aware and editable 3D environments. While previous advancements, such as those detailed in Section 1.4, resolved issues of scale and rendering fidelity, early Neural Radiance Fields (NeRF) and their successors still focused almost exclusively on modeling the plenoptic function, encoding geometry and appearance into continuous volumetric fields or explicit primitives without intrinsic understanding. These methods treated the world as a collection of pixels and rays rather than distinct, identifiable objects, a limitation that hindered advanced applications such as object-level editing, interactive segmentation, and robotics navigation. The evolution toward semantic and editable feature fields addresses this gap by distilling knowledge from powerful 2D foundation models directly into 3D representations, thereby enabling open-vocabulary understanding and language-driven manipulation within the efficient frameworks now available.

A pivotal advancement in this domain involves the distillation of features from pre-trained vision-language models, such as CLIP and DINO, into the parameters of neural radiance fields. Early approaches demonstrated that by optimizing a NeRF to reproduce not only RGB colors but also high-dimensional semantic features extracted from 2D images, one could lift 2D open-vocabulary capabilities into the 3D domain. For instance, methods have been developed to distill open-vocabulary multimodal knowledge into NeRFs using only text descriptions of objects, effectively achieving view-consistent 3D segmentation without requiring manual 3D annotations [27]. This paradigm allows the system to recognize and segment objects based on natural language queries, overcoming the scarcity of large-scale 3D labeled datasets. Similarly, other frameworks have integrated hierarchical embeddings from foundation models like the Segment Anything Model (SAM) to decompose NeRFs into object-centric components, ensuring consistent recognition across varying viewpoints even in the presence of occlusion [28]. These techniques validate that semantic labels can be treated as viewpoint-invariant functions, mapping spatial coordinates to a spectrum of semantic concepts to facilitate robust scene interpretation [29].

However, the computational bottleneck of volumetric rendering in NeRFs limited the interactivity and scalability of these semantic fields, creating a disparity between the rich semantic understanding achieved and the real-time performance required for practical deployment. The emergence of 3D Gaussian Splatting (3DGS), having already proven its utility in handling complex geometry and scale via the mechanisms discussed previously, provided a catalyst for overcoming these limitations. Recent innovations have extended 3DGS beyond radiance reconstruction by attaching arbitrary-dimensional semantic feature vectors to each Gaussian primitive. This approach, exemplified by works that supercharge 3D Gaussian Splatting with distilled feature fields, enables the direct rendering of semantic maps at interactive frame rates [30]. By addressing the disparities in spatial resolution and channel consistency between RGB images and feature maps through specific architectural changes, these methods allow for novel view semantic segmentation and language-guided editing that significantly outperforms implicit NeRF-based counterparts in terms of speed. Furthermore, specialized adaptations like CLIP-GS have introduced strategies to learn compact semantic representations within Gaussians, utilizing self-training mechanisms to enforce 3D coherence and resolve ambiguities arising from view-inconsistent 2D supervision [31].

The transition to explicit semantic fields has also unlocked new capabilities for interactive editing and scene manipulation, leveraging the discrete nature of Gaussian primitives. Unlike implicit fields where editing often requires complex latent space optimization or re-training, explicit Gaussian representations allow for direct modification of semantic attributes. Frameworks such as GaussianEditor leverage semantic tracing to maintain consistency during editing tasks, enabling swift and controllable modifications like object removal or integration while preserving geometric integrity [32]. The ability to propagate 2D user queries into the 3D space has been further enhanced by methods that utilize nearest-neighbor feature matching and bilateral search in joint spatio-semantic spaces, allowing users to interactively segment and composite objects with fine structural details [33]. Moreover, the integration of generative priors has facilitated text-driven synthesis and editing, where language instructions can guide the generation of dynamic scenes or the modification of existing structures without extensive re-optimization [34].

Despite these advances, challenges remain in ensuring the robustness of semantic fields under varying conditions and in generalizing to unseen categories. Techniques addressing view-inconsistent semantics through cross-view self-enhancement and region semantic ranking have been proposed to refine the quality of distilled features, significantly improving segmentation accuracy on standard benchmarks [35]. Additionally, the fusion of multi-modal data, such as RGB-D inputs, through attention-based mechanisms has proven effective in enhancing segmentation accuracy by leveraging depth cues to resolve geometric ambiguities [36]. As the field progresses, the convergence of efficient explicit representations like 3D Gaussian Splatting—now optimized for both scale and semantics—with the rich semantic priors of foundation models is establishing a new standard for 3D scene understanding. This synthesis not only enables zero-shot open-vocabulary segmentation but also paves the way for truly interactive, language-driven 3D environments where users can intuitively query, edit, and manipulate complex scenes in real time.

2 Methodological Frameworks for Dynamic and Non-Rigid Scene Modeling

2.1 Canonical Space Deformation and Time-Conditioned Embeddings

2.1 Canonical Space Deformation and Time-Conditioned Embeddings

The core challenge in reconstructing dynamic 3D scenes lies in disentangling the inherent complexity of non-rigid motion from the underlying static geometry and appearance. Before the emergence of unified 4D primitives, a dominant paradigm addressed this by mapping dynamic observations into a static, time-invariant canonical space. This approach fundamentally relies on the assumption that while the observed scene deforms over time, there exists a reference configuration—often referred to as the canonical frame—where the geometry and texture remain fixed. By learning a deformation field that warps points from the observation space back to this canonical space (or conversely, deforms the canonical representation forward), methods can leverage efficient static representations, such as Neural Radiance Fields (NeRF) or 3D Gaussian Splatting, within the canonical domain while modeling dynamics solely through a separate deformation module. This subsection reviews the theoretical underpinnings and recent advancements in these canonical space deformation models, focusing on backward deformation fields, forward warping strategies, and time-conditioned embeddings designed to handle large-scale, complex non-rigid motions.

The foundational concept involves utilizing a backward deformation field, where for any given time step t and a 3D point in the observation space, a neural network predicts a displacement vector mapping this point to its corresponding location in the canonical space. While effective for moderate motions, early implementations often struggled with large displacements and topological changes, as the backward mapping can become discontinuous or ill-posed during significant occlusion or object movement. To mitigate these issues, recent works have introduced sophisticated regularization and decomposition techniques. For instance, [37] proposes a novel adaptation by decomposing the deformation into a strongly regularized coarse component and a weakly regularized fine component. The coarse component extends the deformation field into the surrounding space, enabling the tracking of objects over long-term, studio-scale motions that previous methods could not handle, thereby ensuring the time consistency essential for downstream tasks like 3D editing and motion analysis.

Complementing the backward approach, forward warping strategies have been explored to improve the smoothness and continuity of motion modeling. While backward flows map observation points to the canonical space, they can suffer from non-smoothness in regions of complex motion. In contrast, [38] argues that estimating a forward flow field, which directly warps the canonical radiance field to other time steps, results in a motion model that is smooth and continuous within the object region. This method utilizes voxel grids to represent the canonical field and employs a differentiable warping process, including average splatting and inpainting networks, to resolve mapping ambiguities, demonstrating superior performance in both rendering quality and motion modeling compared to traditional backward-flow-based dynamic NeRFs.

To further enhance the expressiveness and stability of these deformation fields, researchers have integrated time-conditioned embeddings and latent representations, allowing the deformation network to adapt its behavior based on specific temporal contexts. [39] takes a unique perspective by formulating the scene as a composition of bandlimited, high-dimensional signals, effectively bridging classic non-rigid structure-from-motion (NRSfM) priors with neural rendering. By factorizing time and space, this framework leverages well-studied priors to constrain the neural representation, addressing issues of limited expressiveness and entanglement between light and density fields often seen in pure neural approaches. Similarly, [40] introduces a method to learn object-centric canonical spatiotemporal representations that support spacetime continuity. By explicitly encoding time through mapping input sequences to a spatiotemporally canonicalized space and leveraging Neural Ordinary Differential Equations (Neural ODEs), CaSPR achieves robustness to irregular sampling and generalizes to unseen object instances, facilitating continuous reconstruction and correspondence estimation.

The effectiveness of canonical space mapping is further amplified when combined with sparse control mechanisms and physical priors, offering a middle ground between fully implicit fields and explicit geometric primitives. [41] advances this domain by decomposing motion and appearance into sparse control points and dense Gaussians. Here, a deformation Multi-Layer Perceptron (MLP) predicts time-varying 6-DoF transformations for sparse control points, which are then interpolated to drive the motion of dense 3D Gaussians. This sparse-controlled approach reduces learning complexity and enhances the spatial and temporal coherence of motion patterns, while an "As-Rigid-As-Possible" (ARAP) loss enforces local rigidity, making the representation particularly suitable for editable dynamic scenes. Furthermore, the integration of physics-informed constraints into the canonical deformation framework has shown promise in handling extreme deformations. [42] represents the scene as a time-invariant Signed Distance Function (SDF) coupled with a time-conditioned deformation field, bridging this neural geometry with a differentiable physics simulator. This allows for the estimation of physical parameters and ensures that the learned deformations adhere to physical laws, enabling interactive editing and robust reconstruction from monocular inputs.

Despite these advancements, challenges remain in handling topological changes and ensuring cycle consistency in long sequences. [43] addresses this by factorizing deformation through continuous bijective canonical maps, guaranteeing cycle consistency and topology preservation. This method learns a canonical shape that serves as a flexible and stable space for shape prior learning, demonstrating state-of-the-art performance on highly deformable geometries such as human and animal bodies. Additionally, for scenarios involving severe occlusions, [44] integrates shape priors into a variational optimization framework, penalizing irregularities in the time-varying structure based on occlusion tensors, thus improving correspondence establishment in challenging conditions.

In summary, canonical space deformation and time-conditioned embeddings represent a critical evolutionary step in dynamic 3D reconstruction, establishing a robust framework for separating motion from geometry. By refining the mapping between observation and canonical spaces through decomposition, forward/backward flow hybrids, sparse control, and physics integration, these methods have significantly improved the handling of real-world non-rigid dynamics. However, the reliance on separate deformation networks and the computational cost of querying implicit fields or warping dense grids at every timestep highlight the need for more integrated solutions. These limitations naturally motivate the shift toward explicit 4D spatiotemporal Gaussian representations, where time is treated as an intrinsic dimension of the primitive itself, eliminating the need for complex deformation fields and enabling direct, efficient spatiotemporal slicing.

2.2 Explicit 4D Spatiotemporal Gaussian Representations

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While canonical space deformation models effectively disentangle motion from geometry by warping observations to a static reference, they inherently rely on complex, separate deformation networks and face challenges in maintaining cycle consistency over long sequences. Addressing these limitations, the field has witnessed a paradigmatic shift toward explicit 4D spatiotemporal Gaussian representations. Unlike the previous paradigm which treats time as a conditional input to a deformation field, this approach models spacetime as a unified, continuous 4D volume. By redefining the reconstruction primitive itself, these methods move beyond static 3D Gaussians modulated by time-dependent networks to intrinsically 4D primitives that possess extent and variation across both spatial and temporal dimensions. This unification eliminates the need for explicit backward or forward warping modules, allowing for direct temporal slicing, variable-length video synthesis, and the efficient handling of complex, non-rigid motions without the computational overhead of maintaining separate models for each timestamp.

A cornerstone of this approach is the formulation of the 4D Gaussian primitive. Traditional 3D Gaussian Splatting optimizes position, covariance, and opacity in three-dimensional space; in contrast, explicit 4D methods parameterize anisotropic ellipsoids within a four-dimensional spacetime continuum. The paper titled "Real-time Photorealistic Dynamic Scene Representation and Rendering with 4D Gaussian Splatting" [45] proposes approximating the underlying spatio-temporal volume of a dynamic scene by optimizing a collection of these 4D primitives. In this framework, each primitive is defined by an anisotropic ellipse capable of arbitrary rotation in space and time, coupled with view-dependent and time-evolved appearance modeled via 4D spherindrical harmonics. This conceptual simplicity enables the synthesis of novel views at any desired time through a tailored rendering routine that effectively slices the 4D volume, offering superior flexibility for variable-length video generation compared to methods constrained by fixed temporal embeddings.

Complementing this geometric formulation, the work described in "4D Gaussian Splatting Towards Efficient Novel View Synthesis for Dynamic Scenes" [46] introduces a method that explicitly models dynamics using anisotropic 4D XYZT Gaussians. By treating spacetime as an entirety, this approach avoids the pitfalls of learning complex 6D plenoptic functions directly or scaling deformation modeling for intricate dynamics. The core innovation lies in the temporal slicing mechanism, where dynamic 3D Gaussians are naturally composed by slicing the optimized 4D Gaussians at specific timestamps. These sliced primitives can then be seamlessly projected into images using standard rasterization pipelines. The efficacy of this representation is underscored by its performance; the authors report real-time inference rendering speeds of up to 277 FPS on an RTX 3090 GPU and 583 FPS on an RTX 4090 GPU, demonstrating that explicit 4D representations can achieve both high fidelity and exceptional efficiency even in scenes with abrupt motions.

Further advancing the integration of explicit geometry with neural features, the paper titled "4D Gaussian Splatting for Real-Time Dynamic Scene Rendering" [47] proposes a holistic representation that combines 3D Gaussians with 4D neural voxels. This hybrid architecture leverages a decomposed neural voxel encoding algorithm, inspired by HexPlane, to efficiently construct Gaussian features from the 4D voxel grid. A lightweight Multi-Layer Perceptron (MLP) is then employed to predict Gaussian deformations at novel timestamps based on these features. This strategy decouples the heavy feature storage in the voxel grid from the lightweight geometric deformation prediction, achieving real-time rendering at 82 FPS for 800x800 resolution while maintaining storage efficiency. This demonstrates that explicit 4D representations can be enhanced with neural components to capture high-frequency details without sacrificing the speed benefits of explicit primitives.

Another significant variation in this domain is presented in “Spacetime Gaussian Feature Splatting for Real-Time Dynamic View Synthesis” [48], which enhances 3D Gaussians with temporal opacity and parametric motion to formulate expressive Spacetime Gaussians. This method replaces standard spherical harmonics with splatted neural features, facilitating the modeling of view- and time-dependent appearance within a compact storage footprint. By leveraging training error and coarse depth to guide the sampling of new Gaussians in challenging areas, this approach achieves state-of-the-art rendering quality at 8K resolution with a lite-version model running at 60 FPS. The use of neural features within an explicit 4D framework highlights the versatility of this representation in balancing visual fidelity with computational constraints.

The applicability of explicit 4D Gaussian representations extends beyond general scene reconstruction to specialized generative tasks. For instance, “4DGen Grounded 4D Content Generation with Spatial-temporal Consistency” [49] utilizes dynamic 3D Gaussians constructed within a 4D framework to enable efficient, high-resolution supervision during training. By employing spatial-temporal pseudo labels and consistency priors, this method facilitates grounded 4D content creation where users can specify both geometry and motion. Similarly, “STAG4D Spatial-Temporal Anchored Generative 4D Gaussians” [50] combines pre-trained diffusion models with dynamic 3D Gaussian splatting, introducing an adaptive densification strategy specifically crafted for 4D generation to mitigate unstable gradients. These works illustrate that the explicit 4D Gaussian paradigm is not only robust for reconstruction from multi-view videos but also serves as a flexible backbone for text-to-4D and video-to-4D synthesis, ensuring spatial-temporal consistency that is often difficult to achieve with implicit neural fields.

In summary, explicit 4D spatiotemporal Gaussian representations mark a critical evolution from deformation-based frameworks to unified volumetric primitives. By treating time as an intrinsic dimension of the geometry, these methods overcome the discontinuity and computational bottlenecks associated with separate deformation networks. Whether through pure 4D ellipsoids, hybrid neural voxel encodings, or feature-enhanced spacetime Gaussians, these frameworks provide a unified solution for high-fidelity, real-time rendering of complex dynamic events. However, while these continuous 4D volumes excel in rendering efficiency and temporal coherence, they often operate as monolithic entities where individual object trajectories are not explicitly tracked. This limitation motivates the subsequent exploration of particle-based motion models, which adopt a Lagrangian perspective to track discrete, persistent Gaussian particles, thereby enabling dense 6-DOF tracking and structured motion decomposition.

2.3 Particle-Based Motion Models and Dual-Domain Dynamics

2.3 Particle-Based Motion Models and Dual-Domain Dynamics

While explicit 4D spatiotemporal representations unify space and time into a continuous volume, an alternative and equally transformative paradigm focuses on modeling dynamic scenes through discrete, moving particles. This approach marks a distinct shift from purely volumetric or implicit neural fields toward particle-based motion models, where the scene is conceptualized as a collection of 3D Gaussian primitives with persistent attributes. In this framework, properties such as color, opacity, and scale often remain invariant or evolve smoothly over time, while the spatial positions and orientations of these particles are explicitly tracked. This fundamentally redefines the reconstruction problem: rather than learning a time-varying density field, the goal becomes tracking the trajectories of individual geometric primitives. A seminal work in this direction, [51], proposes a method that simultaneously addresses novel-view synthesis and dense six degree-of-freedom (6-DOF) tracking. By enforcing that Gaussians maintain persistent appearance attributes while allowing them to move and rotate, the method naturally recovers dense scene flow without requiring explicit correspondence or optical flow inputs. The core innovation lies in the application of local-rigidity constraints, which regularize the motion and rotation of neighboring Gaussians, ensuring that they model the same physical area of space consistently over time. This persistence allows for the emergence of dense 6-DOF tracking directly from the optimization of the rendering loss, enabling downstream applications such as dynamic compositional scene synthesis and 4D video editing.

Building upon the concept of persistent particles, recent advancements have introduced dual-domain deformation models to explicitly capture the complex attribute deformations of each Gaussian point over extended sequences. Traditional methods often struggle with long-term consistency or require re-optimization for every frame, leading to computational inefficiencies and temporal flickering. To address these limitations, [52] introduces a novel Dual-Domain Deformation Model (DDDM). This framework explicitly models the time-dependent residuals of Gaussian attributes by utilizing a hybrid representation: polynomial fitting in the time domain captures low-frequency, smooth motions, while Fourier series fitting in the frequency domain accounts for high-frequency oscillations and rapid changes. This dual-domain strategy eliminates the need for training separate models for each frame or introducing additional implicit neural fields to model dynamics, thereby achieving training speeds comparable to static 3DGS. The explicit modeling of deformation for discretized Gaussian points ensures that the system can handle complex scene deformations across long video footage while maintaining ultra-fast rendering capabilities. The efficiency gains are substantial, with reported training speeds up to five times faster than per-frame 3DGS modeling, while simultaneously outperforming previous leading methods in novel view rendering quality.

The distinction between Eulerian and Lagrangian perspectives is critical in understanding the evolution of these particle-based motion models relative to the continuous fields discussed previously. While many deformation-based methods operate in an Eulerian view—focusing on specific locations in space through which matter flows—this perspective makes it intractable to extract the motion of constituting objects or parts. In contrast, particle-based approaches adopt a Lagrangian view, parameterizing scene motion by tracking the trajectories of individual particles. [53] leverages this duality by enforcing cycle consistency between Eulerian and Lagrangian views. Under the Lagrangian view, the method factors scene motion into a composition of part-level rigid motions, facilitating the discovery of semantically meaningful parts based on shared motion patterns. This part-based representation is particularly advantageous for applications requiring object-level manipulation, such as part tracking, animation, and 3D scene editing, as it provides a structured understanding of the dynamic scene that pure pixel-wise flow methods cannot offer.

Furthermore, the integration of sparse control mechanisms with dense Gaussian particles has emerged as a powerful strategy for managing motion complexity within this Lagrangian framework. [41] proposes decomposing the motion and appearance of dynamic scenes into sparse control points and dense Gaussians. In this framework, a significantly smaller set of control points learns compact 6-DoF transformation bases, which are then locally interpolated to yield the motion field for the dense cloud of 3D Gaussians. A deformation MLP predicts time-varying transformations for each control point, reducing learning complexity and enhancing the ability to capture spatially and temporally coherent motion patterns. The system employs an "as-rigid-as-possible" (ARAP) loss to enforce local rigidity and spatial continuity, ensuring that the learned motions are physically plausible. This explicit separation of motion and appearance not only improves reconstruction fidelity but also enables user-controlled motion editing, allowing users to modify the trajectory of objects while retaining high-fidelity appearances.

The robustness of particle-based models is further enhanced by incorporating motion cues from 2D observations to guide the optimization of these moving primitives. [54] highlights that existing methods often overlook rich motion information carried by optical flow, leading to performance degradation and model redundancy. By establishing a direct correspondence between 3D Gaussian movements and pixel-level flow, this framework introduces a flow augmentation method that leverages uncertainty estimates and loss collaboration to guide the optimization. For deformation-heavy scenarios, a transient-aware deformation auxiliary module is employed to handle rapid changes, resulting in significant improvements in both rendering quality and efficiency across multi-view and monocular settings. Additionally, the challenge of reconstructing non-rigid scenes from sparse inputs is addressed by [55], which designs an explicit motion model as a factorized 4D representation. By exploiting spatiotemporal correlations and integrating reliable flow priors, this method achieves fast and compact reconstruction even when dense viewpoint coverage is unavailable.

In summary, particle-based motion models and dual-domain dynamics represent a sophisticated synthesis of explicit geometry and learned deformation fields, offering a complementary perspective to the unified 4D volumes. By treating scenes as collections of persistent, moving 3D Gaussian particles, these methods achieve dense 6-DOF tracking and high-fidelity novel view synthesis. The utilization of dual-domain representations, Lagrangian tracking perspectives, and sparse control mechanisms allows these frameworks to handle complex non-rigid deformations, long-term temporal consistency, and editing capabilities that were previously unattainable with implicit neural representations. However, while these particle-based approaches excel in tracking and motion decomposition, the purely point-based nature of discrete Gaussians can still pose challenges for maintaining strict topological consistency and geometric coherence during extreme deformations. This limitation motivates the subsequent exploration of hybrid architectures that anchor Gaussian primitives to explicit mesh structures, combining the radiance fidelity of splatting with the structural integrity of meshes.

2.4 Hybrid Mesh-Gaussian Anchoring for Geometric Consistency

2.4 Hybrid Mesh-Gaussian Anchoring for Geometric Consistency

While particle-based motion models and dual-domain dynamics offer powerful mechanisms for tracking discrete primitives and capturing complex non-rigid deformations, the purely point-based nature of independent 3D Gaussians introduces inherent limitations regarding topological consistency and geometric coherence. As noted in the preceding discussion, discrete Gaussians can struggle to maintain strict structural integrity during extreme deformations, often resulting in "floating" artifacts, inconsistent density distributions, and a lack of connectivity required for physical simulation or precise mesh extraction. To address these challenges, the field has evolved toward hybrid architectures that anchor Gaussian primitives to explicit geometric backbones. This paradigm synthesizes the topological robustness of deformable meshes or local oriented volumes with the radiance fidelity and differentiable rendering efficiency of Gaussian Splatting, effectively bridging the gap between flexible appearance modeling and rigid geometric constraints.

A primary strategy within this hybrid domain involves binding 3D Gaussians directly to an explicit mesh structure, where the mesh serves as a dynamic scaffold guiding the deformation and distribution of radiance primitives. In such frameworks, Gaussians are not optimized as independent entities but are defined over and bound to the faces of an explicit mesh. This coupling establishes a bidirectional dependency: the rendering quality of the Gaussians informs adaptive refinement strategies, such as mesh face splitting, while the topological constraints of the mesh dictate the splitting and merging of the Gaussian primitives. By enforcing these explicit mesh constraints, the system regularizes the Gaussian distribution, suppressing the formation of poor-quality primitives—such as misaligned or excessively elongated Gaussians—that frequently cause visual artifacts during aggressive deformation. Furthermore, this architecture leverages existing large-scale mesh deformation datasets to drive data-driven Gaussian deformation, ensuring that the radiance field evolves realistically alongside the underlying geometry while maintaining high frame rates suitable for interactive applications [56].

Complementing global mesh anchoring, another sophisticated lineage of research focuses on anchoring temporally shared Gaussians within local oriented volumes to handle non-rigid object reconstruction from monocular inputs. This approach specifically targets the severe ill-posedness of reconstructing dynamic objects where multi-view cues are sparse or non-existent. By introducing a two-stage optimization process, these frameworks first learn a low-rank neural deformation model to establish coarse temporal consistency, which subsequently acts as a regularizer for the fine-grained optimization of 3D Gaussians. Crucially, the Gaussians are anchored within local oriented volumes that deform according to the learned motion field. This "Neural Parametric Gaussians" approach ensures that primitives maintain their spatial relationship relative to the local surface patch they represent, even under complex non-rigid transformations. The result is a representation that preserves view consistency and delivers high-quality photo-realistic reconstructions of deforming objects, effectively overcoming the limitations of previous methods that struggled with novel view synthesis from significantly different camera positions [57].

The necessity of such hybrid architectures is further underscored by the inherent limitations of relying on single primitive types for complex dynamic scenes. While meshes excel at representing solid, smooth geometry, they often fail to capture fuzzy structures like hair or translucent materials. Conversely, point-based splatting captures these details adeptly but frequently introduces artifacts in regions with smooth geometry and sharp textures. To mitigate these individual deficiencies, frameworks like GauMesh have been proposed to bridge the gap between 3D Gaussians and meshes for free-view video rendering. These systems simultaneously optimize mesh geometry, texture, opacity maps, and a set of 3D Gaussians alongside a deformation field. At inference time, the rendering pipeline performs alpha-blending based on merged and re-ordered z-buffers derived from both mesh and Gaussian rasterizations. This allows the system to adaptively select the most appropriate primitive type for different regions of the scene, ensuring that smooth surfaces are rendered with mesh precision while complex, fuzzy details are captured by Gaussians, all without compromising rendering speed [58].

Furthermore, the integration of mesh guidance is critical for achieving time-consistent mesh reconstruction from monocular videos, a task essential for downstream graphics pipelines. Frameworks such as Dynamic Gaussians Mesh (DG-Mesh) utilize Gaussian-Mesh anchoring to encourage an even distribution of Gaussians, which in turn facilitates superior mesh reconstruction through mesh-guided densification and pruning. By applying cycle-consistent deformation between a canonical space and the deformed observation space, these methods can project anchored Gaussians back to a canonical frame, optimizing the representation across all time frames simultaneously. This ensures that the extracted mesh sequence maintains high fidelity and temporal stability, enabling advanced applications such as texture editing on dynamic objects [59]. Similarly, for human avatar modeling, joint learning of mesh deformation and Gaussian textures in UV space has proven effective. By utilizing the UV map embedding to learn Gaussian textures and optimizing pose-dependent geometric deformations via an independent mesh network, these methods significantly enhance rendering quality and prevent the blurry textures often associated with coarse mesh guidance [60].

In summary, hybrid mesh-Gaussian anchoring represents a pivotal evolution in dynamic scene modeling, directly addressing the topological inconsistencies identified in purely particle-based approaches. By tethering the flexibility of Gaussian splats to the structural integrity of deformable meshes or local volumes, these frameworks achieve a synthesis of geometric consistency and radiance realism. Whether through direct face-binding for large-scale deformation, local volume anchoring for monocular non-rigid reconstruction, or adaptive blending for heterogeneous scene content, these methods provide the necessary topological constraints to enable robust physics simulation, reliable animation, and high-fidelity editing. However, while hybrid architectures enforce geometric plausibility, they often rely on deterministic optimization pipelines that can struggle when faced with severe data sparsity, high noise levels, or extreme ambiguity. This limitation motivates the subsequent exploration of generative priors and diffusion-based latent dynamics, which introduce probabilistic modeling to resolve ill-posed reconstruction scenarios where geometric anchors alone are insufficient.

2.5 Generative Priors and Diffusion-Based Latent Dynamics

2.5 Generative Priors and Diffusion-Based Latent Dynamics

While hybrid mesh-Gaussian architectures provide essential geometric scaffolding to enforce topological consistency, they often rely on deterministic optimization pipelines that can struggle when faced with severe data sparsity, high noise levels, or extreme ambiguity in monocular settings. In such under-constrained scenarios, purely geometric or photometric losses may fail to converge to plausible solutions, leading to unstable reconstructions or temporally incoherent artifacts. To overcome these limitations, the field has increasingly integrated generative priors, specifically leveraging the powerful distributional modeling capabilities of diffusion models. These models act as robust regularizers, constraining the solution space to manifolds of physically plausible shapes and motions. By effectively denoising latent representations, diffusion-based approaches recover high-fidelity 4D dynamics where traditional deterministic methods falter, bridging the gap between structural constraints and probabilistic completeness.

A pivotal advancement in this domain is the transition from global latent codes to localized, set-based latent representations capable of capturing complex, non-linear deformation patterns. The work [61] exemplifies this paradigm by introducing a 4D diffusion model designed explicitly for dynamic surface reconstruction from point cloud sequences. Unlike conventional feed-forward networks that struggle with ambiguous observations from noisy or partial inputs, this approach parameterizes 4D dynamics using latent sets rather than singular global vectors. This architectural choice enables the model to learn local shape and deformation patterns independently, significantly enhancing generalizability to unseen motions and identities. By synchronously denoising deformation latent sets and facilitating information exchange across multiple frames through interleaved space-time attention blocks, [61] achieves superior temporal coherence and accurate non-linear motion capture. This method demonstrates that diffusion-based priors can effectively resolve ambiguities in sparse observations by iteratively refining compressed latent representations, ensuring that the reconstructed geometry adheres to probabilistic distributions learned from diverse motion data.

Beyond specific architectural innovations for non-rigid objects, the theoretical framework of diffusion probabilistic fields provides a generalized mechanism for learning distributions over continuous functions defined on metric spaces. The concept of [62] extends standard diffusion modeling to handle data that does not reside on Euclidean grids, such as the continuous fields required for neural rendering and dynamic scene representation. This flexibility is crucial for dynamic 3D reconstruction, where the underlying geometry and appearance are often modeled as continuous functions of space and time. By defining an end-to-end learning algorithm that operates directly on field parameterizations without relying on intermediate latent vector discretizations, [62] enables the modeling of complex spatiotemporal distributions. This approach facilitates the application of diffusion priors to various modalities, including 3D geometry and video, allowing for a unified treatment of static structure and dynamic motion within a single probabilistic framework.

The efficacy of these generative priors is further amplified when applied to inverse problems where ground-truth signals are unobservable, and only indirect measurements are available. In the context of inverse graphics and dynamic scene recovery, the goal is often to sample from a distribution of 3D scenes that align with limited 2D observations. The methodology presented in [63] addresses this by integrating a known differentiable forward model directly into the denoising process. This integration allows the diffusion model to learn the distribution of underlying signals—such as 3D geometry and motion fields—even when training samples of the signals themselves are unavailable. By connecting the generative modeling of observations with the generative modeling of the latent signals, this approach enables the synthesis of consistent 4D scenes from monocular inputs, effectively solving the stochastic inverse problem inherent in dynamic reconstruction.

Furthermore, the application of diffusion models as plug-and-play priors offers a versatile strategy for incorporating complex structural constraints into reconstruction pipelines without retraining the generative model. As discussed in [64], pre-trained diffusion models can serve as independent prior distributions $p(\mathbf{x})$ that are combined with auxiliary differentiable constraints representing data fidelity or physical laws. This modularity is particularly advantageous for dynamic scenes, where constraints such as temporal smoothness, rigidity of specific parts, or conservation of mass can be imposed alongside the generative prior. The ability to perform approximate inference by iterating through the fixed denoising network enriched with varying levels of noise allows for the exploration of high-dimensional solution spaces, yielding reconstructions that are both data-consistent and semantically plausible.

The probabilistic nature of these models also provides a critical mechanism for handling uncertainty and multimodality in dynamic scenarios. Unlike deterministic methods that output a single "best guess," diffusion-based approaches can sample multiple plausible trajectories and shapes, capturing the inherent ambiguity in non-rigid motion. This is closely related to the principles outlined in [65], which validates the use of score-based models for analyzing the posterior distribution of images given noisy measurements. By leveraging the theoretically proven probability functions of diffusion models, researchers can perform variational inference to recover not just a point estimate but a distribution of possible dynamic states. This capability is essential for applications requiring robust tracking and prediction, where understanding the confidence bounds of the reconstruction is as critical as the reconstruction itself.

Finally, the integration of these generative priors extends to the modeling of temporal dynamics as continuous stochastic processes. The framework proposed in [66] defines diffusion models in function space, allowing for the natural handling of irregularly sampled temporal data—a common challenge in real-world dynamic scene capture. By treating time series as discretized measurements of an underlying function and applying noise in the function domain, this approach preserves continuity while learning to reverse the diffusion process to generate smooth, coherent motion fields. This perspective aligns with the broader goal of using diffusion-based latent dynamics to synthesize high-quality sequential samples that adhere to the physical and geometric constraints of the 3D world, thereby overcoming the limitations of traditional discrete-time models.

In summary, the fusion of diffusion models with dynamic 3D reconstruction represents a significant methodological leap that complements the structural rigor of hybrid mesh-Gaussian approaches. By utilizing latent vector sets, continuous field parameterizations, and forward-model integration, these generative priors provide a powerful mechanism for denoising and completing sparse observations where geometric anchors alone are insufficient. The works [61], [62], [63], [64], [65], and [66] collectively illustrate how probabilistic latent dynamics can resolve the ill-posed nature of non-rigid reconstruction. This enables robust, temporally coherent, and physically plausible 4D scene modeling, setting the stage for the subsequent integration of explicit physical laws in fluid dynamics and other complex continuous media.

2.6 Physics-Informed Modeling and Fluid Dynamics Reconstruction

2.6 Physics-Informed Modeling and Fluid Dynamics Reconstruction

While the preceding section established how generative priors and diffusion models can resolve ambiguities in sparse observations through probabilistic latent dynamics, a distinct and complementary paradigm addresses dynamic scenes governed by explicit physical laws. This is particularly critical for fluid dynamics, where the absence of rigid geometric structures, the prevalence of complex topological changes, and the inherent ambiguity of visual observations render purely data-driven or geometric approaches insufficient. Unlike solid objects that maintain coherent surfaces, fluids are dictated by continuous physical principles, specifically the Navier-Stokes equations, which enforce the conservation of mass, momentum, and energy. Traditional neural rendering methods, relying solely on photometric consistency or optical flow, often fail to capture these underlying mechanics, resulting in temporally inconsistent reconstructions or physically implausible velocity fields. To overcome these limitations, recent research has shifted toward physics-informed modeling, embedding governing differential equations directly into the optimization pipelines of neural representations. This ensures that reconstructed scenes not only align with observed imagery but also adhere to fundamental fluid dynamics, enabling robust inference even in partially observable or sparse-view scenarios. This rigorous adherence to physical laws serves as a necessary counterpart to the probabilistic flexibility of diffusion priors, setting the stage for the motion-aware optimization strategies discussed subsequently, which leverage kinematic cues for general non-rigid dynamics.

A primary strategy in this domain involves integrating particle-driven representations with differentiable rendering to infer hidden physical states. In scenarios where only the fluid surface is visible, such as splashing water or flowing liquids, inferring internal velocity and pressure fields remains an ill-posed problem. The work [67] addresses this by proposing a differentiable two-stage network that couples a particle-driven neural renderer with a particle transition model. By incorporating fluid physical properties directly into the volume rendering function, this approach facilitates the unsupervised learning of particle-based fluid dynamics. The system optimizes the transition model to minimize the discrepancy between rendered and observed images, effectively grounding visual data in physical reality without requiring direct supervision of particle positions or velocities. This method demonstrates the feasibility of estimating underlying physics, including viscosity and density variations, purely from sequential visual observations of the fluid surface.

Furthermore, the complexity of real-world fluid interactions, particularly involving static obstacles and unknown lighting conditions, necessitates architectures capable of disentangling static and dynamic components while enforcing strict physical constraints. The framework presented in [68] tackles this by leveraging the Navier-Stokes equations as a regularization term within an end-to-end optimization process. This method utilizes a hybrid architecture to separate static scene elements from dynamic fluid volumes, employing neural networks as ansatz functions for density and velocity. A key innovation is the use of a layer-by-layer growing strategy to progressively increase network capacity, mitigating the ambiguity between density and color in radiance fields. By augmenting time-varying neural radiance fields with physics-informed deep learning, this approach successfully reconstructs fluid interactions with arbitrary obstacles without prior knowledge of geometry or boundary conditions, showcasing the efficacy of physical priors in handling sparse multi-view inputs.

Beyond static reconstruction, the accurate estimation of fluid motion from monocular or sparse video data is critical for simulation and editing applications. Standard optical flow techniques often falter in fluid contexts due to the lack of distinct features and the non-rigid nature of the flow. The approach detailed in [69] introduces a hybrid neural velocity representation that combines a base neural field for irrotational energy with vortex particles to model residual turbulent velocity. This method enforces physics-based losses to ensure the inferred velocity field is divergence-free and correctly drives the transport of fluid density. Similarly, [70] proposes an unsupervised prediction-correction scheme where a PDE-constrained optical flow predictor is refined by a physical corrector. This corrector mimics operator splitting methods used in traditional fluid solvers, ensuring that estimated motion trajectories remain consistent with time-dependent partial differential equations, thereby overcoming the limitations of frame-to-frame optical flow methods.

The integration of physics also extends to inverse problems aimed at recovering full 3D dynamics from 2D observations without ground-truth supervision. The work [71] achieves photo-to-fluid-dynamics reconstruction by integrating a differentiable Euler simulator with a volumetric renderer. This system treats the fluid volume as a NeRF-like representation, allowing gradients to flow from the rendering error back through the differentiable simulator to infer underlying pressure and velocity fields. This end-to-end differentiable pipeline enables the learning of coherent dynamic simulations directly from unannotated videos. Additionally, for systems where data is scarce or noisy, frameworks like [72] demonstrate the ability to infer hidden quantities such as pressure and velocity fields solely from the visualization of passive scalars like dye or smoke. By encoding the conservation laws of mass, momentum, and energy directly into the loss function, these models extract quantitative information in complex domains where direct measurement is impossible.

Advanced methodologies further explore the use of neural operators and reduced-order models to accelerate the solution of governing equations while maintaining fidelity. For instance, [73] highlights the versatility of PINNs in solving inverse flow problems and handling high-dimensional parametrized Navier-Stokes equations, offering a mesh-free alternative to traditional computational fluid dynamics. Moreover, approaches such as [74] embed deep neural networks directly within partial differential equations to learn unknown sub-filter-scale terms, effectively correcting discretization errors and improving the accuracy of large-eddy simulations. These methods collectively illustrate a maturing field where the boundary between neural rendering and physical simulation is blurring, yielding systems that are both visually compelling and scientifically rigorous. By strictly adhering to physical laws, these physics-informed neural rendering pipelines provide a robust foundation for reconstructing complex, non-rigid fluid dynamics, bridging the gap between the probabilistic completeness of generative models and the kinematic precision required for the motion-aware optimization techniques explored in the following section.

2.7 Motion-Aware Optimization and Semantic Flow Integration

2.7 Motion-Aware Optimization and Semantic Flow Integration

While the preceding section detailed how embedding physical governing equations can reconstruct complex fluid dynamics, a complementary and equally critical paradigm for general dynamic scene reconstruction involves leveraging data-driven motion cues. The application of 3D Gaussian Splatting (3DGS) to dynamic sequences presents unique challenges that extend beyond static modeling; standard 3DGS often suffers from temporal inconsistencies, floating artifacts, and inefficient optimization when confronted with complex non-rigid deformations. To mitigate these issues without the computational overhead of solving full partial differential equations, researchers have increasingly turned to motion-aware optimization. This strategy integrates external motion priors—specifically optical flow and scene flow—directly into the optimization pipeline. By leveraging these dense motion fields, the densification, deformation, and regularization of Gaussian primitives can be guided to adhere to the observed kinematic dynamics of the scene, bridging the gap between purely photometric reconstruction and physically plausible motion.

The fundamental premise of motion-aware optimization is that pixel-level correspondences across video frames provide strong geometric constraints that are otherwise absent in monocular or sparse-view settings. Traditional approaches that treat time as an independent dimension or rely solely on photometric loss frequently lead to overfitting in texture-less regions or under-constrained motion estimation. Recent frameworks address this by establishing explicit correspondences between the movement of 3D Gaussians and observed 2D optical flow. For instance, the work on [54] introduces a novel framework that mines useful motion cues to enhance various dynamic 3DGS paradigms. This method establishes a direct correspondence between 3D Gaussian trajectories and pixel-level flow, utilizing a flow augmentation technique that incorporates uncertainty estimation and collaborative loss functions. By doing so, it effectively regularizes Gaussian deformation, preventing the model from learning spurious motions that satisfy photometric errors but violate temporal consistency. Furthermore, for deformation-based paradigms facing difficult optimization landscapes, this approach employs a transient-aware deformation auxiliary module, significantly improving reconstruction quality in regions with complex, non-rigid movements.

Beyond simple motion correspondence, the integration of semantic information with motion estimation has proven vital for disentangling foreground objects from dynamic backgrounds. In scenarios involving multiple moving entities, such as autonomous driving or crowded scenes, treating the entire scene as a single deformable entity often results in blurred boundaries and inaccurate geometry. Semantic flow integration addresses this by combining motion segmentation with dense flow estimation. The methodology presented in [75] demonstrates the efficacy of fusing semantic features with motion clues using Convolutional Neural Networks (CNNs). By integrating optical flow as a constraint with semantic features within a dilated convolution network, this approach achieves robust monocular semantic motion segmentation. When applied to Gaussian Splatting, such semantic masks guide the initialization and updating of Gaussian primitives, ensuring that distinct objects maintain coherent motion fields while allowing for independent deformation. This is particularly crucial for handling occlusions and dis-occlusions, where semantic priors help infer the structure of hidden regions based on the motion of visible parts.

The challenge of estimating accurate scene flow in sparse representations, which are analogous to the centers of 3D Gaussians, has also driven innovations in regularization strategies. Standard scene flow methods often struggle with the inconsistency between sparse 3D points and dense 2D pixels. To bridge this gap, [76] proposes regularizing raw points into a dense 2D grid format, preserving most points in the scene and eliminating the density gap between modalities. This dense representation allows for effective feature fusion and more accurate warping projections, which can be directly adapted to optimize the spatial distribution of Gaussian primitives during training. By ensuring that the underlying motion field is dense and consistent, the Gaussian splats can be deformed more accurately, reducing artifacts caused by incorrect motion interpolation.

Moreover, the reliability of motion estimation is heavily dependent on handling outliers, occlusions, and large displacements. Variational methods and learning-based approaches have increasingly focused on robust regularization to manage these imperfections. The [77] technique offers a data-driven approach using Mixed Norms to estimate motion vectors while accounting for various noise sources and local image properties without prior knowledge of noise distribution. Such adaptive regularization is essential for dynamic 3DGS, as it prevents the optimization process from being misled by erroneous flow vectors in regions with motion blur or reflection. Similarly, [78] highlights the importance of occlusion consistency and transformation consistency as self-supervised constraints. By enforcing these consistency checks, the system can better distinguish between valid motion and noise, leading to more stable Gaussian deformation fields over time.

In the context of non-rigid objects, where simple rigid body transformations are insufficient, advanced motion models are required. [57] addresses the ill-posed nature of reconstructing dynamic objects from monocular videos by introducing a two-stage approach. First, a low-rank neural deformation model is fitted to capture coarse motion consistency, which then serves as a strong regularizer for the subsequent optimization of 3D Gaussians. This hierarchical use of motion priors ensures that the high-frequency details captured by the Gaussians do not drift temporally. Additionally, the concept of using epipolar constraints to refine motion saliency, as explored in [79], provides a mechanism to filter out static background noise and focus optimization resources on truly dynamic regions. By leveraging trajectory-based epipolar distances, the system can robustly fuse motion and appearance features, enhancing the segmentation of moving objects, which is critical for assigning independent deformation fields to different Gaussian clusters.

Finally, the integration of motion cues extends to the handling of large-scale displacements and complex topological changes. Methods like [80] utilize multi-scale hybrid matching and graph-based approaches to handle substantial displacements and non-rigid transformations. Adapting such strategies to 3DGS allows for the initialization of Gaussians in correct temporal positions even when object motion exceeds the typical receptive field of standard optical flow networks. By combining these robust motion estimation techniques with the explicit rendering capabilities of Gaussian Splatting, the field is moving towards systems that can reconstruct highly dynamic, non-rigid scenes with unprecedented fidelity and temporal stability. The synergy between motion-aware optimization and semantic flow integration thus represents a cornerstone for the next generation of dynamic 3D reconstruction algorithms, enabling applications ranging from free-viewpoint video of sports events to precise digital twins of deformable industrial components.

3 Human-Centric Reconstruction, Animatable Avatars, and Articulated Motion

3.1 Parametric Priors and Expressive Body Modeling

3.1 Parametric Priors and Expressive Body Modeling

The reconstruction of high-fidelity, animatable human avatars from sparse or monocular inputs remains one of the most challenging frontiers in computer vision and graphics. The inherent ill-posed nature of this problem, characterized by limited observational data, leads to significant ambiguities in inferring 3D geometry, pose, and non-rigid deformations. To address these challenges, the research community has increasingly integrated parametric body models as structural priors. These models, most notably the Skinned Multi-Person Linear (SMPL) model and its expressive extensions like SMPL-X, provide a low-dimensional manifold that encapsulates the statistical regularities of human shape and pose. By constraining the solution space to this learned manifold, researchers can achieve robust reconstruction even when visual cues are occluded, noisy, or incomplete. This foundational reliance on parametric priors sets the stage for the subsequent deformation mechanisms discussed in the following section, where the rigid skeleton provided by these models serves as the anchor for learning complex non-rigid dynamics.

The evolution of parametric priors began with the widespread adoption of SMPL, which utilizes a linear blend skinning function to map pose and shape parameters to a deformable mesh. However, standard SMPL is limited to minimally clothed bodies and lacks the capacity to represent detailed facial expressions or articulated hands, which are crucial for truly expressive avatars. To overcome these limitations, the community introduced SMPL-X, a unified model that extends SMPL by incorporating fully articulated hands and an expressive face [81]. This extension allows for the simultaneous estimation of body pose, hand gestures, and facial expressions from a single monocular image, facilitating a more holistic analysis of human actions and emotions. The transition to SMPL-X has been pivotal in enabling systems to capture the full spectrum of human expressiveness, moving beyond rigid body tracking to nuanced social interaction modeling.

Despite the advantages of expressive models, directly regressing their high-dimensional parameters from images remains fraught with difficulties, primarily due to the risk of generating physically implausible or unrealistic meshes. The parameter space of models like SMPL is under-constrained, meaning that many valid parameter combinations do not correspond to realistic human configurations. To mitigate this, recent works have focused on learning adversarial priors that restrict the parameter space to regions yielding realistic poses [82]. By training a discriminator to distinguish between realistic and unrealistic pose parameters, these methods effectively regularize the optimization process, ensuring that reconstructed avatars maintain natural anatomical constraints even when derived from sparse 2D keypoints or noisy inputs. Furthermore, advancements such as STAR (Sparse Trained Articulated Human Body Regressor) have sought to improve upon SMPL by defining per-joint pose correctives and learning shape-dependent deformations, thereby reducing the number of parameters while enhancing generalization to diverse body shapes and BMI ranges [83].

The utility of these parametric priors extends beyond mere regularization; they serve as the foundational skeleton for driving complex neural rendering pipelines. In the context of neural radiance fields and Gaussian splatting, parametric models provide the necessary kinematic structure to disentangle pose-dependent deformations from appearance. For instance, frameworks like AvatarReX leverage the structural prior from parametric mesh templates to separately model the body, hands, and face, enabling real-time animation and rendering of expressive full-body avatars [84]. Similarly, methods such as ParDy-Human introduce parameter-driven dynamics into 3D Gaussian Splatting, where Gaussians are deformed according to SMPL vertices to animate avatars trained from single monocular sequences [85]. This synergy allows explicit representations to inherit the temporal consistency and articulation capabilities of parametric models while retaining the high-frequency detail capture of neural rendering, effectively bridging the gap between coarse skeletal motion and fine-grained surface appearance.

Moreover, the integration of parametric priors has proven essential for handling challenging scenarios involving heavy clothing or extreme poses. Traditional parametric models struggle with garments that deviate significantly from the body surface, a limitation that necessitates hybrid approaches combining the robustness of parametric skeletons with flexible implicit or point-based representations. For example, ICON (Implicit Clothed humans Obtained from Normals) exploits the SMPL-X model to infer detailed clothed-human normals, using a feedback loop to refine both the underlying mesh and the clothed surface [86]. Other methods, such as those utilizing Neural Parametric Models (NPMs), learn to disentangle 4D dynamics into latent-space representations of shape and pose, offering a learned alternative to hand-crafted parametric constraints that can better capture wrinkles and fabric dynamics [87]. These hybrid strategies naturally lead to the need for more sophisticated skinning mechanisms capable of modeling the complex, non-linear interactions between the body and loose-fitting attire.

Data efficiency is another critical area where parametric priors excel. Training expressive avatar models typically requires vast amounts of multi-view data, which is expensive and difficult to acquire. By leveraging priors like SMPL-X, methods such as ELICIT can learn human-specific neural radiance fields from a single image, utilizing the 3D body shape geometry prior to guide the optimization of invisible regions [88]. This capability is further enhanced by large-scale synthetic datasets like SynBody, which provide layered human models and accurate 3D annotations to improve the training of both parametric estimation and neural rendering tasks [89]. Additionally, the development of foundation models like SMPLer-X, which scales up expressive human pose and shape estimation using massive datasets and Vision Transformers, demonstrates the potential for creating generalist models that perform robustly across diverse and unseen environments [90].

In summary, parametric priors such as SMPL and SMPL-X have become indispensable tools in the reconstruction of dynamic 3D human scenes. They provide a structured, low-dimensional representation that anchors the learning process, ensuring geometric consistency and physical plausibility. While these models offer a robust skeletal framework, they rely on subsequent deformation layers to capture the nuances of soft tissue and clothing dynamics. As the field progresses towards more expressive, real-time, and data-efficient avatar generation, the tight integration of these parametric models with advanced neural rendering techniques continues to drive significant improvements in the fidelity and controllability of digital humans, paving the way for the detailed exploration of skinning mechanisms and pose-dependent deformations in the next section.

3.2 Skinning Mechanisms and Pose-Dependent Deformation

3.2 Skinning Mechanisms and Pose-Dependent Deformation

Building upon the structural foundations established by parametric priors in the previous section, the critical challenge shifts to defining how these rigid skeletons drive the complex, non-rigid deformations of soft tissue and clothing. While parametric models like SMPL-X provide a robust low-dimensional manifold for pose and shape, accurately simulating the high-frequency dynamics inherent in human motion requires sophisticated skinning mechanisms. Traditional approaches have long relied on Linear Blend Skinning (LBS) as the standard deformation mechanism due to its computational efficiency and compatibility with existing animation pipelines. However, LBS is fundamentally limited by its linearity, often resulting in undesirable artifacts such as joint collapsing, volume loss, and an inability to capture the complex, non-linear deformations inherent in loose-fitting garments or soft biological tissues. Consequently, recent research has pivoted towards learning-based skinning mechanisms that augment or replace standard LBS weights with data-driven, pose-dependent deformation fields. These advanced techniques aim to disentangle rigid bone transformations from non-rigid surface details, ensuring that avatars maintain geometric consistency and physical plausibility across a wide range of unseen poses, thereby bridging the gap between coarse skeletal motion and fine-grained appearance.

A primary direction in overcoming the limitations of standard LBS involves learning pose-dependent corrections directly from data. Methods such as [91] introduce a Pose-conditioned Invertible Network (PIN) that extends the traditional LBS process by learning additional deformations that vary with the pose. This architecture ensures that surface correspondences are preserved across different poses while rectifying the artifacts typically introduced by linear blending. By combining this with a differentiable LBS module, the system achieves an end-to-end pipeline that is significantly faster than state-of-the-art reposing techniques while maintaining high fidelity. Similarly, [92] generalizes the concept of articulation and non-rigid deformation for implicit shape learning. Instead of relying on a fixed template resolution, it learns to map points in space to a canonical space where a learned deformation field models non-rigid effects before evaluating the signed distance function. This allows for the reconstruction of detailed, complex surface geometries and deformations in clothing and soft tissue without the need for explicit template registration, laying the groundwork for the explicit representations discussed later.

To address the specific challenges of loose-fitting clothing, which exhibits dynamic modes difficult to predict with simple skinning, researchers have developed hierarchical and graph-based approaches. [93] leverages hierarchical graphs and multi-level message passing to enable real-time prediction of realistic clothing dynamics that are agnostic to body shape. This method effectively handles changes in topology and material properties at inference time, utilizing a hierarchical message-passing scheme to propagate stiff stretching modes while preserving local details. Complementing this, [94] proposes an anchor-based deformation model where rigid transformations are learned via a set of anchors around the mesh surface. This reduces the complexity of the deformation space learning and ensures physical consistency by constraining anchor transformations. Furthermore, [95] introduces a spectrum-inspired approach that utilizes frequency-control strategies to enhance the generation of high-frequency details in free-swinging garments, addressing the common bias towards low-frequency smoothing in supervised learning.

The integration of physical priors into skinning mechanisms has also emerged as a critical strategy for ensuring realistic motion, moving beyond purely geometric constraints. [96] presents a methodology to automatically obtain Pose Space Deformation (PSD) basis for rigged garments through an unsupervised deep learning framework formulated as an implicit physically based simulator. This approach overcomes the data hunger of supervised methods and avoids the computational expense of traditional physics-based simulations while producing cloth-consistent animations with meaningful wrinkles. In a similar vein, [97] realizes a self-supervised training scheme where physics-based deformation models are recast as an optimization problem, allowing neural networks to learn dynamic garment deformations without ground-truth samples. This results in interactive garments with fine wrinkles and a significant reduction in training time. Additionally, [98] integrates a simulation layer into the training process to provide physics supervision, largely reducing cloth-body intersections and modeling clothing as a separate piece of geometry.

For soft-tissue simulation, which requires capturing subtle, non-linear dynamics driven by muscle movement and inertia, specialized skinning extensions have been proposed to complement the expressive capabilities of models like SMPL-X. [99] introduces a learning-based method that models realistic soft-tissue dynamics as a function of body shape and motion. It employs a novel motion descriptor to disentangle subject-specific features and a highly efficient nonlinear deformation subspace, demonstrating robustness across various motion capture databases. [100] utilizes a recurrent neural network to predict nonlinear, dynamics-dependent shape changes over time, learning to predict variations in skin dynamics across different individuals. Moreover, [101] proposes a reduced-space elasto-dynamic solver based on a novel deformation subspace parameterized by scalar weights derived from a rig-sensitive generalized eigenvalue problem, effectively augmenting rigged animations with secondary motion that is rotation equivariant.

Advanced parameterization strategies have also been explored to improve the generalization of skinning models and prepare them for explicit representation formats. [102] embeds virtual cloth into a tetrahedral mesh parameterizing the volumetric air region surrounding the body. This volumetric parameterization allows the network to automatically capture nonlinear deformations due to joint rotations and collisions, showing significant improvement over surface-based parameterizations. Finally, [103] addresses the issue of skin-collapsing artifacts in bending or twisting motions by proposing neural dual quaternion blend skinning (NeuDBS). This method achieves 3D point deformation that performs rigid transformations without the scale and shear factors often introduced by linearly weighted combination matrices, thereby providing a more robust foundation for reconstructing deformable objects from casual videos. Collectively, these advancements signify a paradigm shift from static, linear skinning weights to dynamic, learned, and physics-informed deformation fields. This evolution in deformation logic is essential not only for implicit representations but also serves as the critical algorithmic backbone for the explicit Gaussian Splatting frameworks discussed in the following section, where these learned deformations are applied to millions of primitives to achieve real-time fidelity.

3.3 Explicit Gaussian Representations for Real-Time Animation

3.3 Explicit Gaussian Representations for Real-Time Animation

Building upon the advancements in pose-dependent deformation mechanisms discussed in the previous section, the field of human-centric 3D reconstruction has undergone a transformative shift with the introduction of 3D Gaussian Splatting (3DGS). This evolution marks a decisive move away from the computational bottlenecks inherent in implicit Neural Radiance Field (NeRF) based avatars, which, despite achieving impressive photorealism, remain constrained by the necessity of querying dense multi-layer perceptrons (MLPs) for every sampled point along a ray. Such processes have historically rendered real-time animation on consumer hardware nearly impossible, creating a gap between high-fidelity reconstruction and interactive deployment. The transition to explicit Gaussian representations directly addresses these latency issues by leveraging differentiable rasterization, enabling frame rates that exceed the requirements for interactive applications such as virtual reality and telepresence. This subsection details the methodological progression where millions of learnable Gaussian primitives are bound to parametric templates, facilitating high-fidelity, pose-controllable avatars that operate at real-time speeds while inheriting the deformation logic established by modern skinning techniques.

A primary strategy in this domain involves anchoring 3D Gaussians to a canonical space defined by parametric body models, such as SMPL, and deforming them into posed space using the skinning techniques outlined previously. This approach allows the explicit geometry of the Gaussians to inherit the structural priors of the underlying mesh while retaining the flexibility to model fine-grained details like clothing folds and hair that deviate from the coarse body template. For instance, [104] demonstrates a framework that encodes Gaussian Splatting in a canonical space and utilizes Linear Blend Skinning (LBS) to transform these primitives into the posed space. By integrating effective pose and LBS refinement modules, this method captures fine human details with negligible computational overhead, achieving training times of merely one to two minutes and rendering speeds up to 189 FPS. Similarly, [105] proposes a coarse-to-fine deformation approach that combines forward skinning with local non-rigid refinement. This end-to-end learning process from multi-view observations yields a 1.5 dB improvement in PSNR over state-of-the-art methods on the THuman4 dataset while maintaining an 80 FPS rendering rate, effectively bridging the gap between quality and speed.

To further enhance the controllability and generalization of animatable avatars, recent works have explored binding Gaussians directly to the surface of deformable meshes or utilizing 2D parameterizations, thereby creating a hybrid representation that serves as a precursor to explicit mesh extraction. [106] introduces a strategy where Gaussians are embedded on a triangle mesh using barycentric coordinates and displacement vectors. This "mesh-embedded" approach disentangles low-frequency motion, handled by the mesh deformation, from high-frequency appearance details managed by the Gaussians. Crucially, this method controls Gaussian rotation and translation directly via the mesh, ensuring compatibility with standard animation pipelines like skeletal animation and blend shapes, resulting in rendering speeds exceeding 300 FPS on modern GPUs. In a complementary approach, [107] parameterizes a learned template on front and back canonical Gaussian maps. By leveraging powerful 2D convolutional architectures, this method learns pose-dependent Gaussian maps that adapt to loose garments, achieving lifelike avatars with dynamic appearances that generalize well to unseen poses.

The challenge of initializing and optimizing millions of Gaussians for dynamic humans has also been addressed through novel regularization and optimization strategies designed to stabilize the explicit representation under complex deformations. [108] introduces as-isometric-as-possible regularizations on both Gaussian mean vectors and covariance matrices. This explicit constraint enhances the model's generalization capability for highly articulated, unseen poses, allowing for training within 30 minutes and real-time inference at over 50 FPS. Furthermore, [109] tackles the computational complexity of learning Gaussian parameters in 3D space by attaching them to a deformable character model and learning their attributes in 2D texture space. This innovation enables the use of efficient 2D CNNs, scaling effectively with the number of Gaussians required for photorealistic rendering and outperforming existing real-time methods by a significant margin.

Specialized applications, such as head avatars and talking heads, have also benefited from this explicit representation shift, offering precise control over facial dynamics that implicit methods struggle to match. [110] rigs 3D Gaussians to a parametric morphable face model, optimizing for explicit displacement offsets to achieve precise animation control via expression transfer. This method facilitates fully controllable avatars that can be driven by video or manual parameter adjustments. Similarly, [111] and its variant [112] bind Gaussians to 3D facial models to achieve intuitive control over facial motions. By employing speaker-specific blend shapes and attention mechanisms to manipulate Gaussian attributes based on audio features, these frameworks deliver precise lip synchronization and stable rendering at speeds up to 130 FPS, far surpassing the capabilities of implicit NeRF-based talking head synthesizers. Additionally, [113] extends the explicit representation with learnable latent features blended with parametric model parameters, accelerating rendering speeds by over ten times compared to baselines while improving quantitative metrics.

The transition to explicit Gaussian representations represents not merely an incremental improvement in speed but a fundamental rethinking of how animatable humans are modeled, serving as a critical bridge between volumetric neural fields and explicit geometric constructs. By binding explicit primitives to parametric templates, these methods successfully decouple the heavy computational burden of volumetric rendering from the animation pipeline, laying the groundwork for the geometry-aware mesh extraction techniques discussed in the following section. Whether through direct mesh embedding, canonical space deformation, or 2D map parameterization, the consensus in recent literature indicates that explicit 3DGS frameworks provide the optimal balance of geometric fidelity, textural richness, and interactive performance. As evidenced by the substantial speedups and quality improvements reported in [104], [106], and [114], the era of real-time, photorealistic, and fully controllable human avatars has been firmly established, paving the way for the conversion of these rich neural representations into editable, watertight meshes.

3.4 Geometry-Aware Reconstruction and Mesh Extraction for Avatars

3.4 Geometry-Aware Reconstruction and Mesh Extraction for Avatars

Building upon the explicit Gaussian representations detailed in the previous section, which successfully decoupled high-fidelity rendering from heavy volumetric computation, the field now faces the imperative of converting these rich neural primitives into robust, geometry-aware mesh constructs. While the preceding approaches excel at real-time animation by binding Gaussians to parametric templates, many downstream applications in physics simulation, virtual production, and interactive gaming demand topologically consistent, watertight surfaces that are directly compatible with standard computer graphics pipelines. Consequently, this subsection explores the critical evolution from volumetric or point-based neural fields to editable, high-fidelity avatars. The focus shifts from merely achieving photorealistic novel view synthesis to enforcing strict geometric consistency, leveraging displacement maps, differentiable rasterization, and hybrid neural-mesh architectures to facilitate the extraction of avatars that are not only visually accurate but also structurally sound.

A primary challenge in this transition is bridging the gap between the flexibility of neural fields and the structural rigor of polygonal meshes. Traditional mesh-based approaches often lack the capacity to capture fine-grained details such as skin pores, wrinkles, and hair, while purely neural approaches frequently obscure the underlying surface geometry, resulting in non-manifold outputs. To address this, modern frameworks employ strategies where a canonical base mesh is optimized alongside neural texture or displacement maps. For instance, the [115] framework demonstrates an efficient pathway to learning personalized animatable 3D head avatars from monocular video. By optimizing a canonical geometry using a mesh representation, [115] enables efficient differentiable rasterization and straightforward animation through learned blendshapes and linear blend skinning weights. Crucially, this approach decouples geometry from appearance by factoring observed colors into intrinsic albedo, roughness, and a neural illumination representation, allowing the resulting avatars to be not only geometrically accurate but also relightable in novel environments. This separation is vital for downstream applications where material properties must be manipulated independently of the mesh topology, a prerequisite for the advanced relighting tasks discussed in the subsequent section.

Further advancing the fidelity of mesh extraction, techniques have been developed to handle the complex non-rigid deformations inherent in human performance capture, ensuring that the transition from neural fields to meshes does not compromise quality. The [116] method addresses the common issue where volumetric rendering methods generate noisy meshes with poor topology. [116] proposes a hybrid approach that takes geometry initialization from neural volumetric fields and subsequently optimizes both the geometry and a compact neural texture representation using differentiable rasterizers. This process ensures the generation of accurate, watertight manifold meshes that are comparable in appearance quality to volume rendering methods but offer an order of magnitude improvement in rendering speed. The resulting meshes are fully compatible with existing graphics pipelines, enabling robust simulation and animation workflows that were previously hindered by the non-manifold nature of distilled neural fields.

In the domain of facial animation, where topological consistency across extreme expressions is paramount, specialized architectures have emerged to ensure robust mesh generation without overreliance on rigid statistical models. The [117] framework, known as ToFu, introduces a geometry inference mechanism that produces topologically consistent meshes across diverse facial identities and expressions without relying on an underlying 3D Morphable Model (3DMM) as a hard constraint. Instead, [117] embeds the topological structure of the face into a feature volume sampled from geometry-aware local features. This coarse-to-fine architecture facilitates dense and accurate predictions while capturing displacement maps for pore-level geometric details. The output includes high-quality assets such as albedo and specular reflectance maps, which are readily usable for production-grade avatar creation and physically-based skin rendering. This approach significantly reduces the need for manual cleanup often required after multi-view stereo reconstruction, particularly in challenging regions like hair and extreme expressions.

The integration of parametric priors with detailed geometric refinement has also proven effective for enhancing the alignment between reconstructed models and image observations, ensuring that extracted meshes faithfully represent specific subjects. In [118], the authors propose constructing dense correspondences between initial human model estimates and image observations to refine predictions. By utilizing renderings of 3D models to predict per-pixel 2D displacements, [118] effectively integrates appearance information to minimize reprojection loss. This method transforms per-pixel displacements into per-visible-vertex displacements, leading to improved 3D accuracy and better image-model alignment, which is essential for creating avatars that faithfully represent the subject's specific geometry rather than a generic statistical average.

Moreover, the extraction of editable blendshapes from video data remains a cornerstone for industrial animation pipelines, serving as a direct bridge between neural reconstruction outputs and traditional rigging systems. The [119] technique reconstructs mesh-based blendshape rigs from single or sparse multi-view videos by leveraging neural inverse rendering. This method parameterizes vertex displacements into differential coordinates with tetrahedral connections, allowing for high-quality deformation on high-resolution meshes. By constructing semantic regulations within this representation, [119] achieves joint optimization of blendshapes and expression coefficients, even handling unsynchronized camera inputs through a neural regressor. The resulting blendshapes are both geometrically and semantically accurate, providing a direct bridge between neural reconstruction outputs and traditional animation rigs.

Finally, the ability to edit and manipulate these extracted geometries is enhanced by differentiable rendering layers that support gradient-based optimization, closing the loop between learning and editing. The [120] presents a differentiable layer compatible with deep neural networks that bridges simplex mesh-based geometry representations with raster images. By using Non-Uniform Fourier Transform for anti-aliased rasterization, [120] facilitates end-to-end training and mesh editing guided by neural network outputs. This capability is crucial for refining extracted avatars, ensuring that the final meshes are not only visually faithful but also topologically sound and ready for dynamic simulation, rigging, and interactive editing in virtual environments. Together, these approaches signify a maturing of the field, moving beyond mere visual reproduction to the creation of robust, geometrically consistent, and fully animatable digital humans. By establishing these reliable geometric foundations, the field is now poised to tackle the next frontier: overcoming data scarcity through generative synthesis, as explored in the following section.

3.5 Data-Efficient Learning and Generative Avatar Synthesis

3.5 Data-Efficient Learning and Generative Avatar Synthesis

While the preceding section detailed methods for extracting robust, geometry-aware meshes from dense video capture, a parallel and rapidly evolving frontier addresses the critical bottleneck of data acquisition. The creation of high-fidelity, animatable 3D human avatars has traditionally been constrained by the requirement for extensive data, often necessitating multi-view camera arrays, controlled studio lighting, and dense motion capture sequences. However, recent advancements in neural rendering and generative artificial intelligence have catalyzed a paradigm shift toward data-efficient learning. This transition enables the synthesis of diverse and realistic avatars from minimal inputs—such as single images, few-shot photographs, or even textual descriptions alone. This subsection explores the frontier of generative avatar synthesis, focusing on methodologies that leverage meta-learning strategies, diffusion priors, and parametric body models to overcome data scarcity while maintaining the geometric consistency and animation capability required for the downstream relighting and material disentanglement discussed in the following section.

A primary direction in data-efficient avatar creation involves leveraging large-scale pre-trained diffusion models to guide the generation of 3D assets from sparse supervision. The integration of vision-language models allows for zero-shot generation pipelines where text prompts serve as the sole input. For instance, [121] proposes a framework that utilizes the CLIP model to supervise neural human generation, enabling layman users to customize avatar geometry, texture, and animation solely through natural language descriptions. By initializing geometry with a shape VAE and refining it via volume rendering guided by CLIP scores, this approach achieves superior zero-shot capabilities, generating unseen avatars with novel animations without requiring any specific training data for the target subject. Similarly, [122] addresses the challenge of creating animatable avatars with specific identities and artistic styles from a single text prompt. It employs a coarse-to-fine training strategy combined with diffusion-based constraints and an explicit warping field tied to parametric human models, ensuring that the resulting neural implicit fields are not only high-quality but also readily animatable.

To further enhance the diversity and stability of text-to-3D generation, recent works have introduced mechanisms to decouple geometry and appearance or to enforce multi-view consistency, addressing artifacts common in generative approaches. [123] tackles the limitation of limited diversity in existing text-to-avatar methods by fine-tuning a 3D generative model, specifically EVA3D, allowing for the generation of richly varied avatars from a single prompt through noise sampling. This method introduces a semantic-aware zoom mechanism and a novel depth loss to ensure high textual fidelity and improved geometry quality. Addressing the infamous "Janus problem" (multi-headed artifacts) and view inconsistency, [124] presents a stable pipeline that conditions a 2D diffusion model on DensePose signals. This ensures robust 3D pose control and view consistency, enabling the progressive synthesis of high-resolution, expressive avatars that surpass previous standards in fidelity. Furthermore, [125] introduces self-evolving constraints to decouple geometry and appearance optimization. By periodically updating a template avatar initialized with human priors and applying lightness constraints to albedo textures, this method produces photorealistic meshes suitable for classic graphics pipelines under arbitrary lighting conditions, thereby laying a solid foundation for subsequent relighting tasks.

Beyond pure text guidance, few-shot learning from unconstrained images represents another critical avenue for data efficiency, bridging the gap between generative priors and subject-specific reconstruction. Traditional methods often fail when provided with only a handful of casual photos due to dynamic articulated poses and limited coverage. [126] investigates reconstruction from few-shot unconstrained photo albums by integrating a skinning mechanism with deep marching tetrahedra (DMTet). This drivable tetrahedral representation adapts to arbitrary mesh topologies, while a two-phase optimization strategy aligns avatar identity with reference images and generates plausible appearances for unseen regions. Similarly, [127] introduces a generative implicit head avatar model designed for few-shot user adaptation. By learning a prior from a large multi-view dataset of expressions and anchoring it to a 3DMM-based neural radiance field backbone, the method effectively handles unstable 3DMM fitting and achieves high-quality personalization from just one or a few images per user. The concept of leveraging priors is further extended in [88], which utilizes both 3D geometry priors from skinned templates and visual semantic priors from CLIP to learn neural radiance fields from a single image, effectively hallucinating plausible content in invisible areas.

Meta-learning and adaptation strategies have also been pivotal in reducing the computational and data burden of avatar personalization, offering efficient pathways to customize generic models. [128] investigates a meta-learning framework to balance speed, flexibility, and quality in stylizing dynamic avatars. By learning an initialization in the outer loop that can quickly adapt to novel styles defined by text or images in the inner loop, this approach avoids the need to re-optimize from scratch for every new style input. In the domain of head avatars, [129] proposes an incremental GAN inversion technique that enhances reconstruction fidelity proportionally to the number of input frames. By incorporating a ConvGRU-based recurrent network for temporal aggregation and a neural texture encoder, it mitigates issues like identity flickering and expression inaccuracy common in one-shot inversion techniques. Additionally, [130] formalizes the objective of consistent character generation through cluster-conditioned guidance, offering a lightweight alternative to expensive tuning pipelines. This method guides denoising trajectories toward target clusters using posterior samples, significantly enhancing content diversity while maintaining character consistency across generations.

The synergy between generative models and parametric priors continues to drive innovation in this field, ensuring that synthesized avatars possess the structural integrity necessary for animation and physical simulation. [131] integrates a 3D morphable model into a multi-view-consistent diffusion approach, enabling the creation of fully 3D-consistent, animatable, and photorealistic avatars from a single image of an unseen subject. This accurate conditioning on articulated 3D models facilitates seamless incorporation of facial expression and body pose control. Moreover, [132] utilizes dual fine-tuned diffusion models separately for the face and body to achieve precise control over intricate details, supported by a pose-consistent constraint to eliminate interference from uncontrolled poses in casually captured images. These collective advancements demonstrate that the field is moving rapidly away from data-hungry reconstruction pipelines toward flexible, generative frameworks. By enabling the synthesis of diverse, animatable, and high-quality human avatars from minimal, everyday data sources, these methods provide the essential geometric and textural foundations required for the advanced inverse rendering and relighting capabilities explored in the subsequent section.

3.6 Inverse Rendering, Relighting, and Material Disentanglement for Humans

3.6 Inverse Rendering, Relighting, and Material Disentanglement for Humans

Building upon the data-efficient synthesis and generative frameworks established in the previous section, which provide the essential geometric and textural foundations for animatable avatars, this subsection addresses the critical next step: achieving physical realism through inverse rendering. While generative priors enable the creation of diverse avatars from sparse inputs, these assets often remain entangled with their original lighting conditions, limiting their versatility in dynamic virtual environments. To transition from static or conditionally lit reconstructions to truly interactive digital humans, it is necessary to decompose observed radiance into intrinsic components—albedo, surface normals, roughness, and illumination. This decomposition, known as inverse rendering, is paramount for relighting, allowing avatars to interact naturally with arbitrary lighting in augmented reality or immersive worlds. The core challenge lies in the ill-posed nature of this problem, exacerbated by the non-rigid articulation of human bodies where pose-dependent effects like self-occlusion, soft tissue deformation, and complex clothing wrinkles introduce significant ambiguity in separating shading from reflectance.

Early approaches to human relighting often relied on simplified lighting models, such as spherical harmonics, which struggle to capture high-frequency details like sharp shadows and specular highlights. Furthermore, many initial methods focused primarily on faces, neglecting the complexities of full-body articulation and clothed geometry. To address these limitations, researchers have developed frameworks that explicitly model the physical image formation process. For instance, [133] proposes a principled framework that decomposes space-time varying geometry and reflectance into neural fields of normals, occlusion, diffuse, and specular maps. By integrating these fields into a reflectance-aware physically based rendering pipeline, this approach enables free-viewpoint relighting from monocular videos under unknown illumination, effectively disentangling the dynamic effects of human motion from static material properties. Similarly, [134] advances this domain by modeling secondary shading effects through explicit Monte-Carlo ray tracing rather than approximating them with learned multi-layer perceptrons. This volumetric scattering approach allows for the recovery of high-quality geometry, albedo, and material properties without supervised pre-training, ensuring that the resulting avatars generalize robustly to novel poses and lighting conditions.

The challenge of disentanglement is further compounded by the need to handle complex, heterogeneous materials found in human subjects, such as skin, hair, and eyes, which exhibit distinct reflectance behaviors. Skin, for example, involves subsurface scattering, while eyes require the modeling of refraction and high-frequency specular reflections. [135] addresses this by introducing a hybrid representation that combines an explicit parametric surface model for the eyeball with implicit deformable volumetric representations for the periocular region. This design allows for the accurate simulation of corneal refraction and specular highlights while maintaining the flexibility to model skin reflection, enabling high-fidelity relighting of the eye region under unseen illumination. In the context of full-body avatars, [136] presents a method that represents actor geometry with a drivable density field, encoding normal, visibility, and materials in a UV-mapped space. This architecture facilitates the learning of pose-dependent appearance effects, such as self-shadows and clothing wrinkles, allowing the avatar to be relit and edited under arbitrary skeletal poses using only multi-view recordings from a static lighting condition.

As the field progresses toward real-time interactive applications like virtual reality, efficiency has become a primary constraint alongside fidelity. Recent advancements in 3D Gaussian Splatting have emerged as a powerful solution to capture sub-millimeter geometric details, such as hair strands and pores, which are often lost in traditional mesh-based approaches. [137] leverages this representation by associating Gaussians with learnable radiance transfer parameters and utilizing global illumination-aware spherical harmonics, achieving real-time relighting with spatially all-frequency reflections. The system supports diverse materials uniformly and has been demonstrated to run efficiently on consumer VR headsets, marking a significant step toward practical deployment. Complementing this, [138] extends 3D Gaussians to dynamic scenes by modeling skinned Gaussians in a canonical space and introducing time-dependent ambient occlusion. This approach not only accelerates training and rendering but also enhances the quality of reconstructions in scenes with complex motions and dynamic shadows, crucial for maintaining visual fidelity during relighting.

Despite the shift toward efficient representations, data efficiency and the ability to learn from limited or unconstrained captures remain active areas of research. Traditional methods often require expensive light stage setups to capture ground truth data for training. However, [139] introduces a tracking-free framework that learns from dynamic image sequences captured under varying lighting conditions in a light stage, obviating the need for accurate surface tracking. By explicitly building on the linear nature of lighting, the network predicts appearance in real-time with a single forward pass. For scenarios where controlled capture is impossible, [140] proposes techniques to model geometry and shadow changes from sparse videos under unknown illumination. This includes an invertible deformation field to solve the inverse skinning problem and a pose-aware part-wise light visibility network to estimate occlusion, resulting in high-quality geometry and realistic shadows under different body poses. Additionally, [141] tackles the ill-posed nature of single-image reconstruction by employing a semi-supervised training scheme that learns from both light stage and in-the-wild datasets. This approach effectively disentangles intrinsic face parameters, producing relightable avatars with richer skin reflectance maps than existing state-of-the-art methods.

Finally, ensuring physical consistency under extreme lighting conditions and complex environmental interactions remains a frontier for improvement. [142] addresses the issue of low-contrast and noisy images by learning to decompose illumination, reflectance, and noise, enabling effective intrinsic decomposition even in challenging low-light multi-view settings. Furthermore, the integration of physics-based priors continues to be a vital strategy for improving disentanglement accuracy. [143] introduces a shadow model using a Gaussian density proxy to replace costly sampling, enabling efficient Lambertian shading and shadow casting for neural characters. This method improves the separation of albedo, shading, and shadows, particularly in outdoor scenes with direct sunlight, thereby enhancing the realism of relighted novel poses. Collectively, these works illustrate a rapidly evolving landscape where the fusion of explicit geometric representations, physical rendering equations, and deep learning priors is driving the creation of human avatars that are not only visually indistinguishable from reality but also physically interactive with the light around them. By establishing these physically grounded, relightable assets, this section provides the necessary foundation for the subsequent evolution toward high-level semantic control and interactive editing, where users can manipulate garment styles and attributes without compromising the underlying physical consistency.

3.7 Semantic Control, Editing, and Virtual Try-On

Building upon the physically grounded, relightable avatars discussed in the previous section, the field has advanced toward systems capable of high-level semantic manipulation and interactive editing. While inverse rendering disentangles light and material to ensure physical consistency under varying illumination, semantic control empowers users to modify the intrinsic attributes of digital humans—such as garment style, accessories, and facial features—through intuitive interfaces like natural language. This transition marks a shift from passive reconstruction to active content creation, where the synergy between semantic understanding and generative models facilitates precise, fine-grained editing without compromising identity or pose consistency. This subsection explores this evolution, highlighting how semantic segmentation, large language models, and generative priors converge to enable realistic virtual try-on and attribute manipulation.

At the core of semantic control lies the ability to accurately delineate and understand the compositional elements of human attire, a task complicated by the non-rigid deformations of clothing and intricate body-garment interactions. Traditional approaches often struggled with these complexities, but recent advancements leverage hierarchical semantic parsing to guide the editing process. For instance, frameworks like [144] utilize Large Language Models (LLMs) as a foundational support to interpret user instructions, engaging in iterative interactions to define editing masks. This system employs semantic segmentation models such as Grounded-SAM and MattingAnything to precisely delineate garments or accessories based on natural language prompts, subsequently leveraging visual foundation models like Stable Diffusion to execute the edits. This pipeline demonstrates the potential of combining language guidance with robust segmentation to automate complex fashion editing tasks, enabling operations such as garment replacement, recoloring, and removal with high fidelity.

Virtual try-on represents a particularly challenging application of this semantic control, requiring the seamless transfer of clothing items onto diverse human bodies while preserving texture details and handling occlusions. Early methods often relied on multi-stage pipelines that separately handled warping and synthesis, leading to artifacts and misalignments. To overcome these limitations, single-stage frameworks have emerged that implicitly learn garment warping and body synthesis simultaneously. The [145] paper introduces a novel semantic-contextual fusion attention module that efficiently fuses garment and person features, addressing misalignment issues through a lightweight linear attention mechanism. Similarly, [146] proposes a deformable attention flow scheme that estimates self- and cross-deformable flows guided only by pose keypoints, enabling end-to-end photo-realistic results by merging feature-level and pixel-level information from different semantic areas. These approaches underscore the importance of attention mechanisms in capturing long-range dependencies between semantic regions, ensuring that the warped clothing adheres naturally to the target pose.

Beyond simple warping, maintaining the structural coherence of the generated outfit under varying poses and lighting conditions is paramount. Generative adversarial networks (GANs) and diffusion models have been adapted to enforce such consistency through semantic constraints. [147] augments the GAN inversion process with semantic, pose-related, and image-level constraints, leveraging the CLIP model to enforce semantic alignment between text descriptions and the generated image. This allows for text-conditioned editing where users can describe desired attributes without providing a reference image of the target item. Furthermore, [148] decomposes the generative process into two stages: first generating a plausible semantic segmentation map that obeys the wearer's pose, and then rendering the final image conditioned on this map. This two-stage approach ensures that the global structure remains coherent even when the local textures are significantly altered based on language descriptions.

The challenge of occlusion, where body parts or other garments obscure the target clothing, has also been addressed through semantic-guided strategies. [149] categorizes occlusions into inherent and acquired types, proposing a semantically-guided mixup module to simulate these effects during training. By blending textures with human parts in a copy-and-paste manner aided by semantic guidance, the model learns to de-occlude and synthesize realistic try-on images. Additionally, [150] introduces a network that adaptively decides whether to generate or preserve image content based on a predicted semantic layout, effectively handling large occlusions and complex poses by inpainting missing regions while preserving visible details.

Expanding beyond 2D images, the integration of semantic control into 3D human generation enables interactive editing of clothed avatars, bridging the gap between the static reconstructions of early methods and the dynamic, editable agents required for immersive applications. [151] presents a system that encodes texture, shape, and textual semantics in a canonical UV-aligned space, allowing for multimodal editing via text, images, or sketches. This framework leverages a pre-trained 3D human diffusion model to provide strong priors for diverse generation tasks, facilitating precise control over 3D clothing attributes. Similarly, [152] addresses the compositionality of humans and objects by learning to decompose and recompose 3D scans in an unsupervised manner, enabling the natural attachment of accessories like backpacks or scarves to diverse identities.

Fine-grained attribute manipulation is further enhanced by techniques that operate in latent spaces or utilize recursive architectures to capture sequential dependencies. [153] models a continuous "semantic field" in the GAN latent space, allowing for smooth, dialog-driven editing of facial attributes, a concept extendable to garment attributes. [154] employs feature-wise linear modulation (FiLM) to relate visual features with natural language representations, enabling localized editing without extra spatial information. Moreover, [155] utilizes a recurrent generation pipeline to sequentially layer garments, capturing complex interactions such as tucking shirts or layering jackets, which are difficult to achieve with single-pass methods.

Finally, the robustness of these systems is bolstered by advanced segmentation datasets and models tailored for clothing, providing the necessary geometric and appearance priors for high-quality editing. [156] introduces a large-scale dataset and a learning-based model for fine-grained 3D clothing segmentation. In the realm of unsupervised learning, [157] decomposes the hard mapping of pose transfer into semantic parsing transformation and appearance generation, refining semantic maps end-to-end to preserve clothing attributes. Collectively, these works illustrate a paradigm shift towards semantically driven, language-guided frameworks that offer unprecedented control over human-centric digital content. While the preceding section established the physical basis for relighting and material disentanglement, these semantic advancements complete the pipeline for fully interactive avatars, paving the way for immersive virtual try-on experiences and personalized customization in dynamic 3D environments.

4 Geometry Alignment, Surface Consistency, and Inverse Rendering in Static Scenes

4.1 From Volumetric Radiance to Explicit Surface Representations

4.1 From Volumetric Radiance to Explicit Surface Representations

The evolution of neural scene representation has witnessed a profound paradigm shift from purely implicit volumetric formulations toward explicit geometric constructs compatible with traditional computer graphics pipelines. While Neural Radiance Fields (NeRF) revolutionized novel view synthesis by modeling scenes as continuous functions of density and radiance, their inherent volumetric nature presents significant bottlenecks for downstream applications requiring explicit topology, such as physics-based simulation, real-time interaction in game engines, and precise geometric editing. The core challenge lies in the fact that NeRFs are optimized for view synthesis fidelity rather than geometric accuracy, often resulting in fuzzy boundaries and a lack of watertight surface guarantees. Consequently, a substantial body of research has emerged focused on distilling these volumetric fields into high-fidelity, explicit triangle meshes, effectively bridging the gap between neural rendering and classical geometry processing. This transition sets the stage for the rigorous geometric frameworks discussed in the subsequent section, where the focus shifts from extracting surfaces from density fields to learning them directly via Signed Distance Functions.

The fundamental limitation of standard volumetric rendering is its computational expense and the ambiguity it introduces regarding surface location. As noted in technical digests of the field, NeRFs rely on a probabilistic notion of visibility achieved by integrating through a cloud of light-emitting particles, which necessitates hundreds of samples per ray to render high-resolution images [158]. This approach, while flexible for representing semi-transparent phenomena like foliage or smoke, is inefficient for solid objects where a single surface intersection suffices. To address this, researchers have developed methods that smoothly transition between volumetric and surface-based rendering. For instance, the concept of “Adaptive Shells” introduces a neural radiance formulation that constructs an explicit mesh envelope to spatially bound the volumetric representation. By learning a spatially-varying kernel size, this method allows the representation to converge to a tight surface in solid regions, enabling rendering with a single sample per pixel while retaining volumetric flexibility for complex structures [159]. Similarly, HybridNeRF leverages the strengths of both representations by rendering the majority of a scene as surfaces using Signed Distance Functions (SDFs), while reserving volumetric modeling only for challenging, semi-opaque regions, thereby achieving real-time framerates without sacrificing visual fidelity [160].

A critical avenue of research involves the direct distillation of trained NeRFs into geometrically accurate meshes. The process typically entails extracting an implicit surface, such as an SDF, from the learned density field and then applying marching algorithms to generate a polygonal mesh. However, naive extraction often yields noisy or non-watertight geometries. To overcome this, NeRFMeshing proposes a compact architecture that distills the volumetric 3D representation into a Signed Surface Approximation Network. This approach facilitates the easy extraction of a 3D mesh that is physically accurate and capable of real-time rendering on diverse devices, effectively converting a view-synthesis model into a simulation-ready asset [161]. Further advancing this capability, IRON (Inverse Rendering by Optimizing Neural SDFs) employs a hybrid optimization scheme where a volumetric radiance field is first used to recover the correct topology, followed by edge-aware physics-based surface rendering to refine the geometry and disentangle materials. This two-stage process ensures that the final output is a high-quality triangle mesh suitable for deployment in existing graphics pipelines [162].

Beyond simple extraction, recent works focus on making these explicit representations differentiable and editable. The transition to explicit meshes enables users to manipulate geometry directly, a task that is notoriously difficult in purely implicit fields. Neural Impostor introduces a hybrid representation incorporating an explicit tetrahedral mesh alongside a multigrid implicit field, allowing for pragmatic deformation and compositing while maintaining complex volumetric appearances [163]. Similarly, Mesh-Guided Neural Implicit Field Editing utilizes a differentiable colored mesh extracted via marching tetrahedra to back-propagate gradients from explicit manipulations to the underlying implicit field, enabling fine-grained editing of both geometry and color [164]. For dynamic scenarios, particularly involving human subjects, techniques like Explicifying Neural Implicit Fields propose utilizing a Neural Explicit Surface (NES) to represent implicit fields, facilitating memory efficiency and rapid rasterization-based rendering that supports pose-dependent deformations [165].

The drive toward explicit surfaces is also motivated by the need for physical plausibility in robotics and embodied AI. PhyRecon stands out as a pioneering approach that integrates differentiable physics simulation with neural implicit representations. By employing a novel algorithm called Surface Points Marching Cubes (SP-MC), it enables an efficient transformation between SDF-based implicit representations and explicit surface points, allowing the model to be optimized with both rendering and physical losses. This ensures that the reconstructed geometry is not only visually accurate but also stable under physical simulation, addressing a critical gap in previous neural reconstruction methods [166]. Furthermore, methods like Delicate Textured Mesh Recovery from NeRF via Adaptive Surface Refinement iteratively adjust vertex positions and face density based on re-projected rendering errors, jointly refining appearance and geometry to bake high-quality textures onto the mesh for real-time applications [167].

In summary, the transition from volumetric radiance to explicit surface representations represents a maturation of neural rendering technologies, moving from passive view synthesis to active scene manipulation and simulation. By distilling implicit fields into watertight meshes through adaptive shells, hybrid optimizations, and differentiable extraction techniques, the community is unlocking the potential for neural reconstructions to serve as foundational assets in broader computer graphics and robotics ecosystems. These developments ensure that the high-fidelity appearance captured by neural fields can be preserved while gaining the structural integrity required for interaction, editing, and physical reasoning, thereby providing a robust foundation for the advanced neural surface representations and Signed Distance Functions explored in the following section.

4.2 Neural Surface Representations and Signed Distance Functions

4.2 Neural Surface Representations and Signed Distance Functions

Building upon the transition from volumetric radiance to explicit constructs discussed in the previous section, this subsection delves into the rigorous mathematical frameworks that define these surfaces directly, with Neural Signed Distance Functions (SDFs) emerging as the dominant paradigm. While the distillation methods reviewed earlier focus on extracting geometry from pre-trained density fields, SDF-based approaches integrate geometric consistency directly into the learning process. By representing scene geometry as the zero-level set of a continuous function—where the value denotes the shortest distance to the surface and the sign indicates interior or exterior status—SDFs provide a mathematically robust foundation for defining watertight surfaces. The foundational work [168] established that neural networks could effectively learn continuous SDFs for entire shape classes, enabling high-quality interpolation and completion from partial inputs while significantly reducing model size compared to traditional discretized representations. This direct learning strategy addresses the “floater” artifacts and ambiguous boundaries inherent in volumetric NeRFs, offering the strict geometric consistency required for downstream applications like physics simulation and precise mesh extraction.

However, learning accurate SDFs from multi-view images or sparse point clouds without direct geometric supervision remains a significant challenge due to the ill-posed nature of the problem. A critical factor in achieving high reconstruction accuracy is maintaining gradient consistency throughout the field. Research indicates that the parallelism of level sets is essential for inferring correct geometry; consequently, methods have been developed to introduce level set alignment losses that propagate the zero-level set information across the entire field, thereby eliminating uncertainties caused by sparse observations [10]. Furthermore, standard SDF formulations are inherently limited to closed, watertight topologies, which restricts their application to real-world objects that may possess open surfaces, such as clothing, wires, or scanned room interiors. To overcome this topological constraint, novel frameworks like [169] and [170] have introduced Unsigned Distance Fields (UDFs). These approaches modify the volume rendering scheme to accommodate unsigned distances, allowing for the robust reconstruction of non-closed shapes with complex topologies while maintaining competitive performance on closed surfaces. Similarly, [171] proposes a validity branch alongside the SDF to estimate surface existence probability, effectively handling open surfaces without sacrificing the ability to extract meshes via standard algorithms like Marching Cubes.

To further enhance the fidelity of surface reconstruction, especially in the presence of noise or sparse data, hybrid architectures and advanced regularization strategies have been proposed. For instance, [172] introduces a dual representation mechanism that mutually regularizes a Multi-Layer Perceptron (MLP) and an implicit moving least-squares function, enabling the learning of noise-resistant SDFs from unoriented raw point clouds without requiring ground truth normals. In scenarios involving semi-transparent or thin objects where traditional SDF assumptions of solid surfaces fail, [9] presents a decoupled representation that separates geometry from opacity. This allows for the analytical calculation of ray-surface intersections, accurately reconstructing thin structures that often confound standard volumetric or solid-surface methods. Additionally, for specialized domains such as CAD modeling, where surfaces are composed of developable patches, [173] incorporates curvature constraints into the loss function to enforce zero Gaussian curvature, ensuring the recovery of sharp features and smooth patches characteristic of engineering designs.

The integration of SDFs with other geometric primitives has also yielded significant improvements in handling complex materials and large-scale scenes. Hybrid approaches that combine the efficiency of voxel grids with the continuity of neural fields, such as [174], address the scalability issues of pure neural methods by employing a coarse-to-fine hierarchical structure suitable for online mapping. Moreover, to tackle the challenges of reflective and specular materials which often lead to geometry-radiance entanglement, [175] utilizes a Gaussian-based representation of normals within an SDF field, guided by polarization priors to disentangle specular reflections from underlying geometry. The evolution of these representations also extends to generative modeling, where [176] leverages diffusion models to generate SDFs for shape completion and single-view reconstruction, demonstrating the versatility of implicit representations in probabilistic frameworks. Finally, ensuring that these learned implicit fields can be reliably converted into explicit assets is crucial; methods like [177] have pioneered differentiable mesh extraction techniques, allowing end-to-end optimization of explicit mesh topology directly from deep implicit fields, thus bridging the gap between implicit learning and explicit deployment in graphics pipelines. These advancements collectively underscore the maturity of neural surface representations in resolving topological ambiguities and enforcing strict geometric consistency. With these rigorous geometric foundations established, the subsequent section explores how these representations facilitate the equally critical task of inverse rendering, disentangling material properties and illumination to enable realistic relighting and material editing.

4.3 Inverse Rendering and BRDF Decomposition in Static Scenes

4.3 Inverse Rendering and BRDF Decomposition in Static Scenes

Building upon the rigorous geometric foundations established by Neural Signed Distance Functions (SDFs) in the previous section, the focus now shifts to the equally critical challenge of disentangling material properties and illumination from observed imagery. Inverse rendering in static scenes aims to recover physically-based Bidirectional Reflectance Distribution Function (BRDF) parameters under unknown lighting conditions, a task that is inherently ill-posed due to the profound ambiguity between shading, shadows, and intrinsic material properties. While Section 4.2 addressed the creation of watertight and topologically consistent surfaces, those geometric representations alone are insufficient for applications requiring material editing or relighting. Consequently, recent advancements have transitioned from traditional optimization pipelines to end-to-end differentiable frameworks that leverage neural implicit representations to jointly estimate shape, reflectance, and light. The core strategy involves factorizing the radiance field into distinct components, thereby enabling free-viewpoint relighting and precise material manipulation.

Early neural approaches addressed this decomposition by distilling volumetric geometry into surface representations while simultaneously solving for spatially-varying reflectance. For instance, [178] introduced a framework that recovers 3D neural fields of surface normals, light visibility, albedo, and BRDFs without supervision. By explicitly modeling light visibility, this method successfully separates shadows from albedo, enabling the synthesis of realistic soft or hard shadows under arbitrary lighting conditions. Similarly, [179] proposed a technique capable of handling unconstrained environmental illumination from image collections captured under varying lighting conditions. This approach utilizes physically-based rendering to decompose the scene into spatially-varying BRDF properties and further converts the learned volume into a relightable textured mesh, effectively bridging the gap between neural representations and traditional graphics pipelines.

A significant limitation in early decomposition methods was the treatment of indirect lighting, which was often oversimplified or ignored, leading to errors in material estimation for indoor scenes. To address the complexity of global illumination, researchers have developed specialized frameworks that model indirect light transport more accurately. [180] proposed representing scene lighting as a fully 5D Neural Incident Light Field (NeILF), which naturally handles occlusions and indirect lights without requiring multiple bounces of ray tracing during optimization. Building upon this, [181] unified incident and outgoing light fields through physically-based rendering and inter-reflections, allowing for the disentanglement of geometry, material, and lighting in a strictly physical manner. This union enables the system to serve not only as a decomposition tool but also as a surface refinement module that enhances reconstruction details. Furthermore, [182] presented a novel approach to efficiently recover spatially-varying indirect illumination by deriving it from the learned neural radiance field, thereby recovering interreflection- and shadow-free albedo that previous single-bounce methods failed to capture.

The challenge of handling glossy and highly reflective objects has also seen significant progress through the integration of spherical Gaussians and tensorial representations, which offer efficient approximations for complex light transport. [183] utilized mixtures of spherical Gaussians to represent specular BRDFs and environmental illumination, coupled with a signed distance function for geometry. This formulation allows for efficient approximation of light transport in scenes with challenging non-Lambertian reflectance. In a similar vein, [184] extended tensor factorization techniques to estimate scene geometry, surface reflectance, and environment illumination. By leveraging a low-rank tensor representation, this method achieves fast and compact reconstruction while accurately modeling secondary shading effects like shadows and indirect lighting under single or multiple unknown lighting conditions. Additionally, [185] tackled the specific difficulty of reconstructing reflective objects where view-dependent specular reflections violate multi-view consistency. By applying split-sum approximation and integrated directional encoding, NeRO accurately reconstructs geometry and BRDF without requiring object masks or known environment lights.

To facilitate deployment in standard rendering engines and improve computational efficiency, hybrid approaches combining explicit meshes with neural caches have emerged. [186] introduced a framework that initializes differentiable marching tetrahedrons from a pre-trained neural field and employs a Neural Radiance Cache to model indirect illumination. This combination allows for single-bounce differentiable Monte Carlo ray tracing to jointly optimize geometry, material, and light, preventing indirect illumination effects from being erroneously baked into materials—a crucial step toward the explicit light transport modeling discussed in the following section. For scenarios involving complex indoor environments, [187] utilizes differentiable Monte Carlo raytracing on neural signed distance fields to jointly recover shapes, incident radiance, and materials, enabling physically-based scene relighting and editing.

Addressing the inherent ambiguity in inverse rendering often requires strong priors or generative models to regularize the solution space. [188] leverages diffusion models as priors for albedo and specular components to regularize the optimization process. By formulating the material prior as a diffusion model, this approach resolves ambiguities arising from the coupling of geometry, materials, and lighting, achieving state-of-the-art performance in material recovery. Moreover, [189] replaces costly illumination integral operations with a learned network query, estimating shape, BRDF, and per-image illumination from objects captured under varying lighting. This method learns deep low-dimensional priors on BRDF and illumination, resulting in considerably better decomposition compared to prior arts.

Finally, the scalability of these methods to large-scale scenes and the handling of specific material properties remain active areas of research. [190] formulated the joint recovery of camera pose, geometry, and svBRDF for scenes exceeding object scale using a distributed optimization algorithm. For specific material classes, [191] proposed a compact neural BRDF representation using spherically distributed primitives and codebooks, achieving real-time rendering with versatile material representation. Additionally, [192] extended the paradigm to heterogeneous volumes with piecewise-constant refractive indices, optimizing radiance fields along curved paths tracked by the Eikonal equation to handle refraction. Collectively, these works demonstrate a robust trajectory towards high-fidelity, physically-consistent inverse rendering. While these methods successfully decompose static scenes into their constituent physical properties, the next section delves deeper into the explicit modeling of global illumination and complex indirect light transport, moving beyond decomposition to full physical simulation.

4.4 Modeling Global Illumination and Indirect Light Transport

4.4 Modeling Global Illumination and Indirect Light Transport

While Section 4.3 established the foundations for disentangling material properties and direct illumination in static scenes, a truly physically consistent reconstruction demands an explicit treatment of global illumination (GI) and complex indirect light transport. In real-world environments, light undergoes intricate interactions, including multiple bounces between surfaces, inter-reflections in concave regions, and subsurface scattering within translucent materials. Early neural rendering approaches, particularly initial iterations of Neural Radiance Fields (NeRF), often implicitly baked these high-order lighting effects into the radiance field. Although this yielded photorealistic novel views under fixed lighting, it fundamentally constrained the ability to edit scene materials, modify lighting conditions, or achieve the clean decomposition of intrinsic physical components necessary for advanced applications. To overcome these limitations, recent advancements have transitioned from implicit baking to explicitly modeling light transport by integrating Monte Carlo path tracing with neural caching mechanisms, thereby enabling physically faithful inverse rendering and relighting capabilities.

A primary bottleneck in differentiable inverse rendering is the prohibitive computational cost and high variance associated with estimating indirect illumination via standard Monte Carlo integration. Tracing millions of paths to converge on a noise-free solution for every optimization step is often infeasible. To address this, researchers have developed neural caching strategies that learn to predict indirect radiance, effectively acting as efficient surrogates for expensive path tracing simulations. For instance, the concept of a Neural Radiance Cache (NRC) has been introduced to model indirect illumination efficiently. By combining an initial Signed Distance Field (SDF) estimate with this cache, frameworks can compute global illumination via single-bounce differentiable Monte Carlo ray tracing while jointly optimizing geometry, material, and light. This approach effectively prevents indirect illumination effects from being erroneously baked into material properties, ensuring a clean separation between albedo, roughness, and lighting [186]. Similarly, other methods propose using a neural network as a radiometric prior, where the loss function includes a term representing the residual of the rendering equation. This enforces physical consistency on the radiance field without requiring the explicit evaluation and differentiation of millions of path integrals, significantly reducing computation time while maintaining high-quality recovery of scene parameters [193].

Beyond simple diffuse inter-reflections, capturing complex light transport phenomena such as caustics, specular chains, and subsurface scattering remains a formidable task. Standard path tracing often struggles to converge in scenes dominated by these effects due to the difficulty of sampling rare but high-contribution light paths. Advanced variance reduction techniques, such as path guiding, have been adapted into neural frameworks to mitigate this issue. By leveraging photons traced from light sources, neural networks can reconstruct high-quality sampling distributions for path guiding from sparse sets of samples. This photon-driven approach allows for highly efficient path guiding at any location in the scene, generalizing well to diverse and challenging testing scenarios involving strong global illumination [194]. Furthermore, deep learning has been applied directly to particle-based rendering methods like photon mapping. By training networks to predict optimal kernel functions for aggregating photon contributions, it is possible to reconstruct complex global illumination effects, such as caustics, with an order of magnitude fewer photons than traditional methods [195]. These techniques are crucial for static scene reconstruction where accurate modeling of specular-diffuse-specular paths is required to resolve sharp lighting features.

The modeling of participating media and translucent objects introduces additional layers of complexity due to multiple scattering events within the volume. Classical models often assume simple planar half-spaces, which fail to capture the heterogeneous scattering properties of real-world objects. Neural representations have evolved to address this by learning Bidirectional Scattering-Surface Reflectance Distribution Functions (BSSRDF) that account for full object geometry and internal heterogeneities. These neural methods can represent the appearance of translucent objects under arbitrary distant lighting, effectively functioning as a form of neural precomputed radiance transfer that captures all-frequency relighting capabilities [196]. In scenarios involving highly scattering media, such as fog or biological tissue, traditional diffusion approximations may lack accuracy. Recent work has explored integrating transient diffusion theory into path construction to generate unbiased full transport of time-resolved radiance, significantly improving the quality and efficiency of rendering in multiple scattering settings [197]. Additionally, neural operators have been employed to learn families of frequency-domain Maxwell equations, enabling ultra-fast parametric simulation of photonic devices and complex light-matter interactions that were previously too costly for iterative optimization loops [198].

To further enhance the fidelity of indirect lighting in static scenes, hybrid approaches combining explicit mesh representations with neural fields have gained traction. These methods often utilize differentiable marching tetrahedrons to extract explicit geometry while relying on neural caches to model the high-frequency details of indirect light that meshes alone might miss. The integration of Monte Carlo sampling with neural caching allows for the decomposition of materials and illumination even in the presence of near-field indirect illumination, which is often blurred by simpler spherical Gaussian approximations [199]. Moreover, for scenes containing mirrors or near-perfect specular reflections, specialized reflection tracing methods tailored for volume rendering have been developed. These methods explicitly model reflection behavior using physically plausible materials and estimate reflected radiance with Monte Carlo methods within the volume rendering formulation, avoiding the artifacts common in standard implicit representations [200].

In conclusion, the evolution from implicit baking to explicit, physics-informed neural simulation has fundamentally transformed the modeling of global illumination in static 3D reconstruction. By synergizing Monte Carlo path tracing with neural caching, photon-driven guidance, and advanced scattering models, modern frameworks achieve a level of physical fidelity that supports robust inverse rendering, material editing, and dynamic relighting. These advancements not only improve the visual realism of reconstructed scenes but also provide the necessary geometric and photometric consistency required for downstream applications. Crucially, the explicit modeling of light transport and occlusion detailed here lays the theoretical and methodological groundwork for the next paradigm shift: integrating these physical principles into the efficient, point-based architecture of 3D Gaussian Splatting, as explored in the following section.

4.5 Relightable Gaussian Splatting and Point-Based Inverse Rendering

4.5 Relightable Gaussian Splatting and Point-Based Inverse Rendering

Building upon the explicit modeling of global illumination discussed in the previous section, the integration of these physical principles into the 3D Gaussian Splatting (3DGS) framework marks a pivotal evolution from novel view synthesis to full inverse rendering. While original 3DGS excels at reproducing appearance under fixed lighting, it initially lacked the physical decomposability required for relighting, material editing, and realistic shadow casting. Inverse rendering aims to disentangle observed radiance into intrinsic components: geometry (surface normals and depth), material properties (albedo, roughness, metallic), and illumination (environment maps and local lights). Extending 3DGS to this domain requires overcoming two fundamental hurdles inherent to its explicit point-based nature: the absence of native, consistent surface normals and the difficulty of modeling occlusion and light transport within a forward-mapping rasterization pipeline. Recent advancements have addressed these challenges by augmenting Gaussian primitives with physical attributes and developing novel optimization strategies that bridge the gap between discrete points and continuous surface physics.

A primary prerequisite for applying the global illumination models described previously is the accurate estimation of surface normals, which are essential for computing lighting interactions such as diffuse reflection and specular highlights. Standard 3DGS optimizes Gaussian covariance to align with image gradients, but this does not guarantee geometric consistency or accurate normal directions, often resulting in "floater" artifacts and noisy surface estimates. To resolve this, researchers have introduced regularization techniques that explicitly enforce normal consistency. For instance, [201] proposes an efficient optimization scheme that incorporates depth-derivation-based regularization to estimate plausible normals directly from the discrete Gaussian distribution. Similarly, [202] introduces a self-regularization method that facilitates the modeling of surface normals without requiring additional supervision, effectively addressing the complexity of estimating orientation in inhomogeneously distributed point clouds. Furthermore, [203] develops a novel normal estimation framework based on the shortest axis directions of 3D Gaussians, utilizing a delicately designed loss function to ensure consistency between the estimated normals and the underlying geometry of the Gaussian spheres. These methods collectively enable the explicit representation to support physically based shading models, a capability previously dominated by implicit neural fields.

Beyond geometry, achieving realistic relighting necessitates the decomposition of view-dependent appearance into material BRDF parameters and lighting conditions, moving beyond the limited expressiveness of Spherical Harmonics (SH). Traditional 3DGS relies on SH to model view-dependent color, which is insufficient for separating intrinsic material properties from environmental illumination. To overcome this, recent works associate each Gaussian with explicit BRDF parameters, such as albedo and roughness, and optimize them alongside the geometric attributes. [204] presents a differentiable point-based rendering framework where each Gaussian point is augmented with normal directions, BRDF parameters, and incident light estimates. This approach divides incident lights into global and local components, allowing for robust lighting estimation and enabling the scene to be relit under novel illumination environments. By leveraging physically-based differentiable rendering, this framework successfully decomposes the radiance field, paving the way for editing and ray-tracing operations solely based on point clouds.

A significant challenge in point-based inverse rendering, particularly when aiming for the high-fidelity global illumination discussed in Section 4.4, is the simulation of shadows and inter-reflections, which are naturally handled in backward-mapping ray tracing but computationally expensive in forward-mapping splatting. To address the occlusion problem, [201] introduces a baking-based occlusion mechanism to model indirect lighting, effectively approximating the shadowing effects that are critical for realism. Taking this further, [204] innovates by introducing a point-based ray-tracing approach built upon a bounding volume hierarchy (BVH). This structure enables efficient visibility baking, allowing the system to compute accurate shadow effects for 3D Gaussian points in real-time. This integration of ray-tracing logic into the point-based pipeline marks a substantial shift, demonstrating that explicit primitives can support complex light transport simulations previously thought to require dense mesh or volumetric structures. Additionally, [202] proposes an approach to simulate ray tracing specifically to reconstruct indirect illumination, further validating the versatility of 3D Gaussians in capturing high-order lighting effects.

The efficacy of these point-based inverse rendering methods is evidenced by their performance against state-of-the-art baselines. [204] demonstrates improved BRDF estimation and novel view rendering results compared to traditional material estimation approaches, showcasing the potential to revolutionize the graphics pipeline with a relightable and editable system. Similarly, [202] substantiates its efficacy through extensive experiments across multiple tasks, highlighting superior performance in relighting and reconstruction compared to methods leveraging discrete meshes or neural implicit fields. Moreover, [203] strikes a commendable balance between efficiency and visual quality, surpassing standard Gaussian Splatting in PSNR on specular object datasets and significantly accelerating optimization times compared to prior works like Ref-NeRF.

In summary, the extension of 3D Gaussian Splatting to inverse rendering has transformed it from a purely visual reconstruction tool into a physically grounded representation capable of material decomposition and dynamic relighting. By integrating normal regularization, BRDF parameterization, and novel ray-tracing approximations, these methods [204; 201; 202; 203] have established a new paradigm. They offer the dual advantages of the high-fidelity, real-time rendering inherent to explicit splatting and the physical interpretability required for advanced graphics applications. However, while these methods successfully handle general reflective surfaces and global illumination, specific challenges remain in reconstructing extreme non-Lambertian materials, such as perfect mirrors, transparent media, and complex subsurface scattering, which require even more specialized handling of light paths as explored in the following section.

4.6 Handling Challenging Materials: Specularity, Transparency, and Participating Media

4.6 Handling Challenging Materials: Specularity, Transparency, and Participating Media

While the point-based inverse rendering techniques discussed in Section 4.5 have successfully enabled relighting and material decomposition for general reflective surfaces, they often rely on approximations that struggle with extreme non-Lambertian behaviors. Perfect mirrors, highly refractive transparent media, and complex subsurface scattering introduce severe ambiguities in geometry recovery because their appearance is governed by intricate light paths rather than simple surface reflection. In these regimes, the assumption of diffuse reflection or low-order spherical harmonic expansion fails, as the observed radiance is heavily dependent on the viewing angle and the geometry of the surrounding environment. Traditional multi-view stereo and early neural rendering approaches, which rely on photometric consistency across views, are frequently invalidated by these phenomena; a single surface point can exhibit drastically different colors in different images, rendering standard correspondence matching ineffective. To address these limitations, recent advancements have shifted towards specialized models that explicitly account for refraction, perfect reflection, and volumetric light scattering, integrating rigorous physical priors into the learning process to disentangle geometry from appearance.

Specular surfaces, such as polished metals or mirrors, pose a unique difficulty due to the dominance of environment reflection over intrinsic surface color. Reconstructing these objects requires distinguishing between the geometry of the object itself and the geometry of the reflected world. Early attempts to model non-Lambertian reflectance often relied on simplified analytical models like Oren-Nayar or Phong, formulating the problem through Hamilton-Jacobi type equations to recover shape from shading [205]. However, these methods often struggle with global illumination effects and complex environmental maps. More recent neural approaches have moved towards learning implicit representations that can handle arbitrary reflectance. For instance, methods like Neural Reflectance leverage coordinate-based networks to parameterize both shape and unknown reflectance, explicitly predicting cast shadows to mitigate artifacts and achieve higher estimation accuracy without ground truth supervision [206]. Furthermore, to handle the ambiguity inherent in reflective surfaces, techniques such as Ref-NeuS utilize anomaly detectors to estimate reflection scores, adapting the photometric loss to reduce ambiguity by modeling rendered color as a Gaussian distribution [207]. This allows for high-fidelity reconstruction of glossy objects where traditional methods would fail to converge to the correct surface normal, bridging the gap left by standard BRDF parameterizations in point-based systems.

Transparent objects present an even more formidable challenge because light paths are bent via refraction, violating the straight-ray assumption fundamental to most rendering pipelines, including the rasterization-based approaches of 3DGS. The appearance of a transparent object is essentially a warped view of the background, making it difficult to establish correspondences or infer depth. Several strategies have been developed to tackle this by explicitly modeling the refractive light paths. One effective approach involves altering the incident light paths, such as immersing the object in a liquid, to triangulate the refraction and recover the surface shape without assuming a parametric form [208]. In the realm of neural rendering, methods like NeTO leverage implicit Signed Distance Functions (SDF) combined with self-occlusion-aware refractive ray tracing to optimize the geometry via volume rendering, successfully reconstructing high-quality shapes even from limited views [209]. Similarly, NeRRF introduces a refractive-reflective field that unifies the modeling of refraction and reflection using Fresnel terms, enabling the synthesis of transparent objects with complex light path changes [210]. Other works have focused on differentiable ray tracing to directly optimize mesh approximations, ensuring silhouette consistency while tracing refractive rays through the object [211]. These methods demonstrate that explicitly simulating the physics of light bending is crucial for faithful reconstruction of glass and liquids, a capability that current point-based splatting methods are only beginning to approximate.

Beyond surface interactions, participating media and translucent materials involve light scattering beneath the surface, a phenomenon known as subsurface scattering. Modeling this requires capturing the bidirectional scattering-surface reflectance distribution function (BSSRDF), which describes how light enters at one point and exits at another. Classical diffusion theory approximations often fail for heterogeneous materials or non-perpendicular incidence. To overcome this, Neural BSSRDF proposes a compact neural representation that learns the appearance of heterogeneous translucent objects as a function of spatial position and light directions, effectively acting as neural precomputed radiance transfer for arbitrary lighting [196]. This approach avoids the computational expense of full path tracing while maintaining high accuracy. For scenarios involving complex multi-bounce transport within volumes, learning-based methods have been developed to approximate random photon paths using conditional variational auto-encoders, enabling efficient sphere-tracing of volumetric subsurface effects [212]. Additionally, Object-Centric Neural Scattering Functions (OSFs) have been introduced to learn radiance transfer for both opaque and translucent objects, approximating the complex subsurface transport without explicitly modeling every scattering event, thus facilitating free-viewpoint relighting [213].

The integration of these specialized models into a unified framework remains an active area of research, often necessitating hybrid architectures that combine the strengths of implicit neural fields for complex light transport with explicit representations for efficiency. Hybrid approaches, such as those combining explicit parametric surfaces for highly reflective components (like the cornea of an eye) with implicit volumetric representations for diffuse regions, demonstrate the power of tailored architectures [135]. Furthermore, the ability to separate reflection and transmission components in wild images using polarization cues highlights the importance of multi-modal data in resolving these ambiguities [214]. As the field progresses, the convergence of differentiable physics, neural implicit representations, and advanced sampling strategies continues to push the boundaries of what can be reconstructed, moving from simple diffuse shapes to complex scenes filled with mirrors, glass, and scattering media. While these advanced physical models achieve unprecedented fidelity, their deployment in real-time applications requires converting these complex learned representations into standard graphics assets, a transition that forms the focus of the subsequent section on mesh extraction and texture baking.

4.7 Mesh Extraction, Texture Baking, and Real-Time Deployment Assets

4.7 Mesh Extraction, Texture Baking, and Real-Time Deployment Assets

While the preceding section detailed how specialized physical models can disentangle complex light transport for challenging materials, the practical utility of these reconstructions hinges on their conversion into assets compatible with standard real-time rendering engines. The transition from learned neural or Gaussian representations to formats usable by game engines, AR/VR headsets, and mobile devices represents a critical bottleneck. Although methods like 3D Gaussian Splatting (3DGS) and Neural Radiance Fields (NeRF) offer unparalleled photographic fidelity, their native formats—whether unstructured point clouds or implicit volumetric fields—are often incompatible with the rigid triangle-mesh pipelines that underpin commercial graphics. Consequently, significant research has focused on strategies to extract explicit geometric surfaces, bake high-frequency appearance details into traditional texture maps, and optimize these assets for low-latency deployment. This process is essential not only for geometric reconstruction but also for ensuring that the resulting assets remain editable and relightable, thereby bridging the gap between the high-fidelity capture discussed previously and the generative manipulation capabilities explored in the following section.

A primary challenge in this domain is deriving watertight, topologically consistent meshes from implicit volumetric fields or unstructured point data. Traditional extraction methods, such as Marching Cubes applied to density fields, often struggle with noise and lack of surface definition in sparse regions, leading to artifacts that hinder downstream applications. To address this, recent advances have moved towards hybrid representations that explicitly model surfaces alongside volumetric data during the optimization phase. For instance, the concept of "Adaptive Shells" introduces a neural radiance formulation that constructs an explicit mesh envelope to spatially bound a neural volumetric representation [159]. This approach allows the system to transition smoothly between volumetric rendering for fuzzy geometry like hair and surface-based rendering for solid objects, significantly accelerating inference while enabling the extraction of a narrow-band mesh suitable for animation and simulation. Similarly, hierarchical hybrid representations have been proposed to leverage implicit multiresolution hash encoding aided by explicit Signed Distance Function (SDF) priors, facilitating fast scene geometry initialization and making the learned geometry more amenable to robust mesh extraction [215]. These methods demonstrate that enforcing geometric consistency during the learning phase is paramount for generating high-quality meshes that require minimal post-processing cleanup.

Once a coherent mesh topology is established, the next critical step is texture baking, which involves projecting the high-dimensional features or radiance values learned by the neural model onto a 2D UV parameterization of the extracted mesh. This step is essential for decoupling geometry from appearance, allowing the use of standard rasterization pipelines while preserving the view-dependent effects captured by the original model. Approaches like "Deferred Neural Rendering" have pioneered the use of neural textures—learned feature maps stored on top of 3D mesh proxies—to synthesize photo-realistic images even when the underlying geometry is imperfect [216]. By training a deferred neural renderer to interpret these high-dimensional feature maps, systems can achieve real-time performance while retaining the detailed appearance captured during neural optimization. Furthermore, for scenarios requiring physically based rendering (PBR), methods such as "Paint-it" utilize deep convolutional re-parameterization to optimize texture maps directly from text descriptions or input images, effectively filtering noisy gradients to produce high-fidelity albedo, roughness, and normal maps compatible with standard graphics engines [217]. This ensures that the baked textures are not merely static images but contain the necessary physical parameters for dynamic relighting, a prerequisite for the semantic editing tasks discussed later.

For human-centric applications, where deformation and animation are required, the integration of parametric priors with Gaussian or neural representations offers a robust pathway to asset generation. The "UV Gaussians" framework exemplifies this by jointly learning mesh deformations and 2D UV-space Gaussian textures, leveraging the structural guidance of a mesh to prevent the blurring often seen in purely point-based human avatars [60]. By embedding Gaussian attributes into the UV space, this method facilitates the direct baking of detailed textures onto a deformable mesh, ensuring that the resulting asset is both photorealistic and fully animatable within standard rigging frameworks. Similarly, video-driven approaches for facial assets employ variational autoencoders to disentangle expression, geometry, and physically-based textures, allowing for the generation of high-resolution dynamic textures that can be retargeted across different identities [218]. These techniques highlight the importance of maintaining a clear separation between the deformation skeleton and the surface appearance to achieve production-ready results that can serve as the foundation for further generative enhancement.

The final stage of this pipeline involves optimizing these extracted assets for deployment on resource-constrained platforms, such as mobile AR devices or web browsers. This requires aggressive compression and the translation of neural decoders into efficient shader code that can run on traditional graphics hardware. The "Neural Assets" approach demonstrates how volumetric object capture can be transpiled into standard shader code, enabling interactive framerates on existing hardware without requiring specialized neural inference engines [219]. By distilling the learned representation into a grid of features and a small MLP decoder that runs within a traditional graphics pipeline, these methods bridge the gap between research prototypes and deployable products. Moreover, for real-time applications involving dynamic lighting, systems have been developed to estimate lighting conditions and generate neural soft shadows as light-direction-dependent textures, ensuring that virtual objects integrate seamlessly into real-world AR scenarios with minimal computational overhead [220]. Together, these strategies form a comprehensive workflow for converting the ephemeral qualities of neural fields into durable, editable, and high-performance assets. By establishing these robust, relightable, and animatable bases, the field is now poised to leverage generative priors for intuitive material synthesis and semantic-guided editing, as will be discussed in the subsequent section.

4.8 Generative Material Synthesis and Semantic-Guided Editing

Building upon the robust, relightable, and animatable assets established in the preceding section, the focus of 3D reconstruction now shifts from passive capture to active, intuitive manipulation. While Section 4.7 detailed the conversion of neural fields into editable mesh proxies with baked physical parameters, the true potential of these assets is unlocked when users can semantically alter their material properties without re-capturing data. Traditional inverse rendering pipelines often struggle to disentangle lighting, geometry, and surface reflectance—particularly for complex, spatially-varying Bidirectional Reflectance Distribution Functions (SVBRDF)—limiting the editability of reconstructed scenes. The integration of generative priors, specifically diffusion models and large language models (LLMs), has emerged as a transformative paradigm to overcome these barriers. This subsection surveys the state-of-the-art in text-driven material generation, semantic-guided editing frameworks, and the creation of neural assets that facilitate real-time interaction within reconstructed static scenes, effectively bridging the gap between high-fidelity reconstruction and creative control.

A primary challenge in this domain is bridging the semantic gap between natural language descriptions and physically accurate BRDF parameters. Early approaches were constrained by limited datasets or required extensive manual tuning, but recent advancements leverage the rich semantic understanding of foundation models to automate this process. For instance, researchers have developed methods to generate parametric BRDFs directly from natural language descriptions, allowing users to specify attributes such as “dull plastic” or “shiny iron” to instantly alter object appearances in 3D environments [221]. This capability is further enhanced by frameworks that utilize latent BRDF auto-encoders, which learn a smooth, continuous distribution of materials from large-scale real-world collections. By predicting latent embeddings rather than raw parameters, these systems ensure the generation of coherent and realistic materials that avoid the artifacts common in direct parameter regression [222]. Furthermore, the introduction of intuitive control spaces allows for the predictable editing of captured BRDF data, mapping high-level perceptual attributes like glossiness or metallicness to underlying mathematical representations, thereby empowering artists to create novel appearances without acquiring new physical samples [223].

Beyond parametric generation, diffusion-based models have revolutionized the creation of spatially-varying materials with unprecedented diversity and controllability, extending the texture baking concepts discussed previously into a generative context. Frameworks such as DreamPBR utilize multi-modal guidance, including text prompts and style images, to generate high-resolution SVBRDF maps that are tileable and suitable for large-scale scene texturing [224]. Similarly, MatFuse introduces a unified approach that harnesses diffusion models for both creation and editing, integrating multiple conditioning sources like color palettes and sketches to provide fine-grained control over material synthesis [225]. These generative capabilities are not limited to synthetic data; recent works like PhotoMat demonstrate the feasibility of training material generators exclusively on real photos captured with handheld devices, effectively closing the domain gap between synthetic training data and real-world appearance [226]. To address the specific needs of industry pipelines, TileGen offers a generative model specifically designed to produce tileable materials conditional on user-provided structural patterns, ensuring that generated textures like brick layouts or leather wrinkles maintain coherent periodicity [227].

Semantic-guided editing extends these generative capabilities to the precise manipulation of existing

3D assets and scenes, leveraging the structured representations established in prior sections. The integration of vision-language models enables the precise identification and segmentation of material regions within a scene, facilitating localized edits. Make-it-Real exemplifies this by exploiting Multimodal Large Language Models (MLLMs) to recognize and describe materials, aligning them with specific components of 3D objects to apply realistic SVBRDF properties automatically [228]. For more granular control, methods like ProSpect leverage the step-by-step generation process of diffusion models to represent images as collections of inverted textual token embeddings, enabling attribute-aware personalization where specific visual properties such as material and style can be edited independently [229]. Additionally, frameworks have been developed to edit material appearance in single images by manipulating high-level perceptual attributes without requiring explicit inverse rendering, utilizing generative models to preserve high-frequency details while altering surface characteristics [230].

The convergence of semantic understanding and generative synthesis also supports the creation of fully editable neural assets, pushing beyond static meshes to dynamic volumetric representations. PartNeRF introduces a part-aware generative model that decomposes objects into locally defined neural fields, enabling independent manipulation of shape and appearance for different object parts without requiring 3D supervision [231]. This concept is extended to scene-level editing in frameworks like DreamEditor, which utilizes text prompts to identify and modify specific regions within neural fields, ensuring consistency in irrelevant areas while generating highly realistic textures and geometry [232]. Similarly, LatentEditor embeds real-world scenes into the latent space of denoising diffusion models, employing a delta score mechanism to guide local modifications with superior precision and speed compared to traditional optimization-based methods [233]. For interactive environments, Neural Assets propose a novel representation that captures volumetric effects and translucent properties, transpiling neural decoders into efficient shader code to allow seamless integration and real-time editing of photorealistic objects within standard graphics engines [219].

Furthermore, the rigorous disentanglement of geometry and appearance remains critical for achieving high-quality, artifact-free editing. Fantasia3D addresses this by proposing a disentangled framework that encodes surface normals for geometry learning while introducing SVBRDF for appearance modeling, resulting in assets that are compatible with popular graphics engines for relighting and physical simulation [234]. In scenarios where 2D priors bake illumination into textures, MaterialSeg3D infers underlying materials from 2D semantic priors, parsing material properties in 3D space to enable reasonable relighting in novel scenes [235]. The ability to edit out-of-domain content is also addressed through differential activation modules that detect semantic changes, allowing for robust material manipulation even when reconstructing from real-world images that differ from training distributions [236]. Finally, the application of procedural modeling techniques, such as those proposed in An Inverse Procedural Modeling Pipeline for SVBRDF Maps, allows for the decomposition of pixel maps into editable procedural graphs, expanding materials to arbitrary resolutions and enabling high-level user control [237]. Collectively, these advancements signify a paradigm shift towards semantic-aware, generative material pipelines that empower users to intuitively synthesize and manipulate the physical appearance of static 3D scenes with unprecedented fidelity and flexibility.

5 Optimization Strategies, Compression, and Scalability for Real-Time Deployment

5.1 Accelerated Initialization and Adaptive Densification Strategies

5.1 Accelerated Initialization and Adaptive Densification Strategies

The deployment of 3D Gaussian Splatting (3DGS) in real-time applications has been revolutionized by its ability to render high-fidelity scenes at interactive frame rates. However, the traditional pipeline for training these models relies heavily on accurate point cloud initialization derived from Structure-from-Motion (SfM) algorithms. This dependency presents a significant bottleneck, particularly in scenarios where SfM fails due to texture-less surfaces, large baseline variations, or insufficient feature matches. Consequently, recent research has pivoted towards developing accelerated initialization strategies that bypass SfM constraints and adaptive densification schedules that dynamically optimize the spatial distribution of Gaussians during the critical early phases of training. These advancements are crucial not only for reducing the time-to-solution and enhancing robustness in uncontrolled environments but also for establishing an efficient geometric foundation before applying the compression techniques discussed in subsequent sections.

A primary challenge in eliminating SfM dependencies is the severe performance degradation observed when 3DGS is trained from randomly initialized point clouds. Standard implementations often suffer from substantial drops in Peak Signal-to-Noise Ratio (PSNR), typically ranging from 4 to 5 dB, when accurate initialization is absent. To address this, novel optimization strategies have been proposed to relax the accurate initialization constraint. For instance, recent work has demonstrated that by conducting a thorough analysis of SfM initialization in the frequency domain and studying 1D regression tasks with multiple Gaussians, it is possible to design training protocols that successfully converge from random starts. Methods such as RAIN-GS (Relaxing Accurate Initialization Constraint for 3D Gaussian Splatting) exemplify this shift, showing that careful optimization can largely improve performance across multiple datasets without requiring precise SfM-derived point clouds [13]. This approach fundamentally challenges the notion that high-quality geometry priors are mandatory, suggesting instead that the optimization landscape can be navigated effectively if the initialization strategy is appropriately relaxed and guided by frequency-aware objectives.

Complementing random initialization approaches, another promising direction involves leveraging volumetric reconstructions from Neural Radiance Fields (NeRF) to initialize Gaussian primitives. This hybrid strategy aims to distill structural information from coarse, low-cost NeRF models to provide a better starting configuration for 3DGS. Research indicates that by employing a combination of improved initialization schemes and structure distillation, it is possible to achieve results equivalent to, or even superior to, those obtained via traditional SfM pipelines [14]. This method effectively bypasses the fragility of feature-based SfM in texture-less regions by utilizing the dense volumetric cues provided by implicit fields. Furthermore, inspired by classical multi-view stereo techniques, progressive propagation strategies have been introduced to guide the densification of 3D Gaussians. Unlike the simple split and clone heuristics used in standard 3DGS, these methods leverage priors from existing reconstructed geometries and patch-matching techniques to generate new Gaussians with accurate positions and orientations. Such approaches, like GaussianPro, have shown significant improvements in large-scale scenes where SfM typically fails to produce sufficient points, yielding notable gains in reconstruction quality [15].

Once initialization is addressed, the efficiency of the training process hinges on adaptive densification strategies. The density of the Gaussian point cloud must be dynamically adjusted to balance geometric fidelity with computational cost. Static densification schedules often lead to over-representation in simple regions and under-representation in complex areas, resulting in inefficient memory usage and suboptimal rendering quality. Adaptive strategies seek to optimize this density during the early training phases by analyzing gradient magnitudes and view-space errors. However, the effectiveness of these strategies is also tied to the underlying optimization mechanics. The behavior of stochastic gradient descent and adaptive optimizers plays a critical role in how quickly the model converges to a stable configuration. For example, understanding the interplay between learning rates and the curvature of the loss surface is essential; poor choices can lead to "catastrophic Fisher explosions" where the local curvature increases dramatically, hindering generalization [238]. Therefore, advanced densification algorithms must be coupled with robust optimization schedules that can adapt to the non-convex nature of the rendering loss.

Moreover, the concept of adaptive scheduling extends beyond simple density control to include the management of learning rates and batch sizes in distributed settings. In elastic distributed training environments, where resource availability fluctuates, dynamic mini-batch strategies are required to maintain stability. Smoothly adjusting learning rates over time helps alleviate the influence of noisy momentum estimation when batch sizes change, ensuring that the densification process remains stable even under varying computational loads [239]. Similarly, adaptive communication strategies in distributed setups can significantly reduce training time by optimizing the frequency of model averaging, which indirectly supports faster convergence of the densification process [240].

The integration of learnable priors further enhances these adaptive mechanisms. Just as geometric priors have been leveraged to improve point cloud compression by representing high-resolution data as a combination of priors and structural deviations, similar principles can be applied to initialize and densify Gaussian fields more efficiently [241]. By encoding topological constraints as part of the initialization, the system can focus its adaptive densification efforts on capturing fine-grained details rather than reconstructing coarse geometry from scratch. Additionally, the use of predictive context priors, which learn to adjust spatial query locations based on local contexts, offers a pathway to more flexible and generalizable surface reconstruction, potentially informing where new Gaussians should be placed to best fit the underlying scene geometry [242].

In summary, the evolution of initialization and densification strategies for 3D Gaussian Splatting marks a transition from rigid, SfM-dependent pipelines to flexible, learning-based frameworks. By combining random initialization with frequency-domain analysis, distilling knowledge from implicit fields, and employing progressive propagation techniques, researchers are successfully mitigating the limitations of traditional methods. Coupled with adaptive optimization schedules that account for loss curvature and distributed training dynamics, these innovations pave the way for scalable, real-time 3D reconstruction capable of handling the complexities of dynamic and texture-deficient real-world scenes. Crucially, by establishing efficient and robust scene representations early in the pipeline, these methods set the stage for subsequent optimization phases, including the quantization-aware training and low-precision representations necessary for deploying these models on resource-constrained edge devices.

5.2 Quantization-Aware Training and Low-Precision Representations

5.2 Quantization-Aware Training and Low-Precision Representations

Building upon the robust geometric foundations and adaptive densification strategies established in Section 5.1, the deployment of dynamic 3D Gaussian Splatting (3DGS) on edge devices and mobile platforms faces a subsequent critical bottleneck: the substantial memory footprint and computational demands associated with storing millions of high-precision floating-point attributes. Even with optimized primitive counts, each Gaussian requires storage for position, covariance, opacity, and spherical harmonic coefficients, often resulting in gigabytes of data for complex scenes. To address these constraints while preserving the real-time rendering capabilities secured by efficient initialization, Quantization-Aware Training (QAT) has emerged as a pivotal strategy. Unlike post-training quantization, which often suffers from significant accuracy degradation due to the non-linear nature of neural rendering pipelines, QAT integrates the quantization process directly into the optimization loop. This allows the model to adapt its representations to low-precision constraints during training, ensuring that the reduction in bit-width does not compromise the visual fidelity achieved through the earlier densification phases.

A fundamental challenge in aggressive low-bit quantization, such as reducing weights and activations to 2-bit or 4-bit precision, is the presence of outliers within the parameter distributions. These extreme values, which can arise even from well-initialized Gaussians, disrupt the uniform mapping of continuous values to discrete quantization levels, leading to large quantization errors that manifest as visual artifacts. To mitigate this, recent advancements have introduced range-restriction mechanisms designed to mold the distribution of model parameters during the training phase. Specifically, the concept of Range Restriction Loss (R2-Loss) has been proposed to actively remove outliers from weights during pre-training, effectively restricting the range of values to ensure a tight distribution shape [243]. By enforcing such constraints, the model becomes inherently more amenable to lower-bit quantization schemes, allowing limited numeric representation powers to be utilized more effectively without the need for complex post-hoc corrections. This approach is particularly relevant for Gaussian Splatting, where covariance matrices and spherical harmonic coefficients must remain within predictable bounds to ensure stable rasterization.

Beyond simple range restriction, the optimization of quantization intervals themselves has become a critical area of research to complement the adaptive nature of 3DGS training. Traditional fixed-interval quantization often fails to capture the nuanced, evolving distribution of neural attributes in dynamic scenes. Consequently, methods that learn to quantize activations and weights via trainable quantizers have gained traction. These approaches parameterize the quantization intervals and obtain their optimal values by directly minimizing the task loss, allowing quantized networks to maintain accuracy comparable to full-precision counterparts even at bit-widths as low as 4-bit [244]. In the context of dynamic scene reconstruction, where temporal consistency is paramount, such adaptive quantization strategies ensure that the compressed Gaussian attributes do not introduce flickering or temporal noise during video synthesis, thereby maintaining the coherence established by the progressive propagation techniques discussed previously.

Furthermore, the stability of the training process in low-precision regimes is often compromised by phenomena such as weight oscillations, where quantized weights fluctuate between grid points, leading to wrongly estimated statistics and increased noise. This instability can undermine the careful balance of density and geometry achieved in Section 5.1. To overcome these oscillations, novel algorithms involving oscillation dampening and iterative weight freezing have been developed. These techniques have demonstrated state-of-the-art accuracy for low-bit quantization of efficient architectures, suggesting that similar stabilization mechanisms could be adapted for the iterative densification and pruning stages inherent in Gaussian Splatting pipelines [245]. Additionally, the integration of regularization techniques, such as activation kurtosis regularization, has proven effective in preventing the migration of quantization difficulties from inputs to weights, a strategy that is vital for maintaining the geometric consistency of 3D primitives under extreme compression

[246].

The exploration of non-uniform quantization schemes also offers promising avenues for optimizing Gaussian representations, particularly for hardware-constrained environments. Given that the distribution of neural parameters is often non-uniform and concentrated around zero, logarithmic or power-of-two quantization techniques can preserve the shape of the weight distribution more precisely than linear quantization. Research into Power-of-Two (PoT) quantization has shown that replacing multiplication operations with bit shifting can vastly reduce computational complexity while maintaining accuracy, a feature highly desirable for real-time rendering on hardware-constrained devices [247]. Moreover, the application of group-based distribution reshaping techniques, which split filters into groups to learn different quantization parameters, allows for a more granular control over the compression process, further enhancing the performance of low-bit models [248].

Advanced frameworks have also begun to incorporate sharpness-aware objectives to guide the model toward flatter minima in the loss landscape, which are inherently more robust to quantization noise. Methods like Sharpness- and Quantization-Aware Training (SQuAT) alternate between sharpness and step-size objectives to encourage convergence near flat minima, consistently outperforming standard quantized models in low-bit settings [249]. This is particularly pertinent for the high-dimensional optimization problems found in 4D Gaussian Splatting, where the loss surface can be highly rugged. By combining these sophisticated training paradigms with hardware-aware design principles, such as per-vector scaled quantization which assigns separate scale factors to small vectors of elements, it is possible to achieve significant memory and energy savings without sacrificing the visual quality required for immersive applications [250]. Ultimately, the synergy between range-restriction losses, adaptive interval learning, and stability-enhancing regularizations forms a comprehensive toolkit for enabling the next generation of ultra-compressed, real-time dynamic 3D reconstruction systems. These QAT methodologies prepare the parameter distributions for the subsequent layer of compression discussed in Section 5.3, where structured sparsity and vector quantization will be employed to further reduce redundancy and exploit correlations across primitives.

5.3 Structured Sparsity, Codebooks, and Vector Quantization

5.3 Structured Sparsity, Codebooks, and Vector Quantization

While Quantization-Aware Training (Section 5.2) effectively addresses the precision constraints of individual parameters by molding their distributions and learning adaptive intervals, the sheer volume of primitives in dynamic 3D Gaussian Splatting (3DGS) necessitates a complementary approach focused on redundancy reduction and structural efficiency. The deployment of high-fidelity 3DGS models on resource-constrained hardware is fundamentally hindered by the massive memory footprint required to store millions of independent Gaussian primitives. Each primitive typically entails a set of high-dimensional attributes, including position, covariance, opacity, and view-dependent color coefficients, often represented in full precision. To address this bottleneck beyond simple bit-width reduction, the community has increasingly turned toward advanced compression paradigms that combine structured sparsity with vector quantization (VQ). These techniques aim to drastically reduce storage requirements by exploiting correlations across primitives while maintaining strict compatibility with modern GPU rasterization pipelines, ensuring that decompression overhead does not negate the rendering speed advantages of explicit representations.

Vector quantization serves as a cornerstone strategy in this domain, operating by mapping continuous high-dimensional attribute vectors to discrete indices within a learned codebook. Instead of storing floating-point values for every Gaussian, the system stores a compact integer index that references a shared codeword, effectively capitalizing on the high redundancy found among neighboring primitives. Recent advancements in model compression have demonstrated that VQ can achieve state-of-the-art compression ratios with minimal accuracy degradation when the codebook design is optimized. For instance, methods utilizing hardware-efficient codebooks based on lattice structures, such as the E_8 lattice, have shown superior performance in extreme compression regimes by exploiting the geometric distribution of weights [251]. While originally developed for large language models, the principle of using symmetric, packing-optimal codebooks translates effectively to the continuous attribute spaces found in 3D reconstruction, allowing for denser packing of representative vectors and reduced quantization error. This approach complements the range-restriction strategies discussed previously by providing a discrete, efficient representation for the now well-behaved parameter distributions.

The efficacy of vector quantization is further enhanced when combined with structured sparsity. Unlike unstructured pruning, which creates irregular memory access patterns detrimental to GPU throughput, structured sparsity imposes regular constraints on which parameters are retained or zeroed out. This regularity allows hardware accelerators to skip computations efficiently without complex indexing overhead. Research into vision transformers has highlighted the potential of GPU-friendly structured sparsity patterns, such as 2:4 fine-grained sparsity, where specific patterns of zeros are enforced to align with tensor core architectures [252]. Applying similar structured masks to 3D Gaussian attributes enables the elimination of redundant primitives or specific components of the covariance matrix while preserving the parallel processing capabilities essential for real-time rasterization. Furthermore, sparse matrix representations like Delta-Compressed Storage Row (dCSR) have been devised to minimize indexing overhead in SIMD-capable environments, offering a blueprint for how sparse Gaussian data could be organized to maximize memory bandwidth utilization on embedded systems [253].

A critical challenge in applying these techniques to 3DGS is the need for joint optimization of the codebook and the sparse structure to prevent quality collapse. Simple post-training quantization often fails to account for the sensitivity of specific Gaussians to quantization noise, a problem exacerbated in dynamic scenes where temporal consistency is required. To mitigate this, sensitivity-aware clustering and quantization-aware training (QAT) strategies are employed. These methods allow the model to adapt its representation during the optimization phase, learning codebooks that minimize reconstruction loss for the specific distribution of scene attributes. Recent work on compressed 3D Gaussian splatting has successfully utilized sensitivity-aware vector clustering to compress directional colors and geometric parameters, achieving compression rates of up to $31\times$ with negligible visual degradation [254]. Similarly, the integration of learnable masks to prune insignificant Gaussians, coupled with grid-based neural fields for view-dependent color, has demonstrated that storage can be reduced by over $25\times$ while enhancing rendering speeds [2].

The synergy between sparsity and quantization is also evident in the development of compositional quantization schemes. Traditional product quantization assumes independence between subvectors, which may not hold for the correlated attributes of 3D Gaussians. Stacked quantizers and additive quantization methods offer a middle ground, exploiting hierarchical structures in codebooks to achieve lower quantization errors than product quantization while remaining computationally feasible [255]. When applied to 3D scenes, these compositional approaches can better capture the complex relationships between position, scale, and rotation, leading to more compact representations. Moreover, the use of low-rank representations in conjunction with vector quantization provides another avenue for compression. By reducing the dimensionality of the feature space before quantization, methods like Low-Rank Representation Vector Quantization (LR^2VQ) can significantly improve clustering performance and final accuracy, suggesting that projecting Gaussian attributes into a lower-dimensional latent space prior to codebook assignment could yield substantial benefits [256].

Hardware considerations remain paramount in the design of these compression schemes, as the decompression process must be lightweight enough to occur on-the-fly during the rasterization pass. Custom hardware accelerators and instruction set extensions have been proposed to handle sparse and quantized data efficiently. For example, specialized vector processors with instructions for sub-byte computation and indirect memory accesses can accelerate the decoding of quantized weights and sparse indices [257; 258]. In the context of GPUs, explicit caching strategies and optimized sparse matrix-vector multiplication kernels are crucial for minimizing data movement costs, which often dominate the energy consumption and latency of inference tasks [259; 260]. By co-designing the compression algorithm with the underlying hardware architecture—whether through structured sparsity patterns that match tensor cores or codebooks optimized for specific memory hierarchies—it is possible to achieve near-lossless compression that facilitates the deployment of dynamic 3D Gaussian scenes on edge devices.

In summary, the convergence of structured sparsity and vector quantization represents a pivotal advancement in making 3D Gaussian Splatting scalable, bridging the gap between the low-precision parameter optimization discussed in Section 5.2 and the spectral efficiency challenges addressed in Section 5.4. By leveraging hardware-efficient codebooks, enforcing GPU-friendly sparsity patterns, and employing sensitivity-aware training protocols, researchers can drastically reduce the memory footprint of dynamic 3D reconstructions. These methods not only address the storage constraints but also ensure that the decompression and rendering processes remain highly parallelized, preserving the real-time performance that defines the appeal of explicit Gaussian representations while laying the groundwork for frequency-aware optimizations that follow.

5.4 Frequency-Domain Optimization and Anti-Aliasing Compression

5.4 Frequency-Domain Optimization and Anti-Aliasing Compression

Building upon the memory-efficient representations established through structured sparsity and vector quantization in Section 5.3, the deployment of Gaussian Splatting and neural rendering systems in real-time applications necessitates a rigorous approach to managing the trade-off between high-fidelity detail preservation and computational efficiency, particularly concerning aliasing artifacts. While the previous section addressed the reduction of parameter redundancy, this section focuses on the spectral integrity of the rendered signal. Aliasing, a phenomenon inherent to the sub-sampling of signals, manifests as distracting distortions when high-frequency geometric or textural details exceed the Nyquist limit of the display resolution. In the context of point-based and volumetric rendering, where primitives are projected onto a 2D screen, the mitigation of these artifacts is critical for maintaining visual stability across varying viewing distances. Traditional spatial-domain approaches often struggle to balance the blurring required to suppress aliasing with the sharpness needed for perceptual quality. Consequently, recent advancements have pivoted towards frequency-domain optimization strategies that explicitly model and manipulate the spectral characteristics of the scene representation. By decomposing signals into distinct frequency bands, these methods enable adaptive level-of-detail (LOD) mechanisms that dynamically adjust the complexity of the rendered output without sacrificing the integrity of essential structural details.

A foundational insight driving these optimizations is the recognition that network architectures often lack the intrinsic capacity to learn effective anti-aliasing filters purely from data, especially when dealing with out-of-distribution scenarios or aggressive down-sampling. Research indicates that incorporating non-trainable blur filters and smooth activation functions at key architectural junctures can substantially mitigate frequency aliasing effects [261]. This principle is directly applicable to the rasterization pipeline of 3D Gaussians, where the projection of anisotropic ellipsoids must be carefully filtered to match the pixel footprint. Without such explicit frequency constraints, the optimization process may overfit to high-frequency noise, leading to shimmering artifacts during camera motion. Furthermore, the stability of upsampling operations, which are frequently employed in super-resolution rendering pipelines to reconstruct high-resolution views from lower-resolution computations, is heavily dependent on maintaining spatial context. Studies have shown that increasing the kernel size of convolutional upsampling operations allows for better aggregation of context, thereby improving prediction stability and reducing artifacts that arise from insufficient regional information [262].

To address the specific challenges of rendering stability in dynamic environments, frequency-consistent adaptation has emerged as a powerful paradigm. This approach ensures that the frequency density of the rendered output remains consistent with the source domain, effectively narrowing the gap caused by incorrect degradation models or varying sampling rates. By estimating degradation kernels and utilizing frequency density comparators, systems can adaptively adjust their rendering parameters to maintain fidelity across different scales [263]. This is particularly relevant for Gaussian Splatting, where the density of points must be modulated based on the viewing frustum to prevent both under-sampling (aliasing) and over-sampling (redundant computation). The integration of such frequency-aware mechanisms allows for the creation of robust rendering pipelines that perform reliably even under real-world conditions where ideal sampling assumptions do not hold.

Compression strategies for neural representations have also begun to leverage frequency decomposition to achieve higher efficiency, offering a natural bridge between the spectral optimizations discussed here and the quantization techniques detailed in Section 5.3. By separating image signals into distinct frequency bands, compression models can apply selective transmission strategies, allocating more bits to perceptually significant low-frequency components while aggressively compressing or discarding high-frequency noise. This frequency-oriented transform facilitates scalable coding and aligns with human-interpretable concepts of image structure, enabling significant bitrate reductions without compromising semantic fidelity [264]. Similarly, learned multi-frequency compression approaches utilize octave convolutions to factorize latent representations into high and low-frequency components, reducing spatial redundancy and improving rate-distortion performance [265]. When applied to 3D Gaussian attributes, such techniques allow for the compact encoding of scene details, where coarse geometric structures are preserved with high precision, and fine textural details are encoded in a residual manner that is robust to quantization.

The concept of continuous Levels of Detail (LOD) further enhances rendering stability by allowing finely tuned adaptations to resource limitations. Unlike discrete LOD schemes that can cause popping artifacts as details switch abruptly, continuous LOD methods encode representations that support smooth transitions across scales. Techniques utilizing summed-area table filtering enable efficient and continuous filtering at various LODs, ensuring that the rendering engine can adapt to the viewer's focus and hardware constraints in real-time [266]. This is complemented by hardware-aware mechanisms such as Dynamic Sampling Rate (DSR), which analyzes the spatial frequencies of a scene to determine the optimal sampling rate for different screen regions. By leveraging temporal coherence, DSR reduces redundant fragment shader computations in low-frequency areas, yielding significant energy savings and speedups while maintaining image quality [267].

Moreover, the integration of super-resolution techniques guided by auxiliary features offers a pathway to recover high-frequency details that might otherwise be lost during compression or low-resolution rendering. By fusing high-resolution auxiliary features with low-resolution renderings through cross-modality transformers, systems can reconstruct sharp edges and fine textures that are critical for visual realism [268]. This is particularly effective in scenarios where direct high-resolution rendering is computationally prohibitive. Additionally, specialized networks designed for rasterized images, such as those predicting offsets for edge enhancement, demonstrate that ultra-efficient architectures can achieve superior visual effects by focusing computational resources on high-frequency discontinuities while using simple interpolation for flat regions [269].

In summary, the convergence of frequency-domain optimization, adaptive LOD mechanisms, and compression strategies forms the backbone of stable, real-time neural rendering. By explicitly addressing the spectral properties of the scene and the limitations of the rendering pipeline, these methods ensure that Gaussian Splatting and related technologies can deliver high-fidelity visuals across a wide range of devices and viewing conditions. The shift from purely spatial heuristics to mathematically grounded frequency analysis represents a critical evolution in the field, enabling the next generation of scalable and robust 3D reconstruction systems. These spectral efficiency gains not only complement the storage reductions achieved in Section 5.3 but also lay the essential groundwork for the communication-efficient distributed training paradigms explored in Section 5.5, where minimizing bandwidth through intelligent frequency-aware compression becomes paramount for federated reconstruction.

5.5 Communication-Efficient Distributed Training and Federated Reconstruction

5.5 Communication-Efficient Distributed Training and Federated Reconstruction

While frequency-domain optimization and compression strategies (Section 5.4) effectively manage the spectral fidelity and storage footprint of individual scene representations, scaling dynamic 3D Gaussian Splatting to real-world environments often necessitates distributed training across heterogeneous edge networks. In such federated settings, the sheer volume of parameters required to represent complex spatiotemporal scenes—often numbering in the millions—creates severe communication bottlenecks. As gradients and model updates are transmitted between diverse client devices and central aggregation servers, bandwidth becomes the primary constraint on scalability. To address this, recent advancements in federated learning (FL) have pivoted towards sophisticated compression strategies that minimize data transmission without compromising the geometric and photometric fidelity essential for high-quality reconstruction.

A cornerstone of these efforts is adaptive gradient quantization, which dynamically adjusts the precision of transmitted updates based on the training phase and specific client capabilities. Traditional approaches often rely on fixed quantization schemes that fail to account for the varying norms of gradients across different training rounds or the hardware heterogeneity of the network. In response, algorithms such as [270] propose adaptive and heterogeneous gradient quantization mechanisms. These methods exploit the changing magnitude of gradient norms to adjust resolution in real-time, assigning lower resolution to slower clients to synchronize training times while maintaining higher resolution for capable devices, thereby reducing total wall-clock training time by over 50% compared to static baselines. Further refining this adaptivity, [271] introduces strategies that modulate the number of quantization levels throughout the training trajectory. This approach recognizes that early training stages require higher precision to navigate the loss landscape, whereas later stages can tolerate coarser quantization as the model converges, effectively lowering the error floor associated with stochastic gradients. Similarly, [272] formalizes a framework for dynamically selecting quantization schemes at each descent step, theoretically bounding the convergence error to ensure that communication savings do not come at the cost of model divergence. The concept of descending quantization is further explored in [273], which argues that as model updates shrink near convergence, the quantization level should decrease rather than increase, achieving significant reductions in communicated bit volume.

Beyond quantization, sparsification and structured compression offer potent avenues for reducing communication overhead by eliminating redundant information. The seminal work [274] demonstrated that the vast majority of gradient exchanges in distributed stochastic gradient descent are redundant. By employing momentum correction, local gradient clipping, and warm-up training, it is possible to compress gradients by factors exceeding 600x without accuracy loss, a finding highly relevant for transmitting dense Gaussian attribute updates. Building on this, [275] proposes a framework where clients train sparse sub-networks dynamically extracted from the full model. This "two-birds-one-stone" approach reduces both on-device computation and network transmission, naturally accommodating local data heterogeneity and introducing a self-ensembling effect that boosts performance in non-IID settings common in multi-view scene capture.

The efficiency of these compression techniques can be further enhanced by leveraging the temporal correlation inherent in iterative optimization processes. [276] exploits the high correlation between gradients in adjacent rounds, utilizing historical gradients to compress current updates via Wyner-Ziv coding without probabilistic assumptions. Additionally, [277] observes that the subspace spanned by gradients across epochs is often low-rank. By propagating only the leading principal components or scalar multipliers for recycled gradient directions, methods like Look-back Gradient Multiplier (LBGM) drastically reduce the dimensionality of transmitted data. For scenarios requiring even more aggressive compression, [278] investigates one-bit compression schemes, such as Adaptive Stochastic Sign SGD, which compress gradients into binary vectors while maintaining convergence guarantees for non-convex objectives typical in neural rendering.

To mitigate the latency introduced by straggling devices in heterogeneous networks, coded computing techniques have emerged as a critical solution. [279] introduces structured coding redundancy into the learning process, allowing the server to reconstruct global updates from a subset of received gradients, effectively neutralizing the impact of slow or unreliable clients. This is particularly vital for dynamic scene reconstruction where input data streams from multiple sensors may arrive asynchronously. Furthermore, [280] proposes a layered gradient compression framework inspired by video streaming, where gradients are split into layers and transmitted over diverse communication channels (e.g., 4G, 5G), optimizing resource utilization and reducing training time in dynamic network environments.

Hierarchical and decentralized architectures also play a pivotal role in scaling reconstruction tasks beyond simple client-server models. [281] addresses the scalability challenges of fog networks by incorporating gradient tracking into semi-decentralized federated learning, removing restrictive assumptions about gradient diversity and improving model quality in highly heterogeneous data distributions. Similarly, [282] combines device-to-server and device-to-device communications, allowing local clusters to reach consensus before global aggregation, which minimizes network resource utilization and enhances robustness against channel fading. Finally, energy efficiency remains a paramount concern for mobile deployment, bridging the gap between communication constraints and the hardware limitations discussed in the following section. [283] formulates a trade-off problem between learning accuracy and energy consumption, deriving optimal compression ratios and computing frequencies to minimize the total energy footprint of edge devices during collaborative training. Collectively, these strategies provide a robust theoretical and practical foundation for deploying communication-efficient, distributed dynamic 3D reconstruction systems, setting the stage for the hardware-specific acceleration techniques required to execute these optimized models on edge platforms.

5.6 Hardware-Aware Deployment and Inference Acceleration on Edge Platforms

5.6 Hardware-Aware Deployment and Inference Acceleration on Edge Platforms

Building upon the communication-efficient distributed training strategies discussed in Section 5.5, the successful realization of dynamic 3D Gaussian Splatting (3DGS) in real-world applications ultimately hinges on efficient execution at the edge. While federated frameworks effectively mitigate bandwidth constraints and enable collaborative learning across heterogeneous networks, the final deployment of these optimized models onto mobile and embedded devices introduces a distinct set of challenges. The transition from server-side aggregation to on-device inference necessitates a paradigm shift from generic software execution to hardware-centric optimization. As mobile and embedded devices become the primary interface for immersive augmented reality and real-time 3D reconstruction, the heterogeneity of modern System-on-Chips (SoCs) presents both opportunities and challenges. Contemporary mobile SoCs integrate diverse processing units, including central processing units (CPUs), graphics processing units (GPUs), and specialized neural processing units (NPUs), each exhibiting distinct power-performance characteristics [284]. Effective acceleration of dynamic scene reconstruction requires sophisticated strategies that not only leverage the raw computational throughput of these components but also intelligently manage the trade-offs between latency, energy consumption, and thermal constraints, directly addressing the energy efficiency concerns that bridge into the sustainability discussions of Section 5.7.

A critical component of hardware-aware deployment is the development of custom kernels tailored to the specific architectural nuances of mobile GPUs. Unlike server-grade GPUs, mobile GPUs operate under strict power budgets and memory bandwidth limitations, making standard inference engines suboptimal. Research has demonstrated that leveraging the ubiquitous mobile GPU through specialized architectures can enable real-time deep neural network inference on both Android and iOS devices, provided the networks are designed to be mobile GPU-friendly [285]. For Gaussian Splatting, which relies heavily on rasterization and attribute interpolation, this implies the need for custom CUDA or OpenCL kernels that minimize register pressure and optimize memory access patterns. Techniques such as hierarchical Roofline analysis and the use of performance tools like Nsight Compute have proven essential in identifying bottlenecks related to low occupancy and complex data access, allowing developers to achieve significant fractions of theoretical peak performance even on constrained hardware [286]. Furthermore, the integration of mixed-precision strategies is vital; support for lower precision computations, such as FP16 or INT8, reduces data movement and increases computational throughput, which is crucial for maintaining high frame rates in dynamic rendering scenarios [287].

Beyond kernel-level optimization, dynamic execution scaling engines play a pivotal role in managing workloads across the heterogeneous resources of an edge device. The stochastic nature of mobile environments, characterized by fluctuating signal strengths, thermal throttling, and competing background processes, demands adaptive scheduling mechanisms. Systems like AutoScale employ reinforcement learning algorithms to continuously learn and select the most energy-efficient execution target—whether CPU, GPU, or a collaborative cloud edge setup—by adapting to runtime variance [288]. Similarly, PolyThrottle utilizes constrained Bayesian optimization to tune hardware configurations, such as GPU and CPU frequencies, in an energy-conserving manner, demonstrating that fine-grained control over hardware elements can yield substantial energy savings without compromising inference accuracy [289]. For dynamic 3D scene reconstruction, where computational load varies significantly with scene complexity and camera motion, such adaptive engines ensure that the rendering pipeline scales efficiently, preventing thermal throttling and extending battery life, thereby laying the groundwork for the broader sustainable training paradigms explored next.

The synergy between different hardware accelerators is another frontier in edge deployment. Frameworks like Synergy illustrate the potential of pipelined, high-throughput inference by leveraging all available on-chip resources, including ARM processors, FPGA fabric, and SIMD engines, through intelligent work-stealing and load balancing [290]. While FPGAs are less common in consumer mobile phones compared to GPUs and NPUs, their inclusion in specialized edge devices for robotics and autonomous systems offers unique advantages in terms of power efficiency and customizable dataflow architectures [291]. The co-design of algorithms and hardware, particularly for memory-bound operations, can lead to dramatic improvements in performance density. For instance, novel architectures that generate weights on-the-fly or utilize in-memory analog computing can alleviate memory bandwidth bottlenecks, a common limitation in mobile deep learning accelerators [292][293].

Moreover, the challenge of concurrent workload management cannot be overstated. In real-world applications, multiple tasks often compete for the same GPU resources. Advanced scheduling strategies, such as FIKIT, which fills inter-kernel idle time with low-priority tasks, offer a mechanism to improve overall GPU utilization and quality of service in multi-tasking environments [294]. This is particularly relevant for dynamic Gaussian Splatting systems that may need to simultaneously perform semantic segmentation, tracking, and rendering. Additionally, the emergence of automated compilers like TVM provides a unified approach to optimizing deep learning workloads across diverse backends, automating the search for optimal operator fusion and memory layouts that are critical for efficient edge deployment [295].

Finally, the journey towards robust edge deployment involves addressing the variability in hardware capabilities across different device generations. Predictive modeling techniques, including machine learning-based approaches for performance estimation, allow systems to anticipate execution times and power consumption across different GPUs and CPUs, facilitating better resource allocation decisions before runtime [296][297]. By combining these hardware-aware optimization pipelines—ranging from custom mixed-precision kernels and adaptive scaling engines to intelligent concurrent scheduling and automated compilation—the community can unlock the full potential of 3D Gaussian Splatting and neural rendering on resource-constrained edge platforms. These advancements not only enable widespread adoption in mobile AR/VR and autonomous robotics but also establish the operational efficiency required to support the sustainable, energy-conscious training ecosystems detailed in the following section.

5.7 Sustainable and Energy-Efficient Training Paradigms

5.7 Sustainable and Energy-Efficient Training Paradigms

While Section 5.6 addressed the critical challenges of deploying dynamic 3D Gaussian Splatting (3DGS) on resource-constrained edge hardware, the environmental cost of *training* these sophisticated models presents an equally pressing concern. The rapid advancement of dynamic 3D scene reconstruction has ushered in an era of photorealistic real-time rendering, yet this progress often comes at a steep ecological price. Training state-of-the-art models for dynamic scenes demands extensive computational resources, leading to exponential increases in energy consumption and a corresponding surge in carbon emissions. As the field evolves toward more complex spatiotemporal representations, the carbon footprint of the training phase has become a critical metric that can no longer be overlooked. The broader artificial intelligence community increasingly recognizes that accuracy and rendering speed are not the sole indicators of success; energy efficiency and sustainability must be integrated as primary evaluation criteria [298]. This subsection critically assesses the energy implications of training dynamic Gaussian models and proposes a paradigm shift toward sustainable AI practices, focusing on data-free distillation, energy-aware scheduling, and holistic lifecycle optimization.

The fundamental challenge lies in the sheer scale of computation required to optimize millions of Gaussian primitives over time. Traditional training pipelines often prioritize convergence speed and visual fidelity without regard for the energy expended during the process. Research indicates that improperly configured training hyperparameters can consume up to five times the energy of an optimal setup while achieving identical accuracy [The Power of Training How Different Neural Network Setups Influence the Energy Demand]. In the context of dynamic reconstruction, where models must simultaneously learn geometry and motion fields, the risk of "energy bloat"—energy consumed that does not contribute to end-to-end throughput—is significant. Frameworks like Perseus have demonstrated in large model training that a substantial portion of energy consumption is extrinsic or intrinsic bloat that can be eliminated through iterative graph cut-based algorithms and smart scheduling [299]. Applying similar principles to 3DGS training could involve dynamically adjusting the density of Gaussians or the frequency of updates based on the marginal gain in reconstruction quality, thereby preventing unnecessary computational cycles before the model is even deployed to the edge devices discussed previously.

A promising avenue for reducing the carbon footprint of dynamic scene reconstruction is the adoption of data-free knowledge distillation. Dynamic scenes often require vast amounts of video data to capture non-rigid motions and temporal consistency. However, collecting, storing, and processing such massive datasets contributes significantly to the overall environmental impact. Data-free distillation allows for the transfer of knowledge from a large, pre-trained teacher model to a compact student model without access to the original training data, utilizing generative pseudo-replay mechanisms instead [Robust and Resource-Efficient Data-Free Knowledge Distillation by Generative Pseudo Replay]. For 3DGS, this approach could enable the distillation of a high-fidelity dynamic scene representation into a lighter, more efficient model that retains essential motion dynamics while drastically reducing the memory and compute requirements for both training and subsequent inference. Furthermore, data-centric approaches have shown that selecting "elite" training samples can yield remarkable energy savings of up to 98% with negligible performance degradation [300]. By identifying and training only on the most informative frames or spatial regions within a dynamic sequence, reconstruction systems can avoid the diminishing returns associated with processing redundant data, directly addressing the data volume challenges inherent in dynamic scenarios.

Beyond algorithmic efficiency, energy-aware scheduling represents a critical strategy for sustainable training operations. The carbon intensity of electricity grids varies significantly across geographical locations and time periods. Traditional training jobs run continuously regardless of the power source, often coinciding with peak carbon intensity periods. Innovative solutions like "Chase" propose observing real-time carbon intensity shifts and controlling GPU energy consumption accordingly, reducing the carbon footprint of training by over 13% with minimal impact on training time [Chasing Low-Carbon Electricity for Practical and Sustainable DNN Training]. Similarly, CarbonScaler leverages cloud workload elasticity to dynamically scale server allocation based on the carbon cost of the grid, achieving up to 51% carbon savings compared to carbon-agnostic execution [CarbonScaler Leveraging Cloud Workload Elasticity for Optimizing Carbon-Efficiency]. For distributed dynamic reconstruction tasks, where training might occur across multiple data centers or even leverage edge devices, incorporating carbon-aware scheduling algorithms ensures that heavy optimization steps are performed when and where renewable energy is most abundant. This approach aligns with the principles of Green Federated Learning, which seeks to optimize collaborative training to minimize carbon emissions while maintaining competitive performance [301].

Moreover, the design of the training environment and model architecture itself plays a pivotal role in sustainability. Studies have shown that selecting the proper model architecture and training environment can reduce energy consumption by over 80% [Do DL models and training environments have an impact on energy consumption]. In the realm of dynamic 3DGS, this suggests a need to explore hybrid architectures that balance explicit Gaussian primitives with implicit neural fields more efficiently, potentially reducing the number of parameters that require gradient updates. Additionally, the concept of “Green Learning” advocates for models characterized by low computational complexity and logical transparency, offering a viable alternative to the increasingly opaque and resource-hungry deep learning paradigms [302]. By integrating energy consumption as a direct loss term or regularization factor during the optimization of Gaussian attributes, future reconstruction systems could be trained to be inherently energy-efficient, creating models that are naturally suited for the low-power deployment strategies outlined in Section 5.6.

Finally, a holistic view of the lifecycle is essential to truly address sustainability. The environmental impact extends beyond the training phase to include hardware manufacturing and long-term deployment. Tools like Carbontracker and EnergyVis provide mechanisms to track and visualize these footprints, enabling researchers to make informed decisions [Carbontracker Tracking and Predicting the Carbon Footprint of Training Deep Learning Models; EnergyVis Interactively Tracking and Exploring Energy Consumption for ML Models]. As the field of dynamic 3D reconstruction matures, it must adopt these sustainable practices not as optional add-ons but as fundamental design constraints. By combining data-free distillation, intelligent scheduling, and energy-aware architectural design, the community can significantly mitigate the environmental impact of reconstructing the dynamic world, ensuring that the pursuit of visual realism does not come at the expense of planetary health. The transition to such sustainable paradigms is not merely an ethical imperative but a practical necessity for the scalable deployment of 3D AI in a resource-constrained future [303].

6 Robustness in Challenging Conditions, Sensor Fusion, and Neural SLAM

6.1 Handling Real-World Imaging Imperfections and Low-Texture Environments

6.1 Handling Real-World Imaging Imperfections and Low-Texture Environments

The transition of 3D Gaussian Splatting (3DGS) from controlled laboratory settings to real-world deployment exposes significant vulnerabilities in its standard formulation. While the original framework achieves remarkable fidelity under ideal conditions characterized by sharp, well-exposed images and rich geometric texture, practical applications frequently involve handheld captures, low-light scenarios, and unstructured environments. These conditions introduce severe imaging imperfections—such as motion blur, rolling shutter distortions, and color inconsistencies—alongside the geometric ambiguity inherent in texture-less surfaces. Addressing these challenges necessitates a fundamental rethinking of the image formation model and the optimization constraints applied during reconstruction, laying the essential groundwork for the robust, multi-modal sensor fusion strategies discussed in the subsequent section.

Motion blur represents one of the most pervasive issues in dynamic or handheld capture scenarios, where the assumption of a static camera pose during a single frame's exposure time is violated. Standard 3DGS treats camera poses as instantaneous points, leading to blurred reconstructions or geometric artifacts when trained on blurry inputs. To mitigate this, recent advancements have introduced probabilistic models that treat the camera pose not as a single transformation but as a distribution over the exposure duration. For instance, [304] proposes modeling motion blur as a Gaussian distribution over camera poses, enabling a unified framework that simultaneously refines camera trajectories and corrects for motion-induced blur. This approach effectively decouples the blur kernel from the scene geometry, allowing the system to recover sharp details even from significantly degraded inputs. Similarly, [305] extends this concept by incorporating visual-inertial odometry (VIO) estimates to model the continuous trajectory of the camera during exposure. By formulating a differentiable rendering pipeline that accounts for non-static poses and utilizes screen-space approximations, this method successfully compensates for both motion blur and rolling shutter effects, which are particularly detrimental in fast-moving handheld video sequences. Furthermore, [306] takes a comprehensive approach by jointly learning Gaussian parameters and recovering camera motion trajectories, explicitly modeling the physical image formation process of motion-blurred images to achieve high-quality reconstruction where implicit methods like NeRF often fail to recover intricate details.

Beyond motion artifacts, defocus blur and varying illumination conditions pose additional hurdles. Real-world datasets often suffer from depth-of-field effects and inconsistent white balancing due to automatic camera settings. [304] also addresses these issues by introducing mechanisms for defocus blur compensation and correcting color inconsistencies caused by ambient light and shadows. Moreover, [307] tackles a broader spectrum of blur types, including downscaling and defocus blur, by estimating per-pixel convolution kernels via a Blur Proposal Network. This network considers spatial, color, and depth variations to propose a quality-assessing mask, enabling the reconstruction of 3D-consistent scenes even when the input images are severely degraded. The coarse-to-fine kernel optimization scheme employed in [307] further ensures robustness against the sparse point cloud initialization that typically plagues Structure-from-Motion (SfM) pipelines operating on blurry imagery.

Parallel to handling blur, reconstructing texture-less environments remains a critical challenge. In regions such as plain walls, ceilings, or large uniform surfaces, the lack of high-frequency gradients leads to ill-posed optimization problems, often resulting in "floaters" or geometric collapse. The standard adaptive densification strategy of 3DGS, which relies on gradient magnitudes, frequently fails in these low-texture zones. To counteract this, researchers have integrated geometric priors and layout constraints. [308] specifically addresses the under-constrained geometry of texture-less planes in panoramic settings by exploiting layout priors. By guiding the optimization with structural information inherent to indoor scenes, this method prevents the formation of floaters and ensures stable modeling of flat regions like floors and walls. Similarly, [309] introduces a pipeline that initializes thin Gaussians aligned with smoothly connected areas observed in point clouds, transferring these geometric characteristics through a carefully designed densification strategy. This approach enforces explicit geometry constraints, preserving the structural integrity of non-textured regions such as furniture surfaces and ceilings.

Regularization techniques leveraging external depth and normal cues have also proven effective in stabilizing reconstruction in texture-poor areas. [5] extends the 3DGS framework by regularizing the optimization with depth information and enforcing local smoothness among neighboring Gaussians. By supervising the geometry of 3D Gaussians with normal cues, this method achieves better alignment with true scene geometry, significantly improving performance on challenging indoor datasets where texture is scarce. Additionally, [17] utilizes dense depth maps from pre-trained monocular estimation models as geometry guides. This strategy mitigates overfitting and floating artifacts in few-shot scenarios, ensuring that the reconstructed geometry adheres to physical constraints even when visual cues are minimal.

The issue of insufficient initialization in low-texture regions is further addressed by progressive propagation strategies. [15] draws inspiration from classical multi-view stereo to guide densification. Instead of relying solely on split and clone operations, it leverages existing reconstructed geometry and patch-matching techniques to generate new Gaussians with accurate positions and orientations, effectively filling gaps in texture-less surfaces where SfM fails. Moreover, [19] refines the densification criterion by considering the number of pixels covered by a Gaussian in each view. This pixel-aware gradient weighting prompts the growth of Gaussians in areas with insufficient initialization, leading to more detailed and accurate reconstructions in challenging regions. Finally, to ensure geometric consistency in the presence of these imperfections, [310] introduces structured constraints that prevent Gaussians from moving independently, utilizing monocular depth estimates for initialization and flow-based losses to maintain coherence, thereby reducing the jumble of artifacts often seen in sparse or low-texture views. Together, these advancements signify a maturing of 3DGS, transforming it from a technique dependent on perfect data into a robust tool capable of handling the messy realities of real-world imaging. However, while these single-modality improvements address specific visual degradations, they often struggle with the scale ambiguity and drift accumulation inherent in large-scale, unbounded environments. This limitation underscores the necessity of integrating complementary sensor modalities, a topic explored in the following section on multi-modal sensor fusion.

6.2 Multi-Modal Sensor Fusion for Robust Localization and Mapping

6.2 Multi-Modal Sensor Fusion for Robust Localization and Mapping

Building upon the strategies to mitigate imaging imperfections and texture scarcity discussed in the previous section, the deployment of 3D Gaussian Splatting (3DGS) in real-world robotic and autonomous systems necessitates a further paradigm shift: moving from single-modality robustness to multi-modal sensor fusion for operation in unbounded, large-scale environments. While the techniques in Section 6.1 effectively address visual degradations like blur and low texture, they often struggle with the fundamental scale ambiguity and cumulative drift inherent to purely visual systems over long trajectories. To overcome these limitations, contemporary research increasingly advocates for the integration of complementary sensor modalities, including LiDAR, Inertial Measurement Units (IMU), depth cameras, and emerging event-based sensors. This multi-modal fusion strategy is critical for ensuring metric scale awareness, mitigating long-term drift, and maintaining operational continuity in GNSS-denied or adverse weather scenarios, thereby laying the essential groundwork for the semantic-aware dynamic understanding explored in the subsequent section.

A primary challenge in unbounded outdoor scenes is the inadequacy of monocular or stereo visual solutions to provide accurate depth over long ranges, which is essential for initializing and constraining Gaussian primitives without scale drift. The integration of LiDAR data offers a direct solution by providing precise geometric structure information. For instance, the MM-Gaussian system demonstrates the efficacy of fusing solid-state LiDAR with camera inputs to resolve inaccurate depth issues inherent in purely visual approaches [311]. By leveraging the geometric rigor of LiDAR point clouds to initialize and regularize the 3D Gaussian distribution, such systems achieve realistic rendering effects while maintaining robust localization trajectories that would otherwise diverge in large-scale environments. This synergy allows the system to utilize pixel-level gradient descent on color information while being anchored by the metric accuracy of range sensors, effectively bridging the gap between high-fidelity rendering and reliable mapping.

Beyond geometric supplementation, the fusion of inertial data is paramount for handling high-frequency motion and temporary visual occlusions, addressing the temporal gaps that static image-based blur compensation cannot fully resolve. In environments where visual features are sparse or lighting conditions change abruptly, IMU measurements provide critical priors for ego-motion estimation. Advanced frameworks have moved towards tightly coupled architectures that integrate visual, inertial, and Global Navigation Satellite System (GNSS) measurements across multiple layers to correct drift dynamically [312]. Although traditional SLAM systems have long utilized such fusion, adapting these principles to neural representations like 3DGS requires novel optimization strategies that can ingest continuous-time trajectory estimates from inertial odometry to guide the spatiotemporal placement of Gaussians. This is particularly relevant in GNSS-denied environments, where the system must rely on the internal consistency of sensor fusion to prevent catastrophic failure. Recent developments in resilient state estimators, such as those utilizing sparse Gaussian Processes to handle asynchronous multi-LiDAR and multi-IMU setups, highlight the importance of continuous-time trajectory modeling which can be directly adapted to deformable or dynamic Gaussian fields [313].

The robustness of multi-modal fusion is further tested under adverse operating conditions, such as night, fog, or rain, where standard cameras and even LiDAR may suffer from performance degradation that exceeds the correction capabilities of standard deblurring algorithms. In these scenarios, the inclusion of radar and event cameras provides a crucial redundancy. Radar sensors, with their ability to penetrate precipitation and measure velocity via the Doppler effect, offer a distinct advantage for perception in low-visibility conditions. Frameworks that incorporate radar data into fusion pipelines have shown significant improvements in object detection and scene understanding during inclement weather, suggesting a similar potential for enhancing the stability of neural maps [314]. Similarly, event cameras, which capture asynchronous brightness changes with microsecond latency, are uniquely suited for handling high-speed motion and extreme dynamic range variations that cause motion blur in standard frames. The ECMD dataset, which includes synchronized event cameras alongside LiDAR and IMU, underscores the growing recognition of event-based sensing for robust SLAM in challenging driving scenarios [315]. Integrating these high-temporal-resolution signals with 3DGS could enable the reconstruction of scenes with fast-moving objects or under flickering illumination, areas where traditional frame-based optimization struggles.

Furthermore, the calibration and alignment of these heterogeneous sensors remain a non-trivial prerequisite for effective fusion, as misalignment in space or time can introduce severe artifacts in the reconstructed Gaussian field that mimic the geometric collapses seen in low-texture regions. Recent works propose segmentation-based frameworks to jointly estimate geometric and temporal parameters between LiDAR and camera suites without requiring explicit calibration labels, leveraging semantic consistency to refine extrinsic and temporal offsets [316]. Such self-calibrating mechanisms are essential for long-term deployment where sensor rigidity cannot be guaranteed. Additionally, the fusion of semantic information from multiple modalities enhances the interpretability of the map, serving as a bridge to the fully semantic systems discussed next. Approaches that fuse 2D image semantics with 3D point cloud parsing using adaptive attention mechanisms demonstrate that multi-modal semantic fusion can significantly improve detection accuracy and boundary definition, which translates to more semantically coherent Gaussian attributes [317].

Finally, the development of comprehensive datasets and benchmarking protocols is vital for advancing this field. Datasets like MUSES, which integrate panoptic annotations with diverse sensor modalities including radar and event cameras under uncertainty, provide the necessary_ground truth_ to train and evaluate uncertainty-aware fusion models [318]. Similarly, aerial datasets like MUN-FRL facilitate research into GNSS-denied navigation for unmanned aerial vehicles, offering synchronized visual-inertial-LiDAR data over large distances [319]. As the community moves towards more complex fusion architectures, the lessons learned from robust sensor fusion in classical SLAM—such as the use of factor graphs for tightly coupled optimization [320] and the mitigation of GNSS Non-Line-of-Sight errors using 3D maps [321]—must be systematically translated into the differentiable rendering domain of 3D Gaussian Splatting. This convergence of classical robotics robustness and modern neural rendering fidelity represents the frontier of reliable, real-time 3D scene reconstruction for autonomous agents, providing the stable, metrically accurate foundation required for the advanced semantic separation and dynamic object tracking detailed in the following section.

6.3 Dynamic Scene Understanding and Semantic-Aware Gaussian SLAM

6.3 Dynamic Scene Understanding and Semantic-Aware Gaussian SLAM

While Section 6.2 established the critical role of multi-modal sensor fusion in ensuring geometric robustness and scale consistency in unbounded environments, the challenge of operating within truly dynamic scenes requires a fundamental shift from passive reconstruction to active semantic understanding. Traditional SLAM algorithms, predicated on the static world assumption, often succumb to tracking drift and mapping artifacts when confronted with moving agents such as pedestrians and vehicles. Although early semantic SLAM approaches mitigated this by filtering dynamic features via 2D segmentation or geometric consistency, these methods typically treated moving objects as noise to be discarded, resulting in maps that were geometrically complete only for the static background. The advent of 3D Gaussian Splatting (3DGS) has ushered in a new paradigm that moves beyond mere outlier rejection. By leveraging explicit neural representations, contemporary systems can now construct rich, semantic-aware maps that explicitly disentangle static infrastructure from non-rigid, moving agents, thereby enabling high-fidelity reconstruction and real-time interaction in complex, evolving environments.

The core of this evolution lies in the accurate separation and coherent modeling of static and dynamic components. Early iterations, such as DS-SLAM, combined semantic segmentation with moving consistency checks to improve localization, while Det-SLAM integrated Detectron2 with ORB-SLAM3 to identify and eradicate dynamic spots using depth and semantic cues [322] [323]. Similarly, VDO-SLAM advanced the field by estimating the full SE(3) motion of rigid objects without prior shape models, and MaskFusion demonstrated the ability to reconstruct multiple moving objects independently by fusing instance-level masks [324] [325]. However, these approaches often lacked the differentiable, explicit medium required for dense, real-time semantic embedding. 3DGS addresses this limitation by allowing semantic features to be optimized alongside geometric and appearance attributes, facilitating a unified representation that supports both precise localization and detailed scene interpretation.

Building on this capability, the emergence of Semantic Gaussian Splatting SLAM systems marks a critical milestone. SGS-SLAM stands as the pioneering system in this domain, incorporating appearance, geometry, and semantic features through multi-channel optimization [326]. By introducing a unique semantic feature loss and a semantic-guided keyframe selection strategy, SGS-SLAM effectively compensates for the limitations of traditional depth and color losses, preventing erroneous reconstructions caused by cumulative errors in dynamic sequences. This ensures that the resulting map retains precise object-level semantic accuracy alongside high geometric fidelity. Further advancing this framework, NEDS-SLAM proposes an Explicit Dense semantic SLAM system utilizing a Spatially Consistent Feature Fusion model to address the variability of pre-trained segmentation heads in dynamic settings [327]. By compressing high-dimensional semantic features into a compact 3D Gaussian representation via a lightweight encoder-decoder and employing a Virtual Camera View Pruning method, NEDS-SLAM mitigates memory consumption while enhancing scene representation quality amidst dynamic interferences. Similarly, SemGauss-SLAM encodes semantic information directly within the spatial layout of the environment, leveraging feature-level losses and semantic-informed bundle adjustment to jointly optimize camera poses and the map for robust tracking in complex scenarios [328].

The resilience of these semantic-aware Gaussian SLAM systems is further amplified by their capacity to handle severe occlusions and integrate open-vocabulary understanding. While traditional methods often fail when camera views are largely obstructed by dynamic entities, modern approaches tightly integrate geometric and semantic cues to track cameras and estimate cluster-wise dense segmentation simultaneously, even under long-term large occlusions [329]. Moreover, the integration of language models has opened new avenues for flexible scene understanding. Systems like LEXIS harness Large Language Models to embed CLIP features into SLAM graph nodes, enabling unified room classification and segmentation [330]. Although initially focused on static topology, the principles of embedding rich semantic descriptors are directly applicable to distinguishing and labeling dynamic agents within Gaussian maps, bridging the gap between low-level perception and high-level cognitive reasoning.

In summary, the convergence of semantic segmentation, dynamic object tracking, and 3D Gaussian Splatting has created a powerful framework for dynamic scene understanding that complements the sensor fusion strategies discussed previously. By transitioning from simple outlier rejection to explicit semantic modeling of both static and dynamic elements, systems like SGS-SLAM, NEDS-SLAM, and SemGauss-SLAM are setting new standards for robustness and fidelity. These architectures not only enhance camera localization accuracy in highly dynamic environments but also generate dense, semantically annotated maps essential for higher-level robotic tasks. As the field progresses towards the deployment challenges outlined in the following section, the fusion of these explicit Gaussian representations with generative priors and physics-informed dynamics will be paramount. The subsequent discussion on efficient map representations and real-time deployment strategies will address how these computationally intensive semantic-density maps can be optimized for resource-constrained mobile platforms, ensuring that the theoretical gains in dynamic understanding translate into practical, real-time autonomy.

6.4 Efficient Map Representations and Real-Time Deployment Strategies for Robotics

6.4 Efficient Map Representations and Real-Time Deployment Strategies for Robotics

While Section 6.3 established how semantic-aware Gaussian SLAM systems can disentangle static infrastructure from dynamic agents to achieve high-fidelity reconstruction, the computational intensity of these dense, semantically annotated maps presents a formidable barrier to real-world robotics. The transition from theoretical robustness to practical autonomy necessitates a delicate balance: maintaining the photorealistic fidelity and semantic richness of 3D Gaussian Splatting while adhering to the strict latency, memory, and power constraints of mobile platforms. Consequently, recent research has pivoted from purely algorithmic advancements to developing efficient map representations and specialized hardware acceleration strategies. This shift is critical for ensuring that the complex optimization and rendering pipelines required for dynamic scene understanding can operate in real-time without compromising mapping accuracy or exhausting onboard resources.

A primary strategy to mitigate the quadratic complexity associated with global bundle adjustment in dense Gaussian representations is the decomposition of the global map into manageable submaps. This approach significantly reduces memory overhead and enables parallel processing, which is essential for scaling to large environments. For instance, the Gaussian-SLAM framework organizes the scene into independently optimized sub-maps, allowing the system to avoid keeping the entire Gaussian cloud in active memory during optimization [331]. This sub-map management is crucial for long-term autonomy, particularly in the dynamic scenarios discussed previously, as it localizes computational load. Similarly, Loopy-SLAM employs a point-based submap generation method that facilitates efficient map corrections and global pose optimization without storing the entire history of input frames, thereby streamlining the data flow for real-time applications [332].

Complementing algorithmic optimizations, the integration of heterogeneous computing architectures, particularly Field-Programmable Gate Arrays (FPGAs), has emerged as a transformative approach for accelerating SLAM pipelines. FPGAs offer the distinct advantage of runtime reconfigurability, allowing robotic systems to dynamically adapt their computational resources based on environmental complexity and task requirements—a capability vital for handling the variable loads introduced by dynamic objects. Research into energy-efficient FPGA-based accelerators demonstrates that exploiting SLAM-specific data locality and sparsity can yield performance improvements exceeding five times that of state-of-the-art software implementations [333]. These accelerators are particularly effective when targeting specific bottlenecks such as feature extraction and matching, where hardware-level parallelism can be fully leveraged. For example, specialized FPGA architectures designed for ORB-SLAM have achieved frame rate improvements of up to 31 times compared to ARM processors while simultaneously enhancing energy efficiency by a factor of 25 [334]. Such hardware-aware designs are indispensable for enabling dense Gaussian SLAM on battery-powered robots, where energy consumption is a critical constraint.

When onboard resources remain insufficient for full SLAM execution, even with sub-mapping and hardware acceleration, client-server optimization frameworks and edge computing paradigms offer a viable solution. In these distributed approaches, the robot (client) processes only a localized portion of the map while offloading global optimization and storage to a server. This strategy ensures accurate real-time state estimation on the device by utilizing summarized map information transmitted from the server, effectively decoupling computational load from memory constraints [335]. Extending this concept, edge computing architectures like RecSLAM facilitate multi-robot collaboration by directing raw sensor data to edge servers for real-time fusion, thereby reducing communication overhead and processing latency compared to traditional cloud-based solutions [336]. These distributed strategies provide a scalable pathway for deploying complex Gaussian representations across robot swarms or in large-scale infrastructure monitoring, bridging the gap between individual robot limitations and collective intelligence.

Furthermore, the efficiency of map representations themselves is being revolutionized by compact modeling techniques such as Gaussian Mixture Models (GMM). Traditional occupancy grids often suffer from high memory usage during construction, but GMM-based approaches like GMMap enable single-pass compression of depth images into local Gaussian distributions, which are then fused into a globally consistent map. This method has been shown to reduce map size by over 56% and memory overhead by 88% while maintaining real-time construction rates on low-power CPUs [337]. The synergy between compact mathematical representations and hardware acceleration is evident in systems that integrate LiDAR SLAM with FPGA accelerators, achieving speedups of nearly 15 times for scan matching tasks while consuming merely 2.4 watts of power [338]. Additionally, the development of intra-FPGA data distribution services, such as fpgaDDS, addresses communication bottlenecks between hardware-mapped nodes, ensuring that the high throughput of accelerated components is not stifled by software communication layers [339].

Finally, the pursuit of real-time deployment extends to the optimization of the entire robotics software stack. Architectures like RobotCore provide a target-agnostic framework for integrating hardware accelerators into the ROS 2 ecosystem, simplifying the deployment of FPGA and GPU-accelerated nodes for perception and mapping tasks [340]. By combining adaptive approximation controllers that dynamically adjust computational precision based on situational awareness with these hardware advancements, systems like SLAMBooster have demonstrated the ability to reduce computation time by 72% while preserving localization accuracy [341]. As Gaussian Splatting continues to evolve, the convergence of sub-map management, compact probabilistic representations, and reconfigurable hardware acceleration will be paramount. These innovations ensure that the high-fidelity reconstruction methods discussed in previous sections can transition from offline benchmarks to robust, real-time robotic applications. However, validating the efficacy of these optimized systems under diverse and extreme conditions requires a rigorous evaluation framework, a topic addressed in the following section through an analysis of benchmarking, datasets, and robustness metrics.

6.5 Benchmarking, Datasets, and Evaluation Metrics for Robustness

6.5 Benchmarking, Datasets, and Evaluation Metrics for Robustness

Transitioning from the efficient map representations and hardware acceleration strategies discussed in the previous section to real-world deployment necessitates a rigorous framework for validation. The evaluation of Simultaneous Localization and Mapping (SLAM) systems, particularly those integrating neural rendering and Gaussian Splatting for dynamic scenes, is fundamentally constrained by the quality, diversity, and realism of available benchmarks. While algorithmic advancements have surged, the community faces a critical bottleneck: the absence of standardized datasets that comprehensively capture the extreme conditions required to test the robustness and resilience of these computationally intensive models. Traditional benchmarks often rely on controlled environments that fail to expose the failure modes of modern systems, leading to an overestimation of performance when deployed on resource-constrained mobile platforms. To address this, recent efforts have shifted towards curating multi-modal datasets that encompass diverse sensor suites, challenging weather patterns, and complex dynamic interactions, thereby providing a realistic testing ground for next-generation localization frameworks.

A primary limitation of legacy datasets is their inability to simulate all-weather and perceptually degraded environments, which are critical stress tests for systems operating under the latency and power constraints highlighted earlier. The [342] addresses this gap by presenting an extremely challenging real-world dataset designed to push SLAM technologies into domains characterized by structureless corridors, varying lighting, and perceptual obscurants such as smoke and dust. This dataset is pivotal for evaluating systems that must maintain operation when visual features are sparse or corrupted, a scenario where neural rendering approaches can leverage multi-modal inputs like thermal and LiDAR data to sustain tracking. Similarly, the [343] provides millimeter-accurate ground truth across diverse settings ranging from construction sites to historic buildings, enabling the precise quantification of trajectory errors that were previously unmeasurable with standard GPS-dependent methods. The high fidelity of such datasets is essential for training and validating models that claim sub-centimeter accuracy while adhering to strict operational limits.

Beyond real-world collections, synthetic and simulated environments offer controllable perturbations that are difficult to reproduce physically, allowing for systematic stress-testing of generalization capabilities against distribution shifts. The [344] utilizes the CARLA simulator to generate data under tunable weather and dynamic conditions, allowing researchers to isolate specific failure factors such as heavy rain or aggressive motion. Complementing this, the [345] introduces a benchmark built on Unreal Engine that includes dynamic weather simulation and ground truth scene segmentation, which is crucial for evaluating semantic-aware SLAM systems that must distinguish between static infrastructure and moving agents. Furthermore, the [346] proposes a novel pipeline for synthesizing noisy data, introducing a comprehensive perturbation taxonomy that transforms clean simulations into challenging scenarios to assess risk tolerance systematically. These synthetic benchmarks are indispensable for evaluating how well neural radiance fields and Gaussian splats generalize before deployment on physical robots.

The diversity of robotic platforms and locomotion styles also necessitates datasets that extend beyond standard vehicular motion, ensuring that efficiency gains achieved through sub-mapping or hardware acceleration translate across different form factors. The [347] covers handheld, wheeled, legged, and vehicular platforms across 38.7 kilometers of diverse environments, addressing the scalability gap in current evaluations. For scenarios involving deformable terrains, the [348] offers unique challenges by capturing data on sandy and deformable grounds, testing the robustness of systems against non-geometric hazards. Additionally, the [349] fills a critical void by incorporating 4D radar and thermal cameras, providing a robust alternative when visual and LiDAR sensors fail due to adverse weather like fog or snow.

However, the mere existence of diverse datasets is insufficient without standardized and insightful evaluation metrics that reflect both geometric accuracy and system efficiency. Traditional metrics such as Absolute Trajectory Error (ATE) and Relative Pose Error (RPE) often fail to capture the nuanced performance degradation in long-term or dynamic operations, nor do they account for the computational overhead critical to real-time applications. The paper [350] argues that a quantitative characterization of dataset operating conditions is a prerequisite for meaningful robustness evaluation, proposing a framework to link dataset properties directly to system resilience. Moreover, [351] highlights the danger of overexposed datasets leading to biased evaluations, urging the community to adopt more rigorous selection criteria. To address specific failure modes, [352] introduces the concept of commutativity, revealing that many state-of-the-art systems exhibit significant bias between forward and reverse traversals, a flaw overlooked by standard metrics.

Innovations in evaluation also include no-reference metrics for scenarios where ground truth is unavailable or unreliable, a common constraint in large-scale or hazardous deployments. The [353] proposes the Mutually Orthogonal Metric (MOM), which estimates trajectory quality based on the geometric consistency of registered point clouds, correlating strongly with full-reference metrics. For dense mapping applications, [354] introduces a multi-faceted metric assessing completeness, artifacts, accuracy, and resolution, moving beyond simple pose estimation to evaluate the structural integrity of the reconstructed map—a key concern for Gaussian Splatting fidelity. Furthermore,

[355] demonstrates the potential of using machine learning to predict localization errors from raw sensor characteristics, offering a proactive approach to integrity monitoring.

Finally, the assessment of robustness must holistically account for the computational and energy constraints of real-world deployment, bridging the gap between algorithmic performance and the hardware realities discussed previously. [356] provides a framework for holistic comparison across functional and non-functional requirements, such as energy consumption and processing time, which are critical for mobile and edge applications. As the field moves towards neural and dynamic SLAM, the integration of these diverse datasets and advanced metrics is paramount. The [357] study emphasizes that while individual perturbations are often addressed, generalization across combined challenges remains understudied, calling for benchmarks that test lifelong operation under unconstrained conditions. By leveraging these comprehensive resources, the community can better gauge the true readiness of Gaussian Splatting and neural rendering techniques for robust, real-world dynamic scene reconstruction, ensuring that theoretical advances translate into reliable robotic autonomy.

7 Semantic Understanding, Generative Synthesis, and Text-Driven Editing

7.1 Distilling Foundation Models for Open-Vocabulary 3D Semantic Fields

7.1 Distilling Foundation Models for Open-Vocabulary 3D Semantic Fields

While the preceding sections have established the geometric fidelity and rendering efficiency of Gaussian Splatting, a critical frontier lies in endowing these representations with rich semantic understanding. The integration of large-scale pre-trained vision foundation models into 3D scene reconstruction has catalyzed a paradigm shift, moving beyond purely geometric recovery toward environments that machines can perceive and interpret. Traditionally, 3D semantic segmentation relied heavily on expensive, dense manual annotations, severely limiting scalability and generalization to unseen object categories. The emergence of powerful 2D foundation models, such as the Segment Anything Model (SAM) and Contrastive Language-Image Pre-training (CLIP), now provides a robust mechanism to transfer rich semantic and spatial knowledge from the 2D domain to 3D representations without requiring explicit 3D labels. This subsection reviews the methodological frameworks that distill these 2D vision-language features into 3D Gaussian Splatting (3DGS) fields, enabling zero-shot semantic segmentation, panoptic understanding, and open-vocabulary scene interpretation.

The core challenge in this domain lies in bridging the modality gap between 2D image features and 3D geometric primitives while maintaining the real-time computational efficiency inherent to Gaussian Splatting. Early attempts to lift 2D features into 3D often utilized Neural Radiance Fields (NeRFs), but these approaches were hampered by slow rendering speeds and continuity artifacts in implicit feature fields. To address these limitations, recent works have pivoted towards explicit point-based representations. A pioneering approach in this direction is Feature 3DGS, which extends the standard radiance field rendering capabilities of 3DGS to arbitrary-dimensional semantic features. By distilling features from 2D foundation models like SAM and CLIP-LSeg directly into Gaussian attributes, this method overcomes the spatial resolution disparities and channel consistency issues that plague naive implementations. The result is a system capable of real-time novel view semantic segmentation and language-guided editing, significantly outperforming NeRF-based counterparts in both training and inference speed [30].

A critical component of distilling foundation models is ensuring view consistency. Since 2D foundation models generate predictions independently for each view, direct projection into 3D space can lead to semantic ambiguities and inconsistent labeling across different angles. CLIP-GS addresses this by introducing a Semantic Attribute Compactness (SAC) approach, which learns compact semantic representations within the Gaussian attributes rather than rendering high-dimensional feature maps, thereby preserving efficiency. Furthermore, it employs a 3D Coherent Self-training (3DCS) strategy that leverages the multi-view consistency inherent in the 3D model itself. By refining pseudo-labels derived from the trained Gaussian model and imposing cross-view consistency constraints, CLIP-GS achieves superior segmentation accuracy on datasets like Replica and ScanNet while maintaining real-time rendering speeds exceeding 100 FPS [31]. Similarly, Semantic Gaussians proposes a versatile projection approach to map various 2D semantic features into a novel semantic component of 3D Gaussians. This method avoids the additional training burdens associated with NeRFs and includes a dedicated 3D semantic network for fast inference, demonstrating significant improvements in mean Intersection over Union (mIoU) and mean Accuracy (mAcc) on standard benchmarks [358].

The synergy between SAM's spatial precision and CLIP's semantic reasoning has also been exploited to enhance 3D instance segmentation. Methods like SAI3D leverage geometric priors alongside semantic cues from SAM to partition 3D scenes into primitives, which are then merged into consistent instance segmentations. This approach utilizes a hierarchical region-growing algorithm with dynamic thresholding to robustly parse fine-grained 3D scenes, outperforming existing open-vocabulary baselines and even surpassing fully supervised methods in class-agnostic settings on challenging datasets like ScanNet++ [359]. In the context of part-level segmentation, PartSLIP++ improves upon earlier heuristic methods by utilizing SAM to generate precise pixel-wise 2D segmentations instead of coarse bounding boxes. It replaces heuristic conversion processes with a modified Expectation-Maximization algorithm that treats 3D instance segmentation as latent variables, iteratively refining them through 2D-3D matching and optimization [360].

Beyond segmentation, the distillation of foundation models enables complex reasoning tasks such as anomaly detection and robotic manipulation. ClipSAM introduces a collaborative framework where CLIP's global semantic understanding is used to localize anomalies, which then serve as prompt constraints for SAM to refine the segmentation masks. This multi-scale cross-modal interaction effectively mitigates the limitations of using either model in isolation, achieving state-of-the-art performance in zero-shot anomaly segmentation [361]. For robotic applications, distilled feature fields combine accurate 3D geometry with rich semantics from CLIP, allowing robots to designate novel objects for manipulation via free-text natural language and generalize to unseen categories with few-shot learning [362].

Furthermore, the field is moving towards unified representations that serve both reconstruction and high-level understanding, laying the groundwork for the generative capabilities discussed in the subsequent section. Uni3DR^2 demonstrates the importance of a unified scene representation by extracting geometric and semantic features via frozen 2D foundation models and a multi-scale 3D decoder. This unified approach not only improves 3D reconstruction fidelity but also enhances the performance of Large Language Models (LLMs) in 3D vision-language understanding tasks, validating the necessity of integrating geometric and semantic representations [363]. Other works like CLIP2Scene and PointCLIP have laid the groundwork for transferring CLIP's knowledge to point clouds through cross-modal contrastive learning and multi-view projections, establishing strong baselines for annotation-free 3D semantic segmentation [364] [365]. These advancements collectively illustrate that distilling 2D foundation models into 3D Gaussian fields is no longer just a supplementary technique but a fundamental strategy for achieving scalable, open-vocabulary, and semantically rich 3D scene understanding. By establishing these semantic priors, recent research is now poised to tackle the next logical step: leveraging such structured understanding to guide the compositional generation of complex, multi-object scenes from natural language.

7.2 Compositional Scene Generation via Layout Planning and Progressive Editing

7.2 Compositional Scene Generation via Layout Planning and Progressive Editing

Building upon the rich semantic priors and open-vocabulary understanding established in the previous section, the generation of complex, multi-object 3D scenes from natural language descriptions represents the next critical frontier in neural rendering. While distilling foundation models enables machines to perceive and segment existing environments, synthesizing coherent scenes with multiple interacting entities from scratch introduces unique challenges. Single-object synthesis has achieved remarkable fidelity through score distillation sampling and diffusion priors; however, scaling these methods to multi-object scenarios often results in "guidance collapse," where objects merge, vanish, or fail to adhere to specified spatial relationships. To overcome these limitations, recent research has pivoted towards compositional frameworks that decouple the generation process into distinct stages of layout planning and progressive local editing. By leveraging the reasoning capabilities of Large Language Models (LLMs) to interpret complex semantic prompts, these frameworks construct editable 3D layouts that serve as structural priors, subsequently refining the scene through iterative, localized optimization steps. This modular approach not only resolves the ambiguities of monolithic pipelines but also lays the essential groundwork for the dynamic, temporally consistent 4D synthesis discussed in the following section.

A central component of this paradigm is the utilization of LLMs as high-level planners capable of decomposing intricate textual descriptions into structured spatial arrangements. Traditional text-to-3D methods often struggle with long prompts containing multiple objects and relational constraints, leading to incoherent outputs. In response, frameworks like [366] demonstrate that LLMs can be effectively prompted to generate plausible layouts for both 2D images and 3D indoor scenes by composing in-context visual demonstrations. This approach allows the model to convert challenging linguistic concepts, such as numerical quantities and specific spatial relations, into precise layout arrangements, significantly outperforming standard text-to-image systems in spatial correctness. Similarly, [367] introduces a method where LLMs extract critical components from lengthy prompts, including bounding box coordinates and detailed object descriptions, to form a foundation for layout-to-image generation. This decomposition strategy ensures that each entity is assigned a specific region and semantic identity before the rendering process begins, mitigating the ambiguity inherent in raw text prompts.

Once a coarse layout is established, the focus shifts to progressive local editing to populate the scene with high-fidelity content while maintaining global consistency. The [368] framework exemplifies this strategy by breaking down the entire generation process into a series of locally progressive editing steps. Crucially, this method constrains content changes to regions determined by user-defined region prompts at each step, preventing the optimization from destabilizing previously generated parts of the scene. To further enhance precision, [368] proposes an overlapped semantic component suppression technique, which encourages the optimization process to focus specifically on the semantic differences between prompts, thereby enabling the creation of precise 3D content for complex semantic interactions. This stepwise refinement is essential for handling scenes where objects must interact physically or occlude one another realistically.

The integration of explicit 3D representations, such as Gaussian Splatting, has further accelerated the adoption of layout-guided generation due to their efficiency and editability. [369] presents a generative framework that combines LLM-derived layouts with a layout-guided 3D Gaussian representation. This system utilizes an object-scene compositional optimization mechanism with conditioned diffusion to collaboratively generate realistic scenes with consistent geometry and texture. By adjusting coarse layout priors extracted from LLMs to align with the generated scene, [369] ensures accurate interactions among multiple objects. Furthermore, [370] advances this concept by introducing a 3D Gaussian Guide composed of semantic primitives and their spatial transformations derived directly from text. This approach employs a progressive scale control during local object generation and models collision relationships at the global level to enhance physical correctness, addressing the training instability often seen in simple blending methods.

Beyond static layouts, interactive and instruction-driven systems are emerging to allow for dynamic scene manipulation, bridging the gap between initial generation and user intent. [371] proposes LI3D, a system that integrates LLMs as 3D layout interpreters, allowing users to flexibly generate and edit visual content through multi-round interactions. By incorporating vision-language assistants to provide generative feedback, such systems can iteratively refine the spatial generation and reasoning processes. For more granular control, [372] employs explicit 3D Gaussian splatting to facilitate local editing within specified bounding boxes, utilizing both text and image prompts to accurately control appearance and location. This dual-modality approach complements the layout planning stage by providing detailed appearance references that pure text descriptions often lack.

The efficacy of these compositional approaches is also evident in urban and large-scale scene generation, demonstrating the scalability of layout-aware architectures. [373] introduces a compositional 3D layout representation comprising semantic primitives with explicit arrangement relationships to handle the vast scale and intricate arrangements of urban environments. By introducing Layout-Guided Variational Score Distillation, this framework conditions the score distillation sampling process with geometric and semantic constraints, successfully scaling text-to-3D generation to drive distances exceeding 1000 meters. Similarly, [374] integrates a semantic graph prior to jointly learn scene appearances and layout distributions, significantly improving controllability and fidelity in indoor scene synthesis compared to methods that implicitly model object relations.

Collectively, these works illustrate a decisive shift from monolithic generation pipelines to modular, layout-aware architectures. By decomposing the problem into semantic parsing, layout planning, and progressive local refinement, researchers are able to generate complex multi-object scenes that adhere to strict spatial and semantic constraints. The synergy between the reasoning power of LLMs for layout interpretation and the rendering efficiency of explicit primitives like 3D Gaussians enables a new class of generative models capable of producing editable, coherent, and physically plausible 3D worlds from natural language instructions. Having established robust mechanisms for static scene composition and layout control, the field is now poised to extend these principles into the temporal domain, tackling the more complex challenge of synthesizing dynamic, four-dimensional content and human-object interactions driven solely by text.

7.3 Text-Driven Synthesis of Dynamic Scenes and Human-Object Interactions

Building upon the layout-aware architectures and progressive editing strategies discussed in the previous section, the frontier of neural rendering has rapidly expanded from static scene composition to the generation of dynamic, four-dimensional content driven by natural language. This evolution represents a paradigm shift where the goal is no longer merely to reproduce observed reality or arrange static objects, but to synthesize plausible, temporally consistent 4D scenes and complex human-object interactions (HOIs) solely from textual prompts. The core challenge in this domain lies in bridging the semantic gap between abstract language descriptions and the intricate spatiotemporal dynamics required for realistic animation. Recent advancements have begun to address this by leveraging large-scale foundation models, explicit Gaussian representations, and sophisticated motion priors to create grounded 4D content that maintains coherence over time.

A significant body of work focuses on decomposing the generative process to ensure both high-fidelity appearance and coherent motion, addressing the limitations of monolithic pipelines. Traditional text-to-4D approaches often struggle with the "three-way tradeoff" between scene appearance, 3D structure, and motion fidelity. To mitigate this, researchers have introduced hybrid optimization strategies that blend supervision signals from multiple pre-trained diffusion models. For instance, [375] proposes a hybrid score distillation sampling technique that alternates between text-to-image and text-to-video models. This approach allows the system to inherit the realistic appearance and 3D structure from image generators while incorporating motion dynamics from video generators, resulting in 4D scenes that are visually compelling and temporally stable. Similarly, [376] leverages dynamic 3D Gaussian Splatting combined with deformation fields, utilizing a compositional generation approach that integrates text-to-image, text-to-video, and 3D-aware multiview diffusion models. This method introduces specific regularization mechanisms to stabilize the distribution of moving Gaussians, enabling the synthesis of vivid dynamic scenes that outperform previous baselines in both qualitative and quantitative metrics.

Beyond general scene dynamics, a critical subset of text-driven synthesis involves the generation of grounded 4D content where user control over geometry and motion is paramount. [49] addresses the limitations of trial-and-error prompt engineering by decomposing the 4D generation task. This framework utilizes static 3D assets and monocular video sequences as key components, allowing users to specify geometry and motion separately. By constructing the 4D representation using dynamic 3D Gaussians, [49] facilitates efficient, high-resolution supervision through rendering, ensuring that the generated content faithfully reconstructs input signals while inferring realistic renderings from novel viewpoints and timesteps. Furthermore, to address the issue of motion scaling and trajectory control, [377] factors motion into global and local components. It represents global motion via rigid transformations along spline-parameterized trajectories and learns local deformations supervised by text-to-video models. This factorization enables the synthesis of scenes animated along arbitrary trajectories, significantly enhancing the realism and extent of generated motion compared to methods limited by fixed bounding boxes.

The synthesis of human-object interactions presents unique challenges due to the need for physical plausibility and precise contact modeling, moving beyond simple object placement to dynamic interplay. Generating synchronized human and object motions from text requires understanding the interdependent nature of these entities. [378] pioneers this space by explicitly modeling the contact between the human body surface and object geometry as a strong proxy guidance. By learning human motion, object motion, and contact in a joint diffusion process inter-correlated through cross-attention, [378] ensures that the generated interaction sequences are physically plausible and coherent. This contact-guided approach allows for the generation of corresponding human motion conditioned on a given object trajectory without re-training, demonstrating robust learning of human-object interdependency.

Expanding on the concept of zero-shot generation, [379] tackles the scarcity of large-scale interaction data by decoupling interaction semantics from dynamics. This framework synergizes knowledge from large language models and text-to-motion models to handle high-level semantics, while employing a dedicated world model to comprehend simple physics and low-level dynamics. This modular design enables [379] to generate text-aligned 3D HOI sequences without direct training on text-interaction pair data, showcasing a powerful capability for generalizing to unseen interaction types. Additionally, [380] adopts a modular design with a dual-branch diffusion model to generate human and object motions simultaneously, enhanced by an affordance prediction model that corrects potential errors in contact areas. This stochastic generation of contacting points diversifies the generated motions, ensuring accurate and close contact between humans and objects.

Compositional approaches have also emerged as a vital strategy for creating complex dynamic scenes involving multiple interacting entities, echoing the layout planning principles seen in static generation but extended to the temporal domain. [381] utilizes Large Language Models to decompose input text prompts into distinct entities and map their trajectories, constructing the scene by positioning objects along designated paths. This method employs compositional score distillation guided by pre-defined trajectories, achieving superior visual quality and motion fidelity. Similarly, [370] leverages the spatial reasoning capabilities of LLMs to derive semantic primitives and their relationships directly from text. By modeling collision relationships at a global level, [370] enhances physical correctness, addressing the training instability often seen in simple blending approaches.

Finally, the integration of motion priors is essential for ensuring temporal consistency and naturalness in generated sequences, laying the groundwork for the interactive manipulation techniques discussed next. [382] decouples motion and appearance to achieve superior video-to-4D generation, introducing adaptive Gaussian initialization and alignment losses to mitigate shape degeneration. This disentanglement facilitates novel applications such as transferring learned motion onto diverse 4D entities based on textual descriptions. Moreover, [383] introduces Canonical Score Distillation to generate non-rigid objects on skeletons extracted from monocular videos, enhancing the authenticity of bones and skinning through inductive priors from diffusion models. These collective advancements signify a maturing field where text-driven synthesis is evolving from simple object animation to the creation of complex, physically grounded, and semantically rich dynamic worlds, setting the stage for robust interactive editing and scene completion in semantic fields.

7.4 Interactive Editing, Inpainting, and Scene Completion in Semantic Fields

7.4 Interactive Editing, Inpainting, and Scene Completion in Semantic Fields

Building upon the text-driven synthesis of complex dynamic scenes and human-object interactions discussed in the previous section, the frontier of neural rendering has pivoted towards empowering users with fine-grained control over these generated or reconstructed environments. While Section 7.3 established methods for creating 4D content from abstract prompts, this section addresses the critical subsequent challenge: interactive scene manipulation. The core objective here is to enable object-level operations—such as deletion, insertion, style transfer, and inpainting—while rigorously maintaining multi-view consistency and semantic coherence. Recent advancements have shifted from purely generative pipelines to robust editing frameworks that leverage semantic distillation and foundation models, ensuring that modifications are not merely superficial texture swaps but structurally and semantically plausible alterations to the underlying 3D representation.

A primary focus within this domain is interactive object-level editing in dynamic scenes, which addresses the limitations of static editing pipelines when applied to time-variant data. Traditional methods often fail to preserve temporal consistency when modifying dynamic sequences. To overcome this, researchers have developed frameworks like [384], which enables interactive editing within dynamic Neural Radiance Fields by incorporating hybrid semantic feature distillation. This approach ensures that spatial-temporal consistency is preserved even after complex operations such as recoloring, transformation, or composition are applied based on user strokes on a single frame. Furthermore, the integration of recursive selection refinement enhances the accuracy of object segmentation within these dynamic fields, a critical step for isolating targets during the editing process. To address the inevitable holes created by removing objects or changing viewpoints, advanced techniques employ multi-view reprojection inpainting, effectively filling missing regions by utilizing information from surrounding frames and views.

Beyond dynamic NeRFs, the application of these principles to explicit representations and complex urban environments highlights the versatility of semantic-guided manipulation. In scenarios involving cluttered urban scenes, conventional inpainting techniques often fail to generate reliable boundaries due to the ambiguity of multiple overlapping semantics. Novel deep learning models have been proposed to manipulate these scenes by removing user-specified portions and coherently inserting new objects, such as cars or pedestrians. These methods leverage semantic segmentation to model both the content and structure of the image, guiding the generation of new objects to ensure they are semantically consistent with the surrounding environment [385]. By designing decoder blocks that combine segmentation and generation tasks, these systems achieve superior performance in reconstructing complex structures compared to multi-stage approaches that rely solely on contour information.

Scene completion, particularly in the context of filling occluded regions, has also benefited immensely from the integration of multi-modal cues and semantic priors. The task of Semantic Scene Completion (SSC) aims to predict complete 3D geometry and semantics from limited observations, a challenge exacerbated in sparse or occluded environments. Innovative frameworks have emerged that utilize local Deep Implicit Functions to produce continuous scene representations without the resolution constraints of voxelization. By encoding raw point clouds into a latent space at multiple spatial resolutions, these methods can assemble a global scene completion function from localized patches, effectively handling extensive outdoor scenes with high geometric fidelity [386]. Moreover, the synergy between 2D segmentation and 3D completion has been exploited to improve performance in data-scarce regimes. Approaches like [387] seamlessly fuse structural depth data with semantic priors derived from 2D segmentation networks, utilizing novel 3D data augmentation strategies to enhance generalization in the absence of large-scale labeled 3D datasets.

The interplay between instance-level understanding and scene-level completion further refines the quality of inpainted regions. Methods such as [388] decouple instances from a coarsely completed semantic scene to guide the reconstruction of fine-grained shape details. Through an iterative mechanism of scene-to-instance and instance-to-scene completion, these frameworks effectively integrate fine-grained instance information back into the 3D volume, significantly improving accuracy in regions where semantic categories are easily confused. Similarly, for indoor environments, dual-stream Generative Adversarial Networks (GANs) have been tailored for free-form 3D scene inpainting. Unlike traditional completion tasks that deal with sensor limitations, these methods address large, diverse missing regions mimicking human drawing trajectories. By fusing geometry and color information in a dual-stream generator, these models produce distinct semantic boundaries and resolve interpolation issues that plague simpler context-based methods [389].

Style transfer within these semantic fields adds another layer of complexity, requiring the preservation of object identity while altering appearance attributes. Language-guided approaches have enabled users to modify 3D indoor scenes using natural language phrases, mapping vertex coordinates to RGB residues and optimizing them via vision-language models to achieve visually pleasing results consistent with human priors [390]. Additionally, localized style transfer techniques allow for the stylization of specific target objects based on textual descriptions without affecting the background, utilizing large language models to parse stylization goals and localized loss functions to constrain the editing region [391]. In the realm of 2D-assisted 3D editing, methods have also been developed to remove objects from 3D scenes guided by multiview segmentation, spreading user annotations across views to ensure 3D consistency before applying depth supervision and perceptual losses for high-quality inpainting [392].

Finally, the robustness of these editing and completion systems is often bolstered by addressing occlusion and visibility explicitly. Systems that decompose scenes layer-by-layer can infer underlying occlusion relationships and automatically learn which parts of objects need completion, interleaving instance segmentation and scene completion tasks to disentangle complex occluded relationships [393]. Furthermore, bounding box-based detectors have been utilized to recover depth ordering and detect occlusions without extensive training, providing a faster alternative for ensuring semantic consistency in edited scenes [394]. Collectively, these advancements demonstrate a clear trajectory towards highly interactive, semantically aware, and geometrically consistent editing pipelines. However, the efficacy of such interactive manipulation is fundamentally dependent on the quality of the semantic masks used to isolate objects. This dependency sets the stage for the next section, which explores a parallel revolution in generating these prerequisites: advanced, training-free segmentation paradigms that leverage the implicit knowledge of foundation models to achieve fine-grained, part-level understanding.

7.5 Advanced Segmentation Paradigms: Part-Level and Training-Free Queries

7.5 Advanced Segmentation Paradigms: Part-Level and Training-Free Queries

Building upon the semantically aware editing and scene completion frameworks discussed in the previous section, the field has witnessed a parallel revolution in how semantic masks—the fundamental prerequisites for such manipulations—are generated. The intersection of generative AI and 3D perception has catalyzed a paradigm shift, moving away from supervised learning on closed vocabularies toward training-free, open-world capabilities. This evolution is particularly evident in the transition from coarse object-level labeling to fine-grained part-level segmentation, where the inherent knowledge embedded within pre-trained diffusion models is leveraged to dissect scenes without the need for expensive manual annotations. Traditional segmentation frameworks often struggle with the scarcity of pixel-level or point-level labels, especially for rare categories or complex articulated parts. However, recent advancements demonstrate that text-to-image diffusion models, trained on massive internet-scale datasets, implicitly encode rich spatial and semantic correspondences that can be unlocked for discriminative tasks through novel attention mechanisms and optimization strategies.

A cornerstone of this new paradigm is the utilization of attention maps within diffusion architectures to derive segmentation masks without additional training. The insight driving these methods is that to generate photorealistic images faithful to text prompts, diffusion models must learn precise object shapes and their semantic locations. Research has shown that self-attention maps effectively characterize object boundaries and shapes, while cross-attention maps indicate semantic alignment with text tokens [395; 396]. By aggregating these attention signals across different layers and time steps of the denoising process, it is possible to construct high-fidelity segmentation masks for arbitrary concepts described in natural language. For instance, training-free approaches have been developed that extract these attention maps to perform open-vocabulary segmentation, achieving competitive performance against supervised baselines by simply exploiting the model's internal generation process [397]. These methods often involve refining textual prompts or optimizing token embeddings to enhance the clarity of the resulting attention maps, allowing for the segmentation of objects that were never explicitly seen during any segmentation-specific training phase [398; 399].

Extending these 2D capabilities to the 3D domain introduces unique challenges regarding view consistency and geometric coherence, yet the principles of leveraging foundation models remain robust. In the context of 3D point clouds and meshes, advanced paradigms are emerging that integrate diffusion-derived features with geometric reasoning to achieve part-level segmentation. One significant direction involves the distillation of 2D vision-language features into 3D representations. By lifting features from models like CLIP or segmentation-aware diffusion models into 3D Gaussian fields or point clouds, systems can perform zero-shot semantic segmentation and panoptic understanding [400]. This allows users to query a 3D scene with diverse, free-form prompts to retrieve specific objects or parts, such as "the wooden leg of the chair" or "the red handle of the tool," effectively bridging the gap between high-level language and low-level 3D geometry and providing the precise masks required for the interactive editing pipelines described previously.

Furthermore, the capability for reasoning-based part segmentation represents a frontier in interactive 3D understanding. Recent works have introduced tasks where models must segment 3D object parts based on implicit textual queries that require world knowledge and logical reasoning, rather than explicit category names [401]. This involves training or prompting large multimodal models to comprehend complex instructions and output corresponding segmentation masks, thereby enabling more intuitive human-robot interaction and content editing. In scenarios where labeled data is completely absent, unsupervised object discovery frameworks utilize diffusion models to synthesize data or extract saliency maps, effectively acting as a teacher for downstream segmentation networks [402]. These approaches often employ techniques like "AttentionCut" or inversion methods to map real images back into the diffusion feature space, bridging the structural gap between generative and discriminative models [402].

The versatility of these training-free paradigms is further enhanced by their ability to handle variable granularities and complex scenes. Methods have been proposed that allow for interactive segmentation in 3D point clouds, where users can iteratively refine masks by clicking on objects of interest, with the system adapting in real-time using pre-trained priors [403]. Additionally, the integration of layout planning and scene graph guidance enables the decomposition of complex text prompts into editable 3D layouts, facilitating multi-object scene generation with precise spatial relationships [404]. By leveraging bounded attention mechanisms, these systems can prevent semantic leakage between similar subjects, ensuring that each object instance is distinctly segmented even in crowded environments [405].

Moreover, the application of diffusion models extends to challenging domains such as camouflaged object segmentation and medical imaging, where visual cues are subtle. By learning multi-scale textual-visual features, these models can distinguish foreground objects from backgrounds that share similar textures or colors, a capability that is crucial for robust 3D reconstruction in uncontrolled environments [406]. The use of prototype memory banks and contrastive learning strategies further enhances the efficiency of these models in weakly-supervised settings, allowing them to generalize from extremely limited annotations [407].

In summary, the emergence of advanced segmentation paradigms utilizing diffusion attention maps marks a transformative period in 3D computer vision. By harnessing the implicit geometric and semantic knowledge of pre-trained generative models, researchers are developing systems that offer fine-grained, part-level understanding and training-free open-world querying. These approaches not only reduce the dependency on large-scale annotated datasets but also provide the critical semantic infrastructure necessary for the interactive editing, autonomous navigation, and semantic-rich 3D content creation discussed throughout this survey, paving the way for more intelligent and adaptable virtual environments.

8 Emerging Frontiers: Neuromorphic Computing, Physics-Informed Modeling, and Future Directions

8.1 Neuromorphic Perception and Event-Based Dynamics for Robust Sensing

8.1 Neuromorphic Perception and Event-Based Dynamics for Robust Sensing

The rapid advancement of dynamic 3D scene reconstruction, particularly through explicit primitives like 3D Gaussian Splatting, has traditionally relied on standard frame-based cameras. While these sensors provide rich photometric data, they are fundamentally limited by motion blur, rolling shutter artifacts, and fixed temporal resolution, which severely degrade reconstruction quality in high-speed scenarios. These limitations create a critical bottleneck: without precise, high-frequency motion cues, the deformation fields governing dynamic Gaussians often suffer from instability and artifact propagation. To overcome these physical constraints, the integration of neuromorphic perception—specifically the synergy between event-based cameras and Spiking Neural Networks (SNNs)—emerges as a vital precursor to robust modeling. This paradigm shift offers a bio-inspired solution to capture the high-frequency dynamics that conventional sensors miss, providing the dense, temporally precise motion priors necessary to stabilize and guide the subsequent physics-informed simulations discussed in the following section.

Event-based cameras operate asynchronously, emitting spikes only when pixel-wise luminance changes exceed a threshold, thereby offering microsecond-level latency, high dynamic range, and negligible motion blur. However, processing this sparse, irregular data stream requires architectures that diverge from standard Convolutional Neural Networks (CNNs). Spiking Neural Networks, with their event-driven computation and intrinsic temporal memory, serve as the ideal computational counterpart. Recent research demonstrates that SNNs can effectively extract spatio-temporal features from raw event streams, outperforming traditional Analog Neural Networks (ANNs) in tasks requiring fine-grained temporal resolution. For instance, hybrid architectures that combine the asynchronous processing capabilities of SNN layers with the training stability of ANN layers have shown significant improvements in optical flow estimation, achieving substantial reductions in end-point error while lowering energy consumption [408]. Such hybrid models are particularly relevant for dynamic scene reconstruction, where accurate flow fields are a prerequisite for initializing and updating the positions of moving Gaussian primitives, laying the groundwork for more complex trajectory predictions.

A primary challenge in deploying SNNs for reconstruction tasks is the “spike vanishing” phenomenon in deep networks, which hinders the propagation of temporal gradients. Innovative solutions have been proposed to address this, such as adaptive neuronal dynamics and learnable leakage parameters that preserve signal integrity across deep layers. The [409] framework exemplifies this by utilizing surrogate gradient-based backpropagation to train deep SNNs from scratch, resulting in models that consistently outperform similarly sized ANNs in optical flow accuracy while offering dramatic reductions in computational energy and parameter count. These efficiency gains are crucial for real-time applications, suggesting that SNN-based motion estimators could run on edge devices to provide immediate, low-latency feedback for Gaussian densification and pruning, ensuring the geometric fidelity required before applying physical constraints.

Beyond pure optical flow, the ability of SNNs to perform direct regression on asynchronous inputs opens new avenues for handling complex camera motions and rolling shutter distortions. Traditional methods often require synchronizing events into frames, which reintroduces latency and information loss. In contrast, frameworks like [410] demonstrate the feasibility of predicting continuous 3-DOF angular velocities directly from irregular event streams without pre-processing. This capability is vital for correcting rolling shutter artifacts in high-speed captures, a common failure mode for standard Structure-from-Motion (SfM) pipelines used to initialize Gaussian Splatting. By integrating such direct regression modules, future reconstruction systems can self-correct pose estimates in real-time, ensuring the geometric consistency necessary for the stable application of physical laws in downstream modeling.

Furthermore, the integration of sensor fusion strategies enhances robustness in challenging lighting conditions where event cameras alone may struggle with sparse data density. Approaches like [411] leverage the complementary nature of frame and event data, using SNNs to process asynchronous events and ANNs for static frames. This dual-stream processing ensures dense motion coverage, which is essential for reconstructing texture-less regions or handling occlusions in dynamic scenes. Similarly, [412] highlights the potential of treating flow estimation as a sequential problem using recurrent SNNs, achieving temporally dense predictions at frequencies far exceeding standard video rates. This high temporal bandwidth allows for the precise interpolation of Gaussian states between sparse camera exposures, effectively smoothing the trajectory of dynamic objects and reducing jitter in the reconstructed 4D field, thus providing a cleaner state representation for physical simulation.

The hardware efficiency of SNNs further accelerates their adoption in robotic and embodied AI contexts, where power budgets are strict. Studies such as [413] and [414] illustrate that principled model simplification and hardware-aware design can yield real-time optical flow pipelines with orders-of-magnitude reduced complexity. These efficient backbones can be embedded directly into robotic platforms to support Simultaneous Localization and Mapping (SLAM) systems enhanced by Gaussian Splatting. For example, [415] validates the utility of SNNs in reactive control loops, demonstrating reliable collision avoidance through event-driven processing. Extending this to 3D reconstruction implies that robots could build and update dense Gaussian maps of dynamic environments on-the-fly, reacting to moving obstacles with minimal latency and generating the high-quality motion data required for physics-aware modeling.

Finally, emerging techniques in unsupervised learning and attention mechanisms within SNNs promise to reduce the dependency on labeled ground truth, which is scarce for high-speed dynamic events. Unsupervised hierarchical architectures [416] and attention-enhanced SNNs [417] enable the network to focus on salient motion patterns and learn robust representations without extensive supervision. As the field moves toward integrating these neuromorphic perception modules with explicit 3D representations, the convergence of event-based sensing and Gaussian Splatting stands to revolutionize dynamic scene understanding. By securing robust, high-fidelity motion priors through neuromorphic sensing, we establish a solid foundation for the next evolutionary step: embedding fundamental physical laws directly into the dynamics of these primitives to ensure long-term consistency and generalizability.

8.2 Physics-Informed Gaussian Dynamics and Latent Space Simulation

8.2 Physics-Informed Gaussian Dynamics and Latent Space Simulation

Building upon the robust, high-frequency motion priors established by neuromorphic perception in the previous section, the next critical evolutionary step is to ensure that the trajectories derived from these sensors adhere to fundamental physical laws. While event-based sensing provides the precise temporal data necessary to initialize dynamic Gaussians, purely data-driven deformation fields often fail to generalize to unseen physical conditions, leading to artifacts such as inter-penetration, energy drift, and unstable long-term predictions. To address this, emerging frameworks are shifting from black-box kinematic models to physics-informed neural representations. This subsection examines the convergence of Graph Neural Ordinary Differential Equations (Graph Neural ODEs) with deformable Gaussian fields, creating a paradigm where the motion of Gaussian primitives is governed by conservation laws rather than mere visual interpolation.

The core challenge in modeling dynamic scenes lies in treating Gaussian primitives not as independent entities, but as nodes in a interacting graph that respects physical symmetries. Traditional approaches often lack the inductive biases required to preserve invariants like momentum and energy. Recent advancements propose coupling Graph Neural Networks (GNNs) with Neural ODEs to create continuous-time dynamics models that inherently respect these constraints. For instance, frameworks like “Learning the Dynamics of Particle-based Systems with Lagrangian Graph Neural Networks” demonstrate that learning the Lagrangian of a system directly from trajectories provides a strong inductive bias, significantly outperforming baseline methods in systems with constraints and drag while offering zero-shot generalizability to larger system sizes [418]. In the context of Gaussian Splatting, where splats act as particles, this approach ensures their trajectories remain physically plausible under external forces, effectively bridging the gap between the high-fidelity motion cues provided by neuromorphic sensors and stable geometric evolution.

Furthermore, the incorporation of symmetry and equivariance into these models is crucial for robust generalization across varying viewpoints and object orientations. Physical laws are invariant to translations and rotations, yet many neural architectures struggle to enforce these properties strictly. The development of “Subequivariant Graph Neural Networks” addresses the challenge of modeling systems where symmetry is partially broken by external fields, such as gravity, by relaxing strict equivariance to subequivariance. This allows the model to learn physical interactions between objects of diverse shapes and sizes while maintaining theoretical guarantees of universal approximation [419]. When applied to dynamic Gaussian fields, such architectures ensure that scene deformation remains consistent regardless of the camera perspective, a critical requirement for the autonomous navigation and robotic applications hinted at in the preceding discussion on neuromorphic SLAM.

In addition to spatial interactions, the temporal evolution of dynamic scenes must be modeled with high fidelity to prevent error accumulation over long horizons, a necessity for leveraging the dense temporal data from event cameras. Neural ODEs provide a natural framework for this by defining the state derivative as a neural network, allowing for continuous integration over time. However, standard Neural ODEs can suffer from instability when modeling stiff systems or those with long-range dependencies. Innovations such as "Modulated Neural ODEs" introduce time-invariant modulator variables to separate static factors of variation from dynamic states, improving the ability to generalize to new dynamic parameterizations and perform far-horizon forecasting [420]. Similarly, "Anamnesic Neural Differential Equations with Orthogonal Polynomial Projections" propose modeling the latent continuous-time process as a projection onto a basis of orthogonal polynomials, enforcing long-range memory and preserving a global representation of the underlying dynamical system [421]. These techniques are essential for deformable Gaussian fields, where the history of deformation influences future states, preventing unrealistic collapse or expansion of the scene geometry.

The explicit enforcement of conservation laws serves as another cornerstone of physics-informed Gaussian dynamics, offering superior data efficiency and reliability compared to implicit learning. "Variational Integrator Graph Networks" unify energy constraints with high-order symplectic variational integrators and graph neural networks, demonstrating that combining these inductive biases leads to coupled learning of generalized position and momentum updates, which is vital for accurate long-term prediction [422]. Moreover, "Exact conservation laws for neural network integrators of dynamical systems" utilize Noether's Theorem to inherently incorporate conservation laws into the network architecture, ensuring that quantities like energy and momentum are preserved exactly rather than approximately [423]. In the context of Gaussian Splatting, embedding such constraints ensures that the total energy of the splat field remains bounded and that momentum transfer during collisions is handled realistically, preventing the "floating" or "sinking" artifacts common in purely geometric reconstruction methods.

For systems involving stochastic dynamics or non-conservative forces, such as fluids or soft bodies with friction, more specialized frameworks are required to complement the deterministic models discussed above. "Graph Neural Stochastic Differential Equations for Learning Brownian Dynamics" combine SDEs with GNNs to learn Brownian dynamics directly from trajectories, theoretically proving the conservation of linear momentum and demonstrating superior performance in systems at finite temperatures [424]. This is particularly applicable to reconstructing participating media or fluid-like phenomena using Gaussian primitives, where thermal noise and random fluctuations play a significant role. Additionally, "Thermodynamics-informed graph neural networks" extend the Hamiltonian formalism to the GENERIC structure to model dissipative dynamics, achieving high accuracy in both Eulerian and Lagrangian descriptions of fluid and solid mechanics [425].

Finally, the scalability of these physics-informed models is critical for real-world deployment, especially when handling the millions of primitives required for high-fidelity reconstruction. "Multipole Graph Neural Operator for Parametric Partial Differential Equations" introduces a multi-level graph neural network framework that captures interactions at all ranges with linear complexity, addressing the computational bottlenecks associated with long-range dependencies in large particle systems [426]. By integrating such efficient operators with Gaussian Splatting, it becomes feasible to simulate complex, large-scale dynamic scenes while maintaining physical consistency. The synergy between these advanced neural ODE and GNN frameworks and the explicit nature of Gaussian Splatting paves the way for a new generation of dynamic 3D reconstruction systems that are not only visually compelling but also physically grounded. However, realizing the full potential of these systems requires more than just architectural innovations; it necessitates a fundamental shift in how these models are trained. As we move toward deploying these physically consistent models on edge devices, the limitations of traditional backpropagation become apparent, setting the stage for the biologically plausible learning mechanisms and adaptive topologies discussed in the following section.

8.3 Biologically Plausible Learning Mechanisms and Adaptive Topologies

8.3 Biologically Plausible Learning Mechanisms and Adaptive Topologies

While the previous subsection established the importance of embedding physical laws into Gaussian dynamics to ensure mechanical consistency, a complementary challenge lies in the learning mechanisms that drive these systems. The trajectory of dynamic 3D scene reconstruction is increasingly converging with neuromorphic computing principles, necessitating a fundamental shift from static, offline backpropagation to dynamic, brain-inspired learning paradigms. Traditional deep learning relies heavily on gradient descent algorithms that require non-local information transmission and the storage of extensive activation histories, rendering them incompatible with the constraints of real-time, energy-efficient edge devices—a limitation that becomes acute when scaling to the millions of primitives required for high-fidelity Gaussian Splatting. In contrast, biologically plausible learning mechanisms, such as Spike-Timing-Dependent Plasticity (STDP), offer a pathway to online adaptation where synaptic weights are updated based solely on local pre- and post-synaptic activity. This transition is critical for enabling autonomous systems to learn and adapt to changing environments without the prohibitive memory and energy costs associated with Back-Propagation Through Time (BPTT). The core challenge lies in bridging the performance gap between these local, event-driven rules and the global optimization capabilities of standard deep learning, a gap that recent research has begun to close through novel architectural designs and hybrid learning strategies.

A primary focus of this paradigm shift is the refinement of STDP to handle complex temporal patterns and supervised tasks essential for distinguishing dynamic foreground objects from static backgrounds. While classical STDP is inherently unsupervised and struggles with deep network architectures, recent advancements have introduced supervised variants that maintain biological plausibility. For instance, the Stabilized Supervised STDP (S2-STDP) rule integrates error-modulated weight updates that align neuron spikes with desired timestamps, effectively guiding the learning process without requiring non-local gradient propagation [427]. Furthermore, the introduction of architectures like Paired Competing Neurons (PCN) enhances this by fostering intra-class competition, which improves classification accuracy and hyperparameter robustness on standard benchmarks [427]. These developments suggest that dynamic 3D reconstruction models could eventually employ similar competitive mechanisms to efficiently segregate and track moving entities within a scene in real-time.

Beyond simple plasticity rules, the concept of adaptive topologies represents a significant frontier that directly addresses the need for flexible scene representation. Biological neural systems do not possess fixed connectivity; they constantly rewire themselves to optimize resource allocation. Implementing such structural plasticity in hardware allows networks to evolve their connectivity patterns during training, optimizing for sparsity and efficiency. Research on analog neuromorphic systems has demonstrated strategies where pre- and post-synaptic partners are constantly rewired while maintaining a constant neuronal fan-in, effectively optimizing the network topology with respect to the training data [428]. This capability is particularly relevant for Gaussian Splatting scenarios where the density of primitives needs to adapt dynamically to scene complexity. By mimicking these biological processes, systems can achieve a level of flexibility that static architectures cannot, allowing for the efficient representation of sparse or rapidly changing 3D environments through dynamic densification and pruning of Gaussians.

The integration of three-factor learning rules further enhances the viability of online learning on neuromorphic hardware, providing the necessary bridge between local plasticity and global objectives. Unlike traditional two-factor STDP, which depends only on spike timing, three-factor rules incorporate a global modulatory signal, such as a reward or error estimate, to gate synaptic updates. The Event-based Three-factor Local Plasticity (ETLP) rule exemplifies this by using pre-synaptic traces, post-synaptic voltages, and projected labels as update triggers, achieving competitive accuracy with significantly reduced computational complexity compared to BPTT [429]. Similarly, the Online Training Through Time (OTTT) algorithm provides a theoretical bridge between BPTT and online learning, deriving update rules that resemble three-factor Hebbian learning while requiring only constant memory costs regardless of sequence length [430]. These methods are pivotal for deploying reconstruction algorithms on resource-constrained devices where storing long temporal histories is infeasible, thus enabling continuous map updates in unbounded environments.

Moreover, the robustness of these biologically inspired systems to hardware imperfections and noise offers a distinct advantage for embedded deployment. Neuromorphic substrates often suffer from device mismatch and stochastic behavior, which typically degrade the performance of conventional algorithms. However, biologically plausible learning rules can inherently compensate for these variabilities. For example, on-chip learning has been shown to mitigate the effects of fixed-pattern noise by adapting synaptic weights to match the specific excitability of individual neurons, effectively turning hardware non-idealities into features for exploration [431]. Additionally, the incorporation of astrocyte-inspired mechanisms has been proposed to enable self-repair in SNNs, allowing networks to maintain functionality even when a significant percentage of neurons or synapses fail [432]. This fault tolerance is essential for long-term deployment of autonomous agents in unstructured environments, ensuring that perception systems remain operational despite sensor degradation or computational faults.

Finally, the move towards hybrid learning frameworks combines the strengths of global error-driven learning with local, neuroscience-oriented plasticity, setting the stage for the embodied intelligence discussed in the following section. Brain-inspired meta-learning paradigms have been developed to meta-learn local plasticity rules while receiving top-down supervision, enabling multiscale synergic learning that excels in few-shot and continual learning scenarios [433]. Such approaches allow systems to rapidly adapt to new scenes or objects with minimal data, a capability that mirrors human perception. As these mechanisms mature, they promise to revolutionize dynamic 3D reconstruction by enabling systems that learn continuously, adapt their topologies on the fly, and operate with the energy efficiency of biological brains. This evolution from fixed, offline training pipelines to adaptive, online learning forms the foundational algorithmic layer required for the next generation of neuromorphic SLAM and autonomous navigation systems.

8.4 Embodied Intelligence, Autonomous Navigation, and Neuromorphic SLAM

8.4 Embodied Intelligence and Neuromorphic SLAM with Dense Representations

Building upon the biologically plausible learning mechanisms and adaptive topologies discussed previously, the convergence of embodied intelligence, autonomous navigation, and neuromorphic computing represents a critical paradigm shift in how robotic systems perceive and interact with dynamic environments. While Section 8.3 established the algorithmic foundations for online, energy-efficient learning through local plasticity rules and structural rewiring, this subsection explores their practical deployment in mobile agents transitioning from structured industrial settings to unstructured, real-world scenarios. The demand for perception systems that are simultaneously low-latency, robust to sensor imperfections, and computationally frugal has never been more acute. Traditional frame-based vision systems coupled with deep neural networks often falter under the high computational overhead required for real-time dense mapping and localization on resource-constrained edge platforms. In this context, Spiking Neural Networks (SNNs) emerge as the natural hardware-software co-design solution, mimicking the event-driven processing of the biological brain to enable the continual learning and ultra-low power consumption necessitated by the adaptive strategies outlined earlier. The deployment of such spiking systems for real-time robot navigation has already yielded significant successes, particularly in visual place recognition (VPR) and obstacle avoidance, laying the essential groundwork for a new generation of neuromorphic SLAM systems that bridge the gap between theoretical plasticity and physical embodiment.

A primary application where these bio-inspired principles manifest is Visual Place Recognition, which serves as the backbone for loop closure detection in SLAM pipelines. Recent advancements have demonstrated that SNNs can achieve performance comparable to state-of-the-art convolutional networks while operating at fractions of the energy cost and latency, directly leveraging the efficiency gains promised by local learning rules. For instance, the introduction of temporally encoded SNNs has allowed for training within minutes and inference in milliseconds, making them highly suitable for deployment on compute-constrained robotic systems [434]. By utilizing temporal coding schemes that determine spike timing based on pixel intensity rather than rate coding, these systems have achieved spike efficiency improvements of over 100%, enabling inference speeds exceeding 50 Hz on standard CPUs [434]. Furthermore, addressing the scalability challenges inherent in dynamic environments, modular ensemble approaches have been developed where compact, region-specific spiking networks collectively manage large-scale contexts. This effectively overcomes the computational infeasibility faced by previous single-network SNN implementations on extensive datasets [435]. These modular systems incorporate regularization techniques to suppress hyperactive neurons, ensuring robust place recognition even as the map size grows sub-linearly, a direct reflection of the adaptive resource allocation strategies discussed in the context of structural plasticity [436].

Beyond place recognition, neuromorphic principles are being integrated directly into navigation controllers and mapping strategies, extending the concept of adaptive topologies to physical movement. Bio-inspired architectures leveraging Spike-Timing-Dependent Plasticity (STDP) have enabled robots to learn and map uncharted territories without initial all-to-all connectivity, demonstrating robustness even under partial sensor failure [437]. This capability is crucial for embodied intelligence, where agents must adapt to unknown environments dynamically, mirroring the self-repair and fault-tolerance mechanisms highlighted in prior sections. The integration of sequence matching and ensemble methods in SNN-based VPR has further enhanced the responsiveness and accuracy of localization tasks, providing a viable path toward energy-sensitive robotic applications [438]. Moreover, the synergy between place categorization and specific place recognition has been exploited to create hybrid systems that use semantic context to inform navigation, significantly improving robustness against local environmental changes such as lighting variations or transitions between indoor and outdoor spaces [439].

While current neuromorphic SLAM systems have largely relied on sparse representations or topological graphs, the integration of dense, explicit scene representations like 3D Gaussian Splatting (3DGS) offers a compelling frontier that unifies the efficiency of spiking computation with high-fidelity geometric modeling. 3DGS has revolutionized neural rendering by enabling real-time, high-fidelity reconstruction through differentiable rasterization, yet its computational demands for training and optimization remain a bottleneck for embedded deployment. The potential synergy lies in combining the event-driven efficiency of SNNs—optimized via the local learning rules described in Section 8.3—for feature extraction and tracking with the geometric richness of Gaussian Splatting for dense semantic mapping. Recent works have already begun to explore Gaussian Splatting for navigation, demonstrating that 3DGS maps can support safe, real-time path planning and robust state estimation by interpreting the splats as a point cloud [440]. Similarly, frameworks like GaussNav have shown that 3DGS representations retain textural features essential for instance image-goal navigation, outperforming traditional Bird's Eye View maps [441].

The future trajectory of neuromorphic SLAM involves tightly coupling these dense Gaussian representations with spiking processing pipelines to realize the full potential of the adaptive, online learning frameworks previously discussed. One promising direction is the use of event cameras, which naturally align with SNN architectures, to drive the incremental update of Gaussian maps. This would allow for continuous, low-latency mapping where only changing regions of the scene trigger updates to the Gaussian parameters, drastically reducing energy consumption compared to frame-based optimization and embodying the "event-based three-factor" learning principles. Additionally, the semantic capabilities of modern Gaussian Splatting systems, which distill foundation model features into 3D fields for open-vocabulary understanding [442], could be enhanced by neuromorphic front-ends that pre-process visual streams for salient object detection. Systems like SemanticGaussians and SGS-SLAM have already established the feasibility of semantic Gaussian mapping [358; 326], but integrating spiking neural networks could enable these systems to operate perpetually on battery-powered agents. By leveraging the temporal precision of spikes to guide the densification and pruning of Gaussians, future neuromorphic SLAM systems could achieve a level of autonomy and efficiency that bridges the gap between biological navigation strategies and high-fidelity synthetic perception. This evolution sets the stage for the next logical leap: moving from efficient perception and mapping to higher-order cognitive functions, where neuro-symbolic convergence and generative world modeling will enable agents to not only navigate but also reason about the dynamic worlds they inhabit, as explored in the following section.

8.5 Neuro-Symbolic Convergence and Generative World Modeling

8.5 Neuro-Symbolic Convergence and Generative World Modeling

Building upon the efficient, event-driven perception and dense geometric mapping enabled by neuromorphic SLAM and Gaussian Splatting, the frontier of dynamic 3D scene reconstruction is now expanding toward a synergistic integration of neuro-symbolic artificial intelligence. While the previous paradigms excel at low-latency sensing and high-fidelity rendering, purely data-driven neural approaches often struggle with long-horizon planning, causal reasoning, and strict adherence to physical laws in complex, changing environments. The emerging paradigm of neuro-symbolic convergence seeks to bridge this gap by combining the perceptual richness of deep learning with the logical rigor, interpretability, and physical consistency of symbolic reasoning. By integrating foundation model priors with intuitive physics prediction, this approach aims to achieve reliable decision-making and the generation of physically consistent 4D scenarios that transcend simple pattern matching.

A central challenge in this domain is enabling agents to perform "mental simulation" to predict future states of dynamic environments, a capability critical for embodied AI and autonomous navigation. Recent research suggests that the neural mechanisms underlying human mental simulation are best modeled by systems that predict future states within the latent space of pretrained foundation models optimized for dynamic scenes [443]. By embedding symbolic constraints into these latent spaces, neuro-symbolic frameworks can guide the generative process to ensure that predicted trajectories are not only visually plausible but also physically grounded. This allows models to reason about the consequences of actions and interactions among multiple agents. For instance, frameworks that decompose scenes into object-centric representations can leverage symbolic rules to manage interactions, ensuring that predictions respect conservation laws and collision constraints, thereby enhancing the reliability of long-term forecasting [444].

The integration of symbolic knowledge is particularly vital for addressing the issue of reasoning shortcuts, where purely neural models might achieve high accuracy by exploiting spurious correlations rather than learning true causal structures. Such shortcuts can lead to catastrophic failures when models encounter out-of-distribution scenarios or novel physical configurations. Neuro-symbolic predictors mitigate this risk by mapping sub-symbolic inputs to higher-level concepts and performing logical inference on these intermediate representations, thus enforcing consistency with prior knowledge [445]. However, identifying and mitigating these shortcuts remains an active area of research, as standard mitigation strategies like reconstruction losses or concept supervision have inherent limitations. Advanced strategies, such as concept-level continual learning, are being developed to ensure that the semantics of acquired concepts remain stable over time, preventing the erosion of symbolic knowledge during adaptation to new tasks [446].

In the context of 4D scenario generation, neuro-symbolic approaches facilitate the creation of worlds that are both semantically rich and dynamically coherent. By utilizing large language models to decompose complex text prompts into editable 3D layouts, these systems can generate multi-object scenes with precise spatial and temporal relationships. This compositional generation is further enhanced by incorporating probabilistic abduction and execution mechanisms, which allow the system to infer hidden rules from observed data and execute them to predict future states [447]. Such capabilities are essential for applications requiring high-level cognitive functions, such as robotic instruction following, where a robot must infer goal predicates from natural language and synthesize feasible action sequences [448].

Furthermore, the fusion of neural perception with symbolic physics engines enables the development of "intuitive physics" models that can generalize to novel situations much like humans do. While memory-based neural models can mimic human judgments in familiar contexts, simulation-based models that incorporate explicit physical reasoning demonstrate superior generalization to unseen scenarios and align better with human perceptual illusions and judgment asymmetries [449]. This suggests that future world models should not rely solely on end-to-end learning but should instead adopt hybrid architectures where neural frontends extract object properties and symbolic backends simulate their evolution. For example, models that learn layered representations of the physical world can use allocentric viewpoints to simplify dynamics prediction, effectively separating viewpoint complexity from physical interaction [450].

The potential of neuro-symbolic convergence extends to complex event detection and temporal reasoning, where traditional neural architectures often fail due to limited context windows. By integrating symbolic finite-state machines with neural encoders, systems can recognize complex event patterns in multimodal data streams with significantly higher accuracy than purely neural counterparts [451]. This synergy allows for the modeling of long-duration events and the reasoning about causal chains that span extended time horizons. Additionally, the application of neuro-symbolic methods to spatio-temporal reasoning provides a framework for grounding abstract concepts in physical reality, enabling AI agents to navigate and act in space and time with a level of sophistication previously unattainable [452].

Looking forward, the trajectory of dynamic scene reconstruction points toward generative world models that are inherently neuro-symbolic. These models will likely leverage the scalability of foundation models for perception while relying on symbolic engines for verification, planning, and physical consistency. The ultimate goal is to create autonomous agents capable of robust operation in unstructured environments, where they can not only perceive and predict but also reason about the "why" and "how" of dynamic changes. As the field progresses, addressing challenges such as the formal linkage between reasoning shortcuts and loss function optima, and developing efficient algorithms for probabilistic abduction in high-dimensional spaces, will be crucial [445]. The convergence of these disciplines promises to unlock a new generation of AI systems that possess both the adaptability of neural networks and the reliability of symbolic logic, paving the way for truly intelligent and trustworthy dynamic scene understanding.

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[356] SLAMBench2 Multi-Objective Head-to-Head Benchmarking for Visual SLAM

[357] Robust SLAM Systems Are We There Yet

[358] Semantic Gaussians Open-Vocabulary Scene Understanding with 3D Gaussian Splatting

[359] SAI3D Segment Any Instance in 3D Scenes

[360] PartSLIP++ Enhancing Low-Shot 3D Part Segmentation via Multi-View Instance Segmentation and Maximum Likelihood Estimation

[361] ClipSAM CLIP and SAM Collaboration for Zero-Shot Anomaly Segmentation

[362] Distilled Feature Fields Enable Few-Shot Language-Guided Manipulation

[363] Unified Scene Representation and Reconstruction for 3D Large Language Models

[364] CLIP2Scene Towards Label-efficient 3D Scene Understanding by CLIP

[365] PointCLIP Point Cloud Understanding by CLIP

[366] LayoutGPT Compositional Visual Planning and Generation with Large Language Models

[367] LLM Blueprint Enabling Text-to-Image Generation with Complex and Detailed Prompts

[368] Progressive3D Progressively Local Editing for Text-to-3D Content Creation with Complex Semantic Prompts

[369] GALA3D Towards Text-to-3D Complex Scene Generation via Layout-guided Generative Gaussian Splatting

[370] DreamScape 3D Scene Creation via Gaussian Splatting joint Correlation Modeling

[371] Towards Language-guided Interactive 3D Generation LLMs as Layout Interpreter with Generative Feedback

[372] TIP-Editor An Accurate 3D Editor Following Both Text-Prompts And Image-Prompts

[373] Urban Architect Steerable 3D Urban Scene Generation with Layout Prior

[374] InstructScene Instruction-Driven 3D Indoor Scene Synthesis with Semantic Graph Prior

[375] 4D-fy Text-to-4D Generation Using Hybrid Score Distillation Sampling

[376] Align Your Gaussians Text-to-4D with Dynamic 3D Gaussians and Composed Diffusion Models

[377] TC4D Trajectory-Conditioned Text-to-4D Generation

[378] CG-HOI Contact-Guided 3D Human-Object Interaction Generation

[379] InterDreamer Zero-Shot Text to 3D Dynamic Human-Object Interaction

[380] HOI-Diff Text-Driven Synthesis of 3D Human-Object Interactions using Diffusion Models

[381] Comp4D LLM-Guided Compositional 4D Scene Generation

[382] SC4D Sparse-Controlled Video-to-4D Generation and Motion Transfer

[383] AnimatableDreamer Text-Guided Non-rigid 3D Model Generation and Reconstruction with Canonical Score Distillation

[384] 4D-Editor Interactive Object-level Editing in Dynamic Neural Radiance Fields via Semantic Distillation

[385] Semantic-Guided Inpainting Network for Complex Urban Scenes Manipulation

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[387] Data Augmented 3D Semantic Scene Completion with 2D Segmentation Priors

[388] Semantic Scene Completion via Integrating Instances and Scene in-the-Loop

[389] Free-form 3D Scene Inpainting with Dual-stream GAN

[390] Language-guided Semantic Style Transfer of 3D Indoor Scenes

[391] Soulstyler Using Large Language Model to Guide Image Style Transfer for Target Object

[392] OR-NeRF Object Removing from 3D Scenes Guided by Multiview Segmentation with Neural Radiance Fields

[393] Visiting the Invisible Layer-by-Layer Completed Scene Decomposition

[394] BBBD Bounding Box Based Detector for Occlusion Detection and Order Recovery

[395] Diffusion Model is Secretly a Training-free Open Vocabulary Semantic Segmenter

[396] EmerDiff Emerging Pixel-level Semantic Knowledge in Diffusion Models

[397] FreeSeg-Diff Training-Free Open-Vocabulary Segmentation with Diffusion Models

[398] Open-Vocabulary Attention Maps with Token Optimization for Semantic Segmentation in Diffusion Models

[399] SLiMe Segment Like Me

[400] OpenMask3D Open-Vocabulary 3D Instance Segmentation

[401] PARIS3D Reasoning-based 3D Part Segmentation Using Large Multimodal Model

[402] DiffusionSeg Adapting Diffusion Towards Unsupervised Object Discovery

[403] Interactive Object Segmentation in 3D Point Clouds

[404] SceneGenie Scene Graph Guided Diffusion Models for Image Synthesis

[405] Be Yourself Bounded Attention for Multi-Subject Text-to-Image Generation

[406] Leveraging Open-Vocabulary Diffusion to Camouflaged Instance Segmentation

[407] COARSE3D Class-Prototypes for Contrastive Learning in Weakly-Supervised 3D Point Cloud Segmentation

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[409] Adaptive-SpikeNet Event-based Optical Flow Estimation using Spiking Neural Networks with Learnable Neuronal Dynamics

[410] Event-Based Angular Velocity Regression with Spiking Networks

[411] Fusion-FlowNet Energy-Efficient Optical Flow Estimation using Sensor Fusion and Deep Fused Spiking-Analog Network Architectures

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[427] Paired Competing Neurons Improving STDP Supervised Local Learning In Spiking Neural Networks

[428] Structural plasticity on an accelerated analog neuromorphic hardware system

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[432] On the Self-Repair Role of Astrocytes in STDP Enabled Unsupervised SNNs

[433] Brain-inspired global-local learning incorporated with neuromorphic computing

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[435] Ensembles of Compact, Region-specific & Regularized Spiking Neural Networks for Scalable Place Recognition

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[437] Gridbot An autonomous robot controlled by a Spiking Neural Network mimicking the brain's navigational system

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[442] FMGS Foundation Model Embedded 3D Gaussian Splatting for Holistic 3D Scene Understanding

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